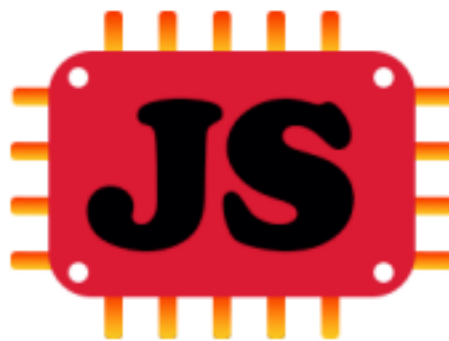




ELECTRONICS

JFET

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List of topics covered in this note



1. Introduction to JFET
2. Modes of Operation of JFETs and their Characteristic curve
3. Equations and Problem solving on JFETs
4. Introduction to MOSFETs
5. Modes of Operation of Enhancement type MOSFETs and their Characteristic curve
6. Governing Equations and Problem solving (E-type MOSFETs)
7. Modes of Operation of Depletion type MOSFETs and their Characteristic curve
8. Governing Equations and Problem solving (D-type MOSFETs)
9. MOSFETs as Amplifier (Theory & Problem solving)



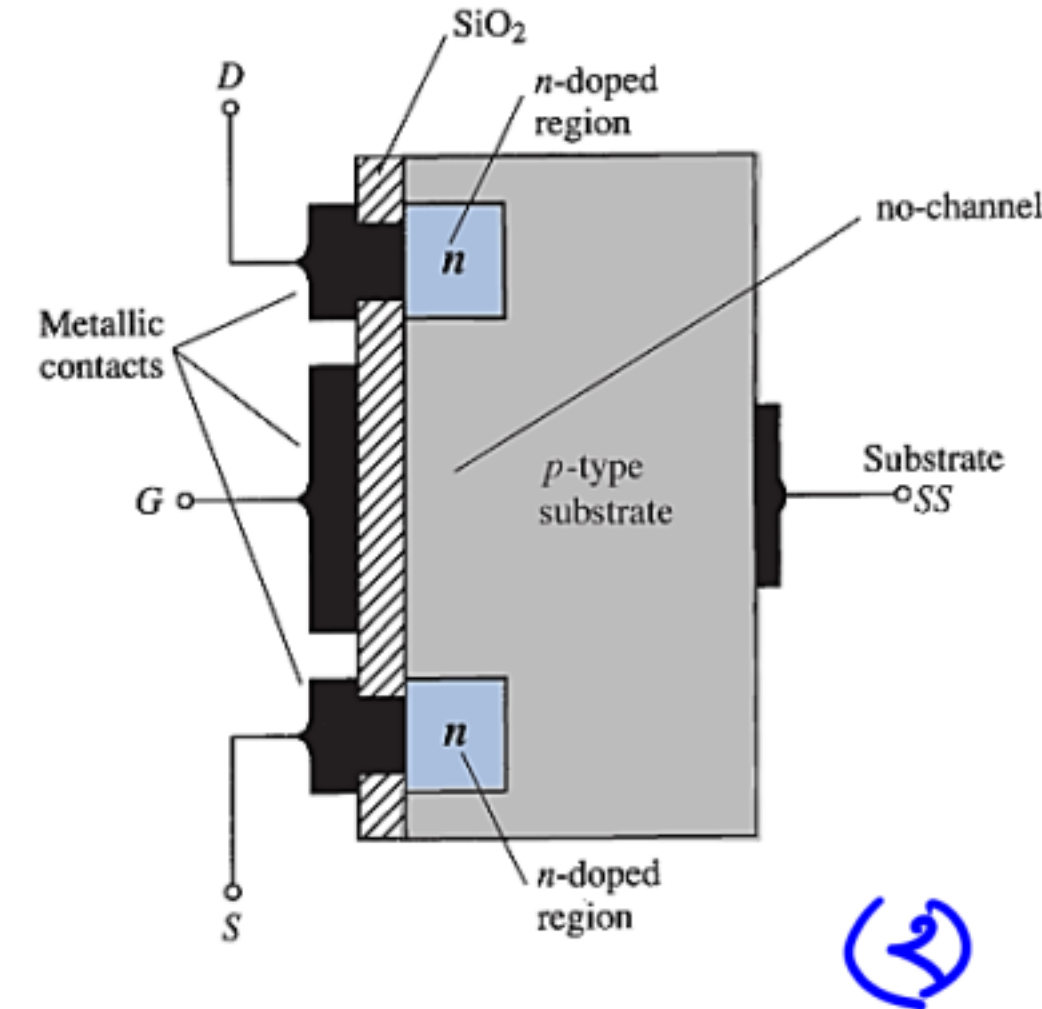
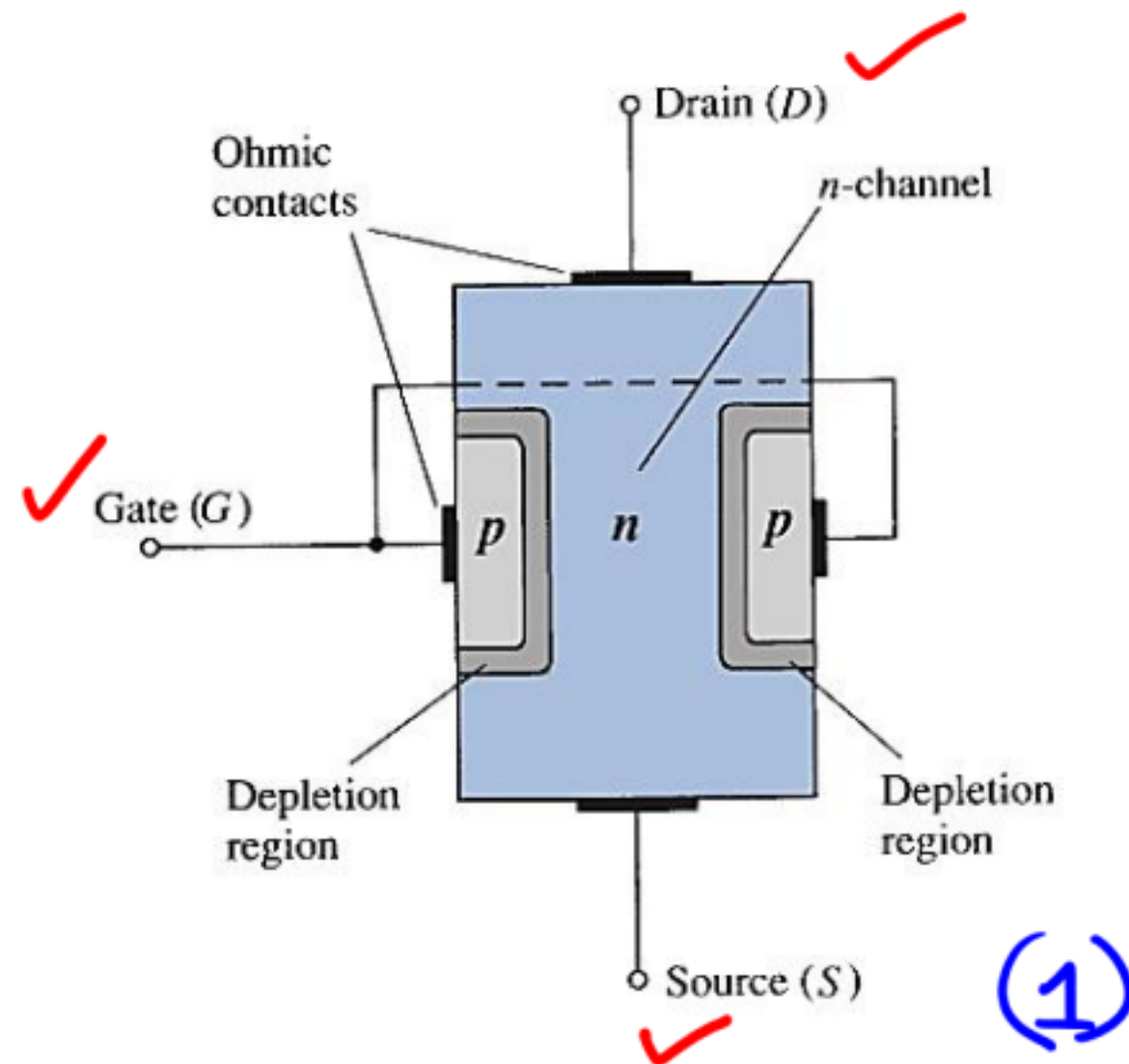
Concept of FET



- FET stands for “Field Effect Transistor” *(viva)*
- It is so named because in FETs, charge/ current flow is affected by the Electric Field.

FETs are of two types:

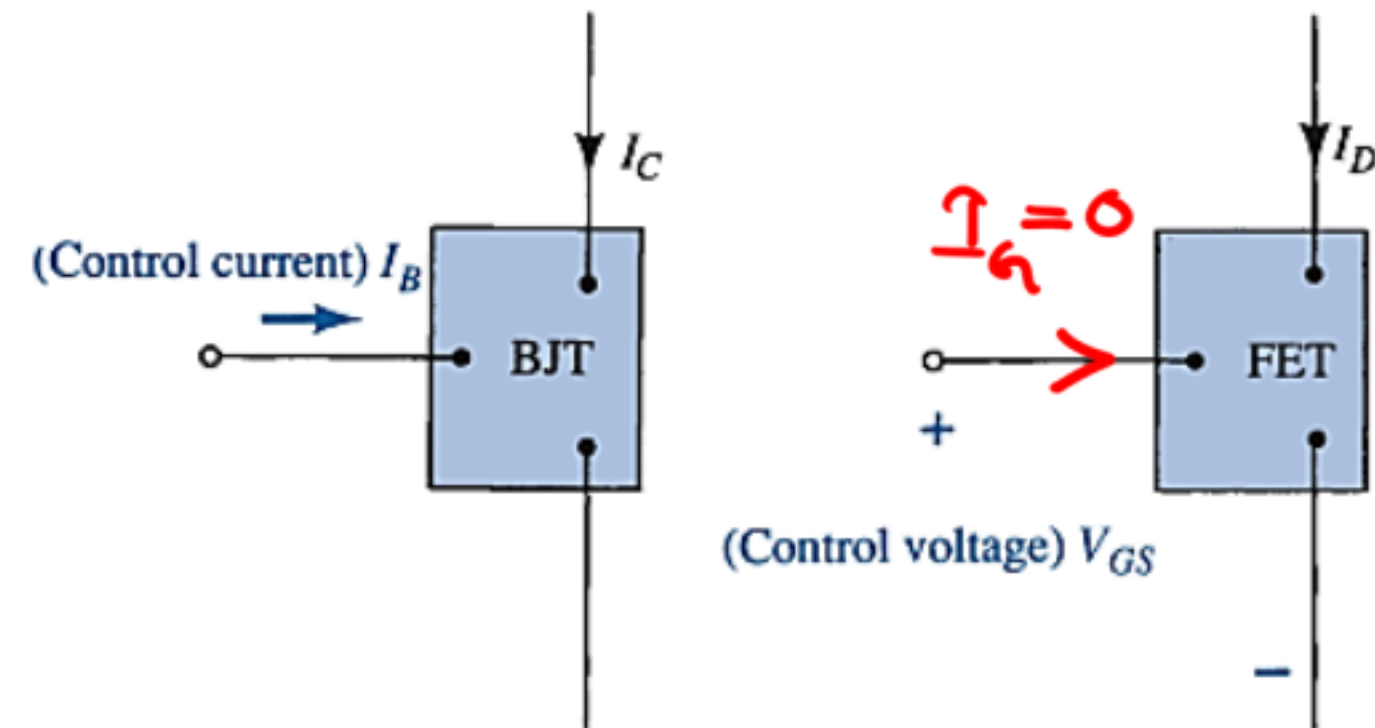
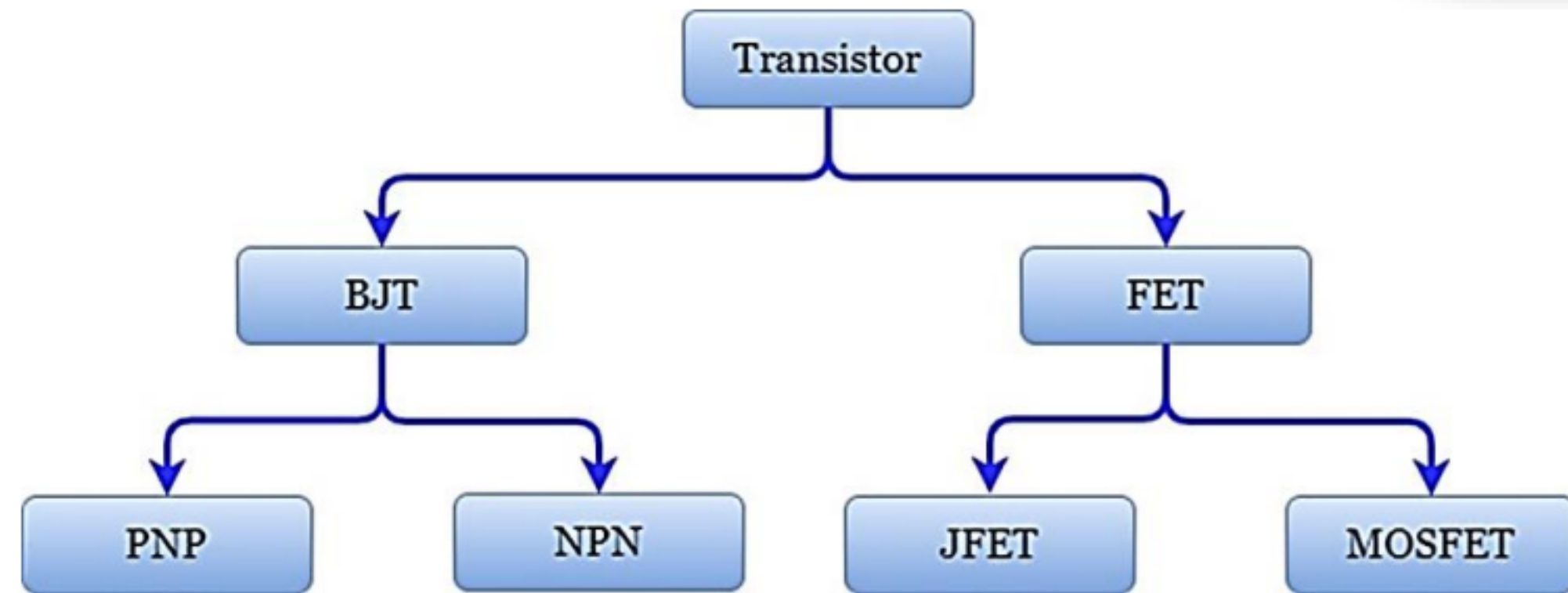
1. JFET (Junction FET)
2. MOSFET (Metal Oxide Semiconductor FET)



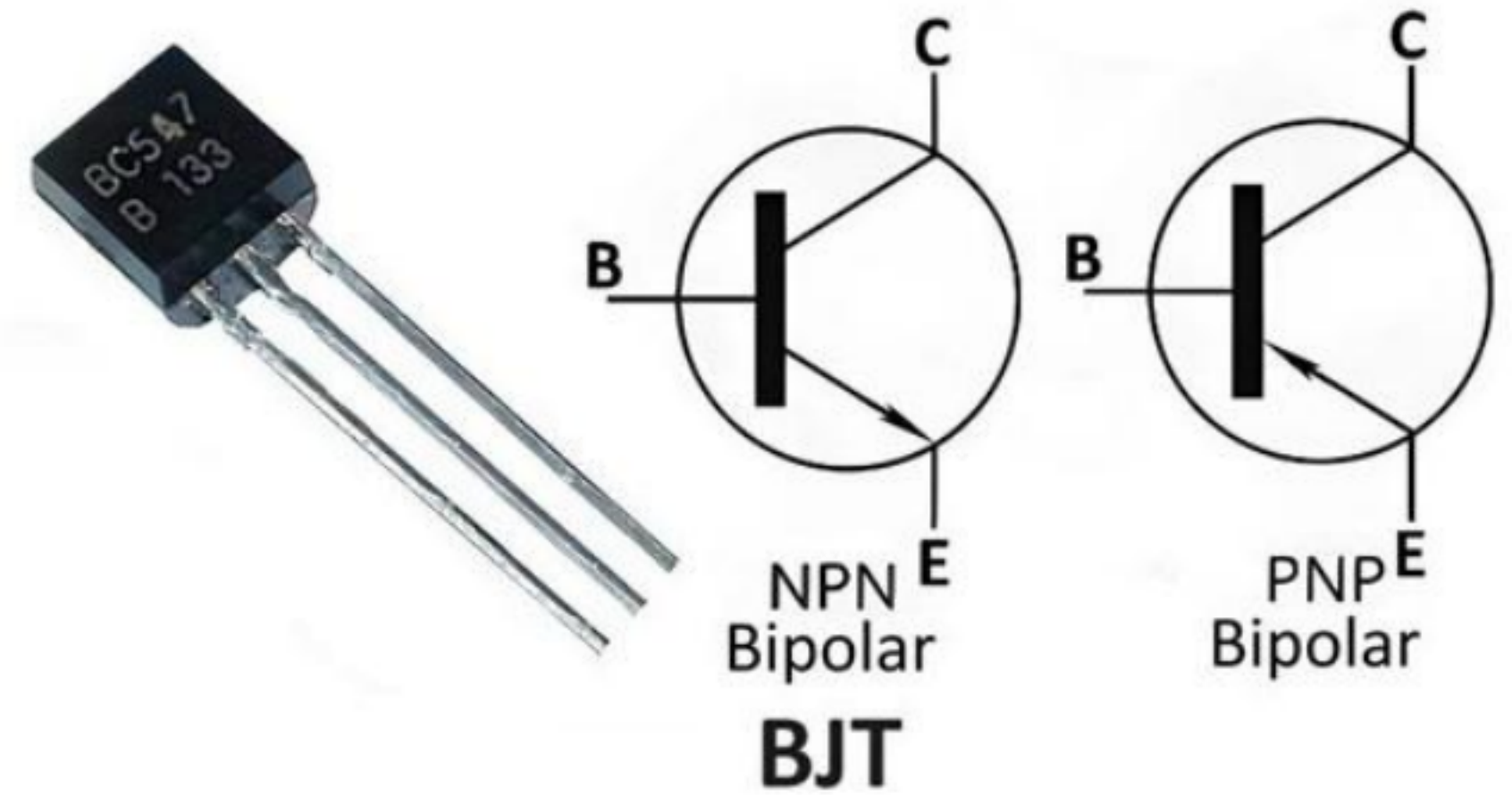
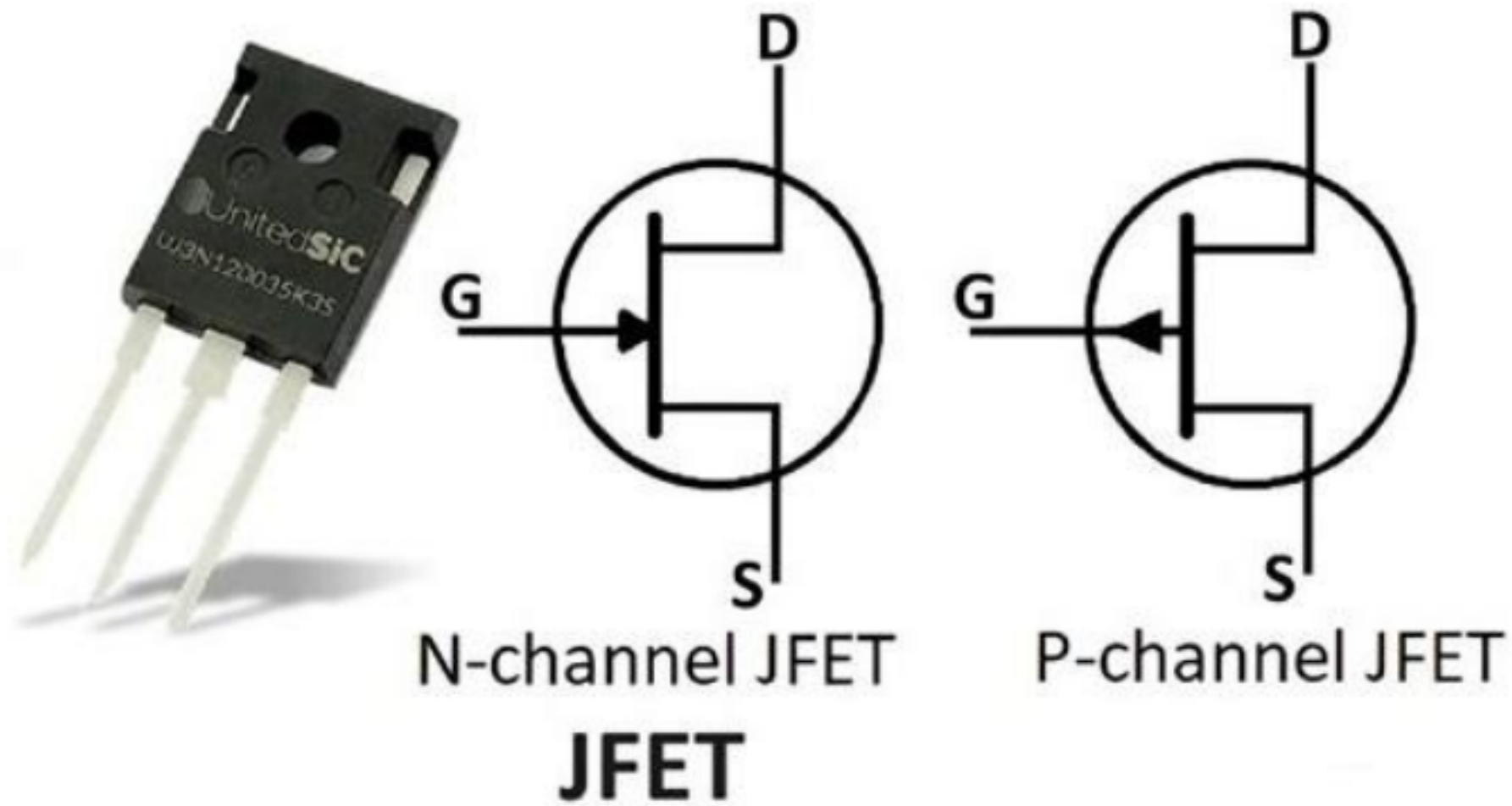
JFET



- JFET = Junction Field Effect Transistor
- It is a 3-terminal device. (Gate, Source, Drain)
- It is also called “**majority carrier device**”
- FETs have high input impedance
- FETs are more temperature-stable than BJTs
- FETs are smaller in size compared to BJTs making them more useful in IC (integrated-circuit) chips.



JFET chip vs. BJT chip



Types of JFET



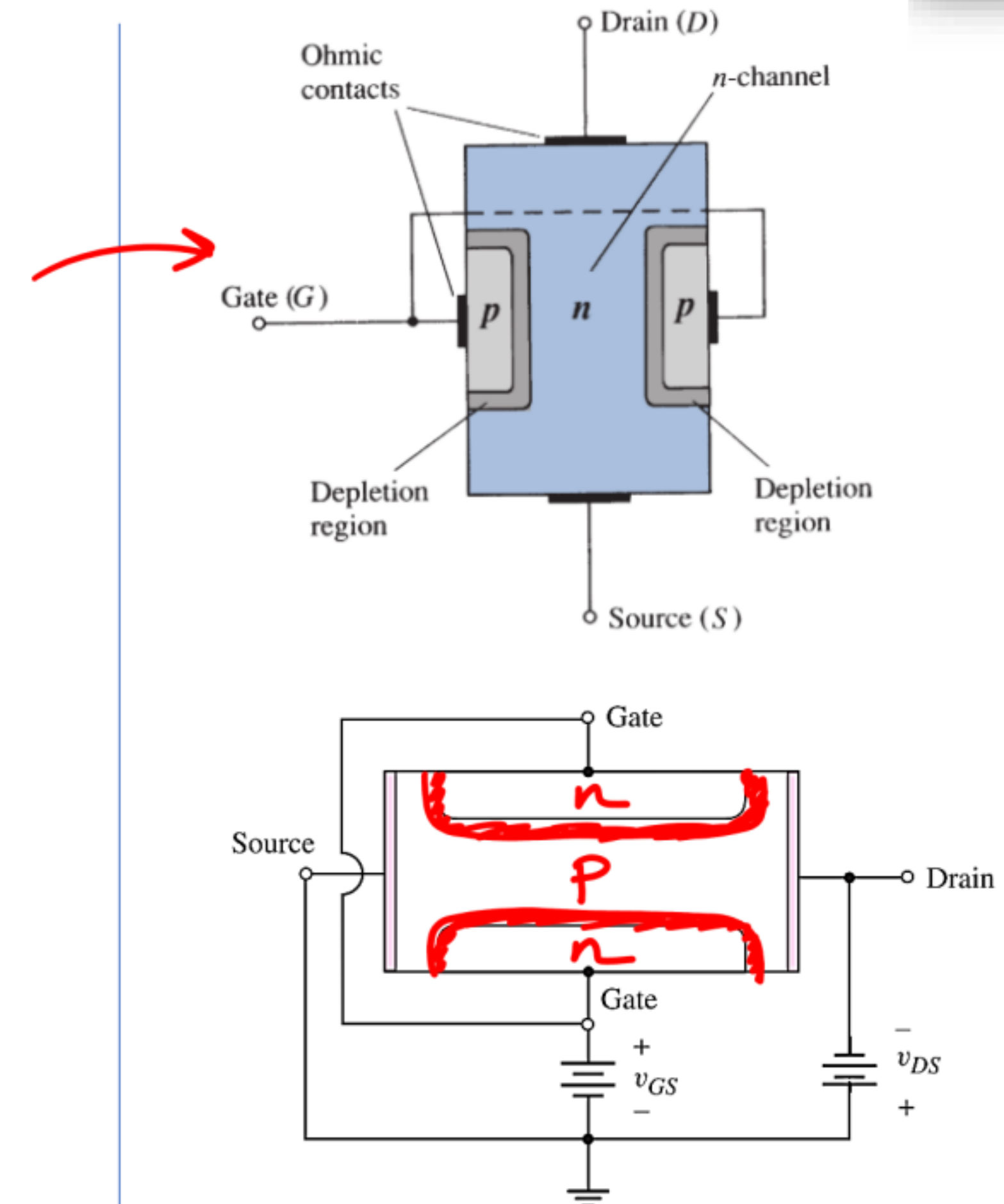
JFETs are classified into two categories.

1. N-channel JFET :

- The channel is n-type.
- Majority charge-carrier “electrons” flows through the channel.
- There are two (p⁺)-doped regions at the two sides.

2. P-channel JFET :

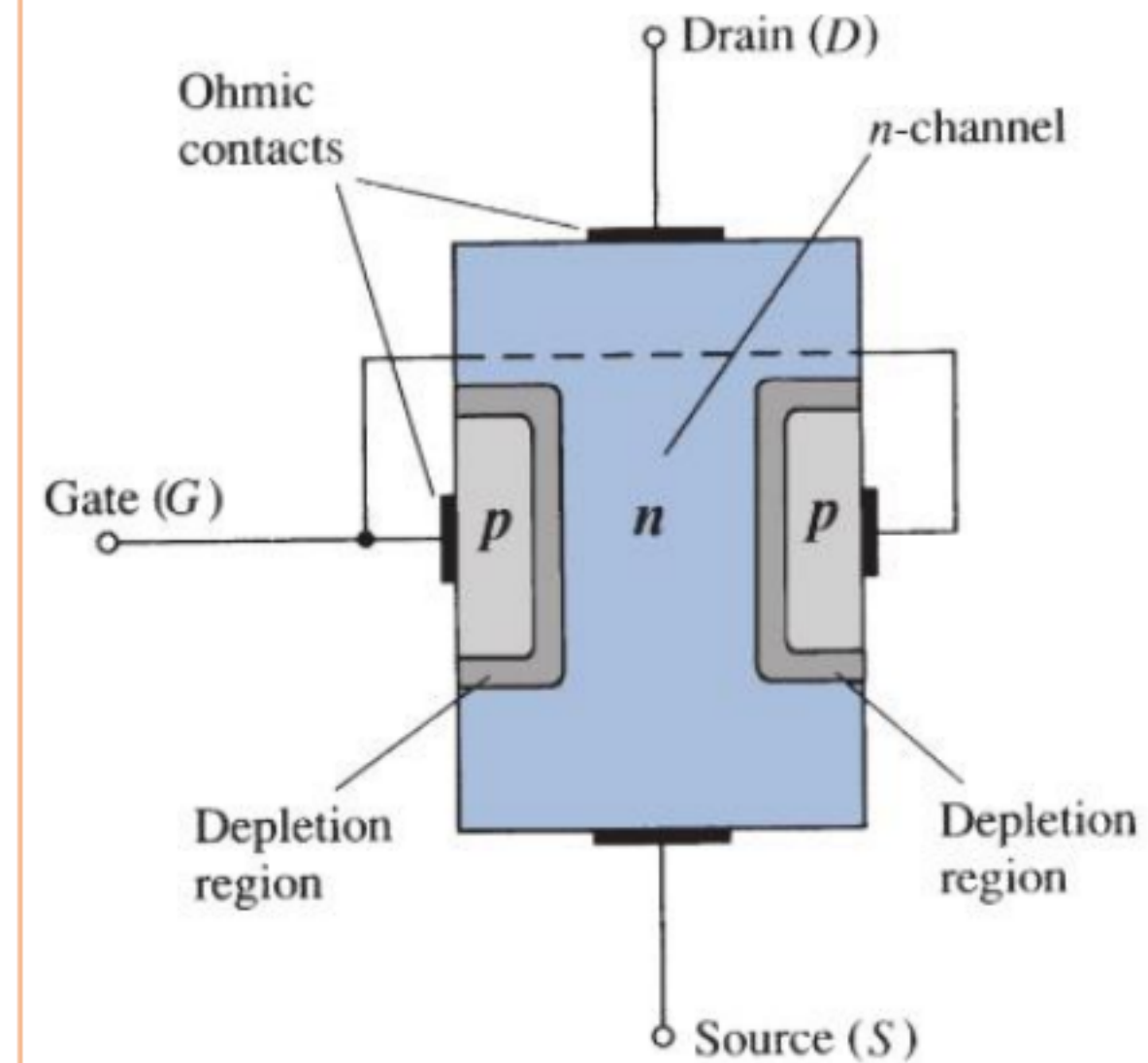
- The channel is p-type.
- Majority charge-carrier “holes” flows through the channel.
- There are two (n⁺)-doped regions at the two sides.



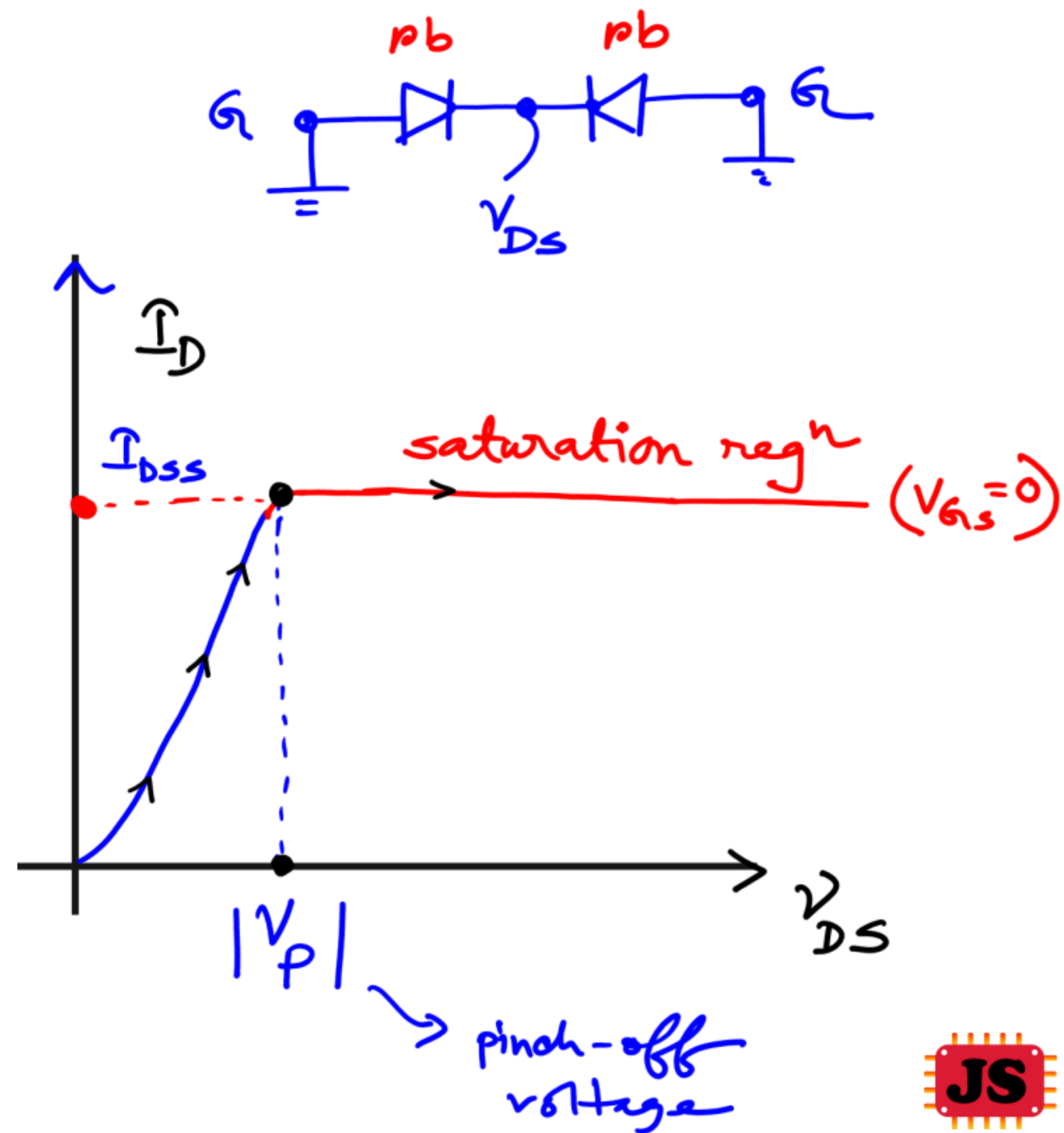
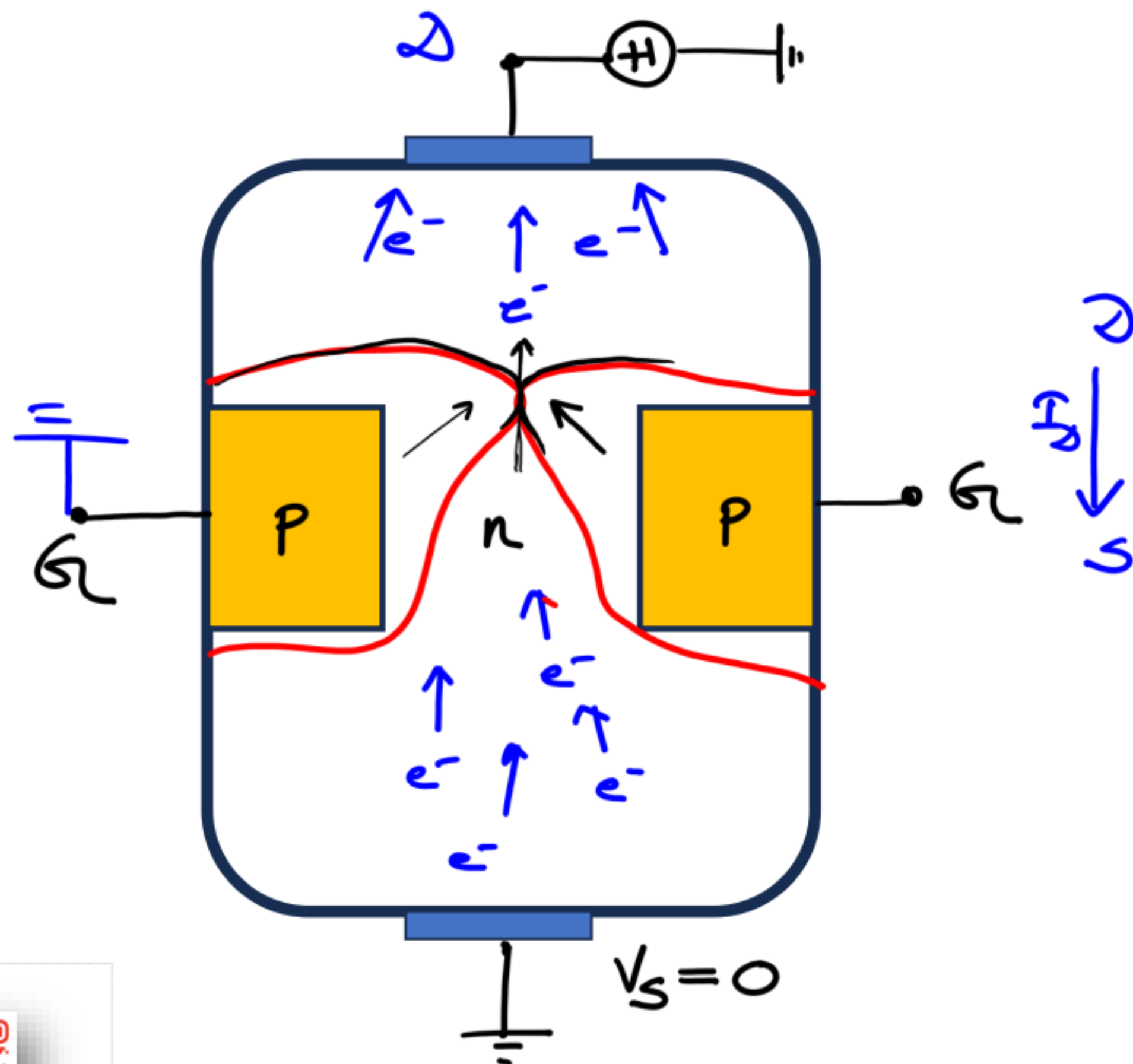
Introduction to N-JFET



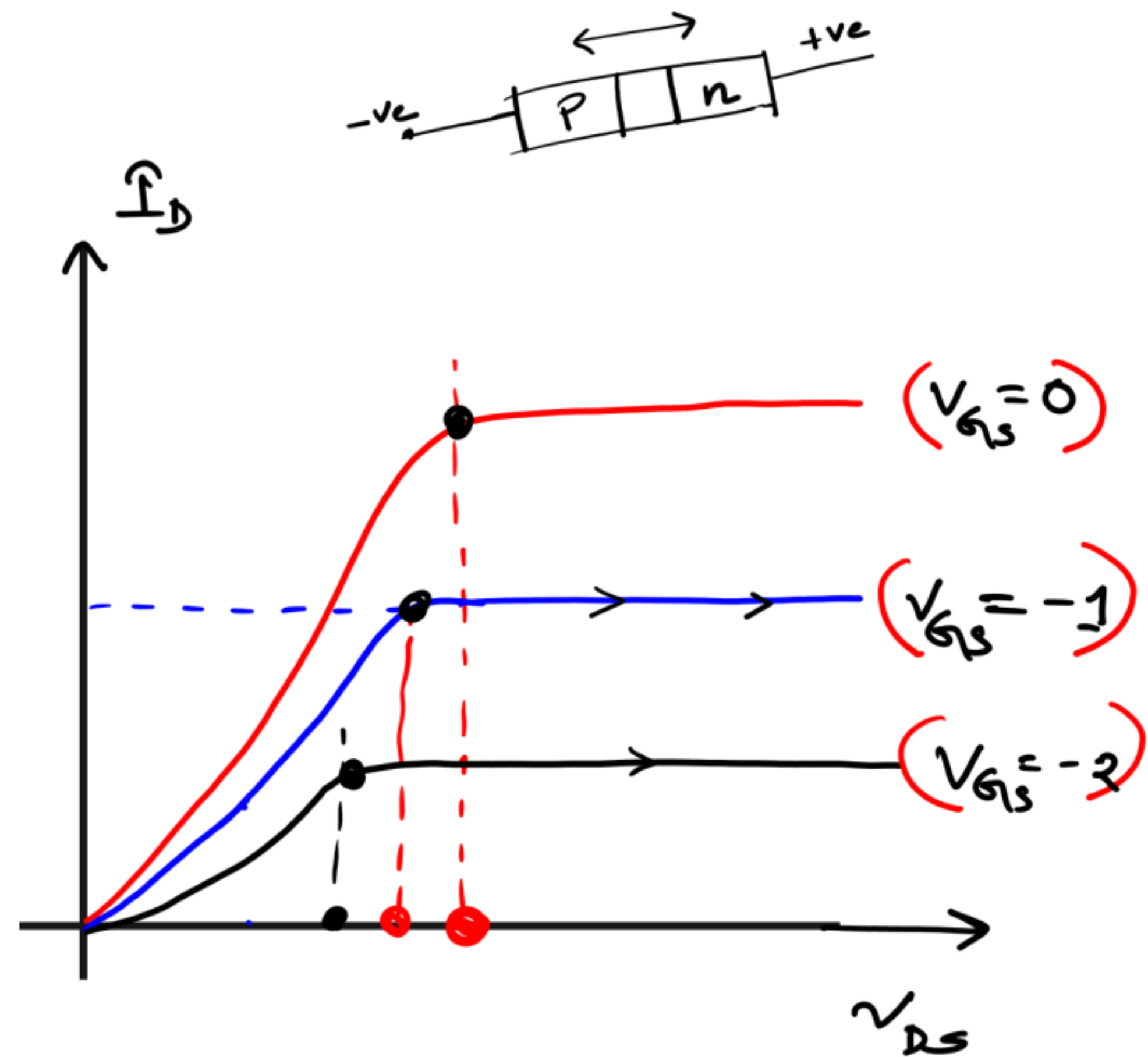
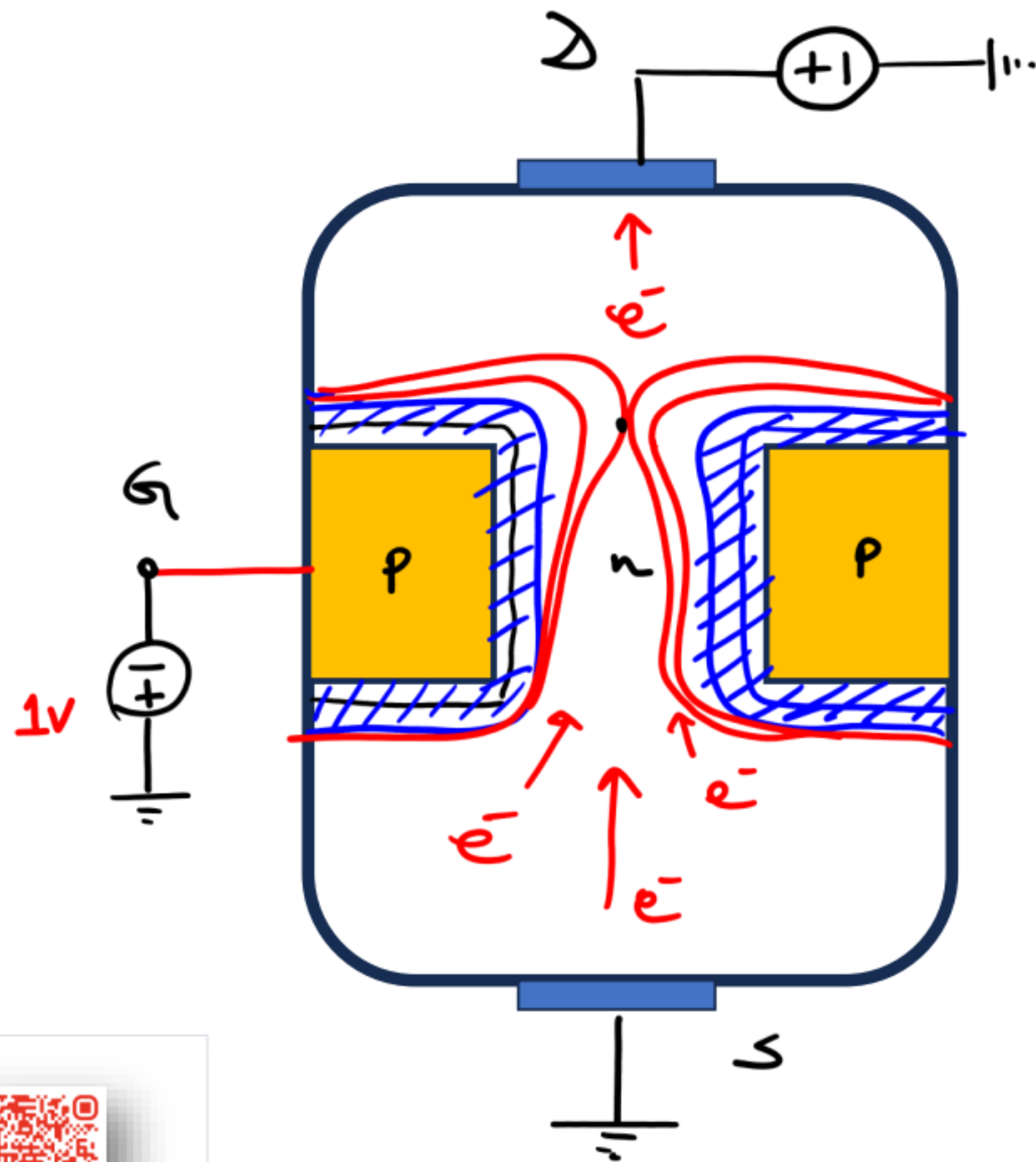
- In N-JFET, the channel where the current flow is “n-type”
- The n-channel is surrounded by two p-type layers
- With no bias voltages at any nodes, there are two depletion regions in the JFET. (in between the 02 p-n junctions)
- Depletion regions are created whenever a p-type & an n-type semi-conductor are joined. (Refer to our Diode lecture)



Understanding the mode of operation of n-JFET



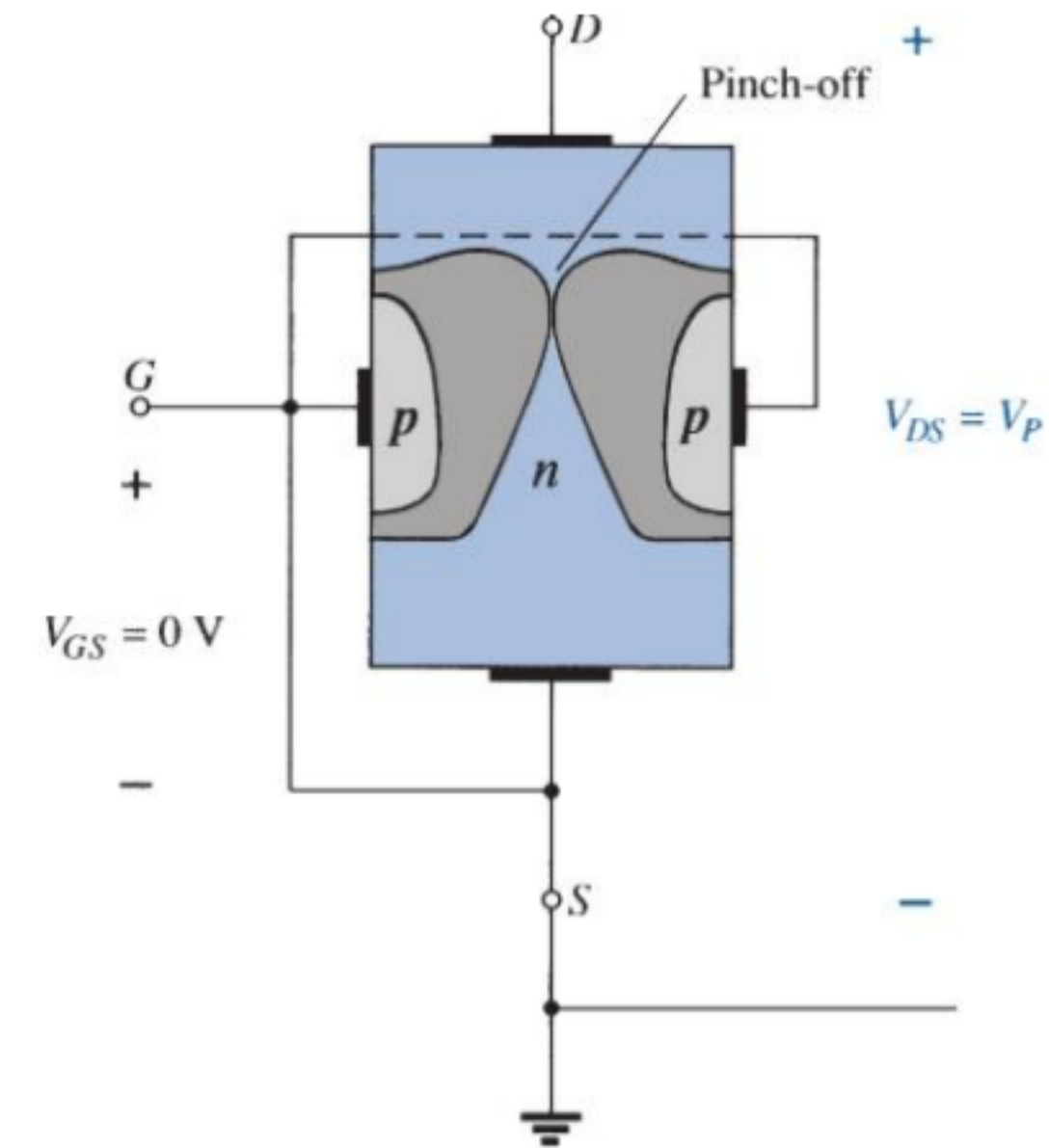
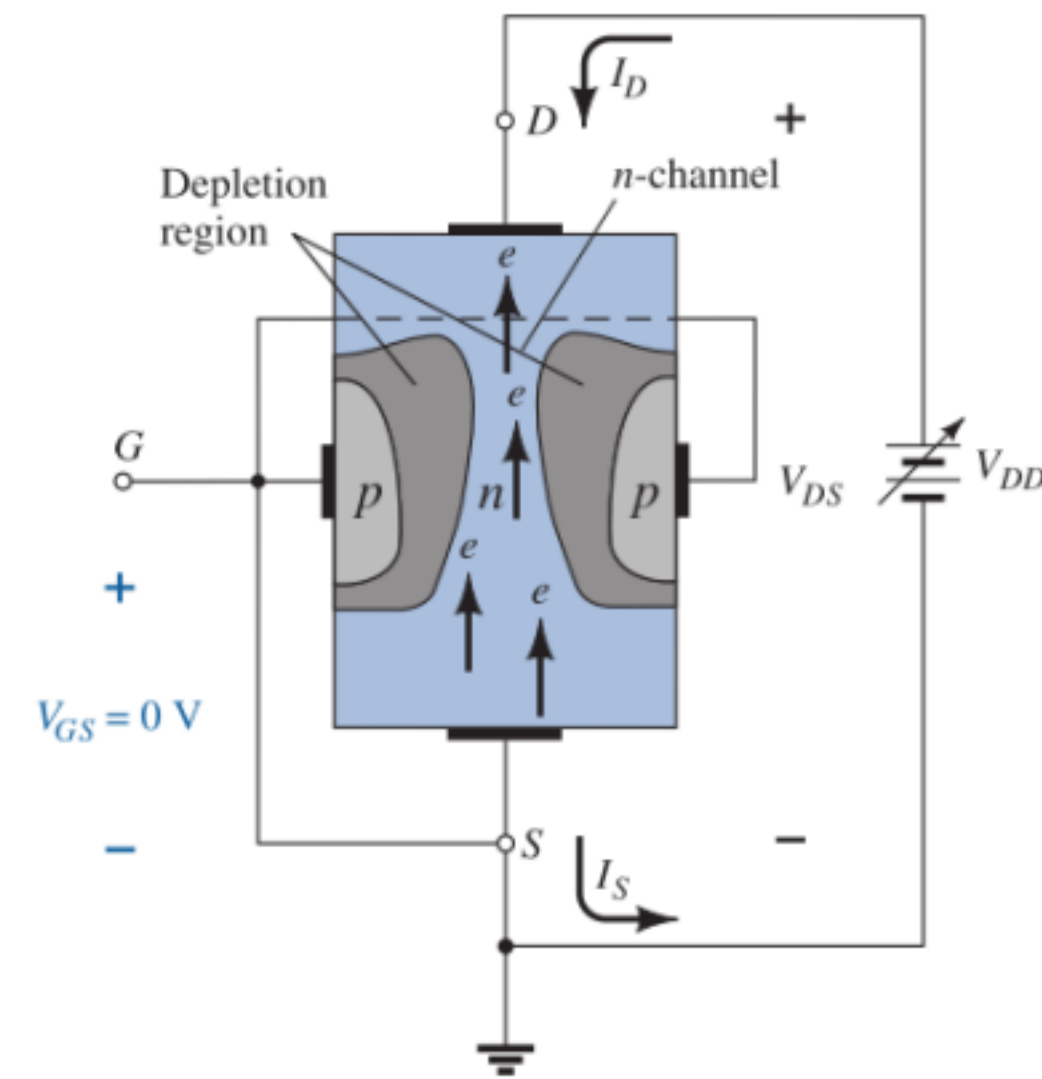
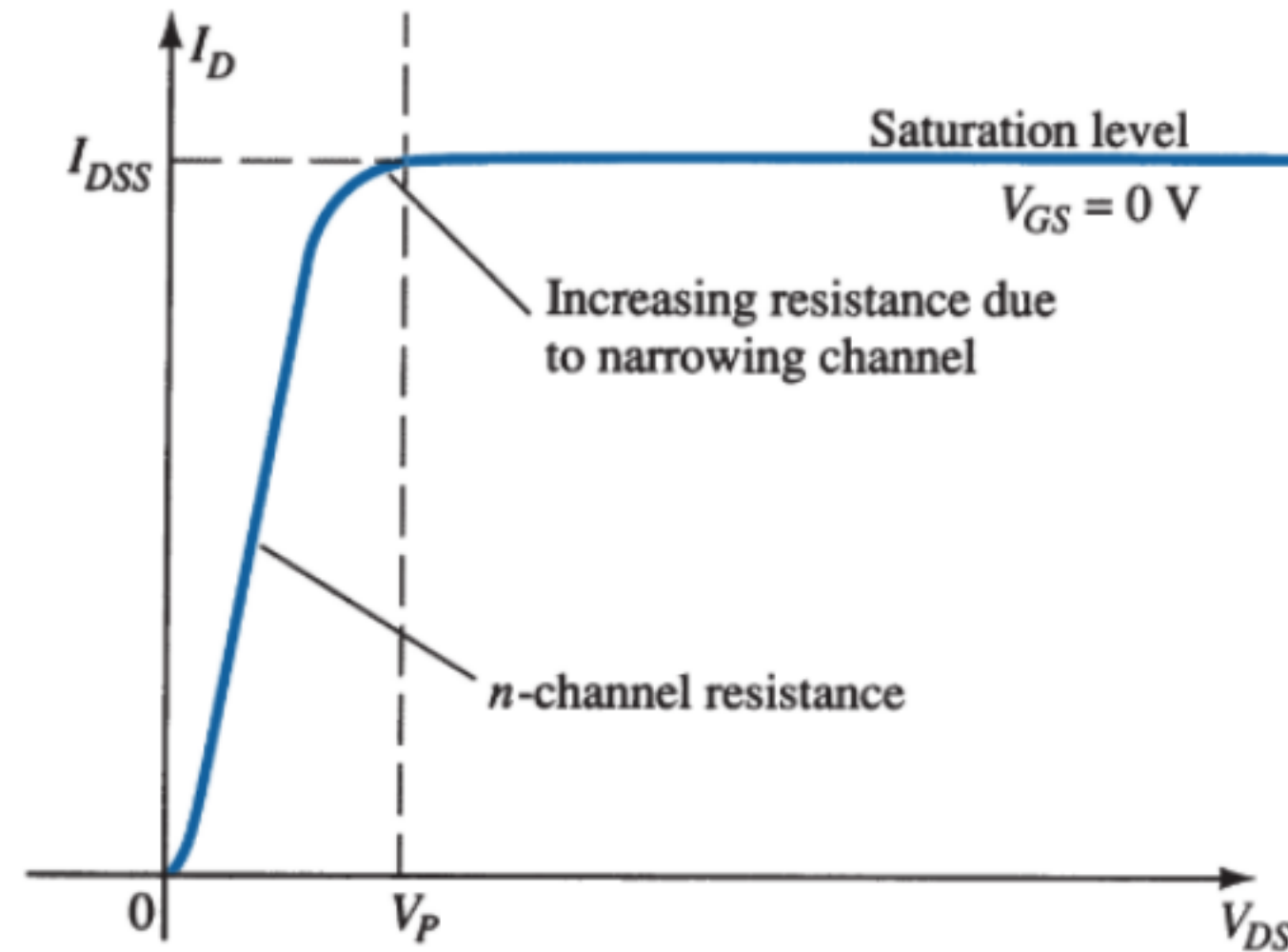
Understanding the mode of operation of n-JFET



Mode of operation in N-JFET (1)

Case 01: $V_{GS} = 0$ & $V_{DS} > 0$

- Due to voltage between drain and source, current starts to flow
- The depletion region also increases in size decreasing the n-channel width
- Current will continue to increase with increasing V_{DS} until the channel is pinched off!
- The maximum value of I_D is termed as I_{DSS}
- Current I_D reaches I_{DSS} when voltage V_{DS} reaches $|V_p|$
- V_p (pinch-off voltage)

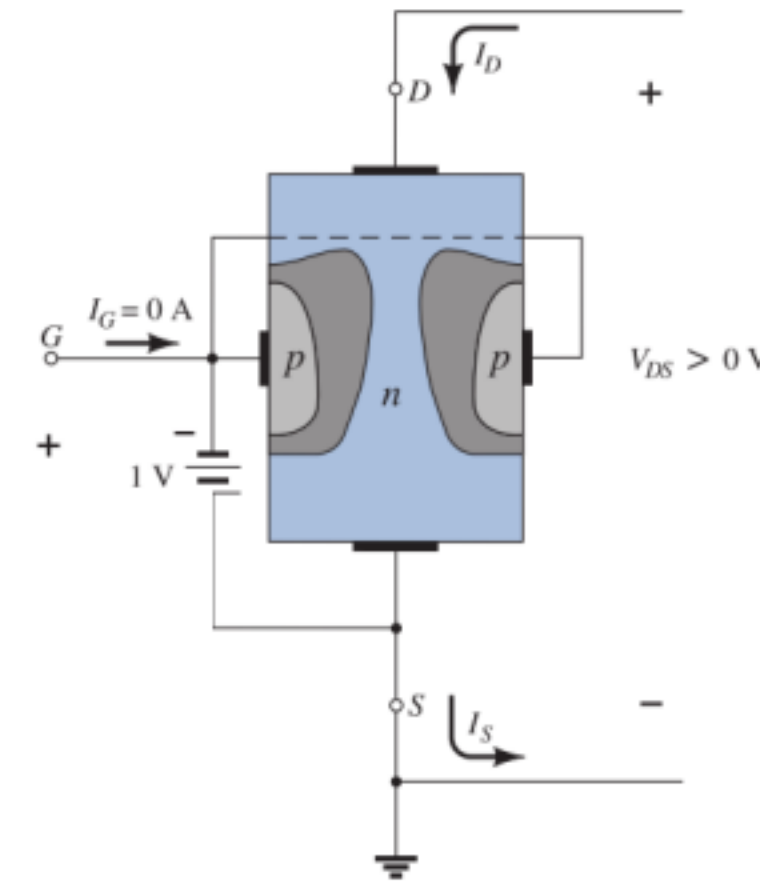
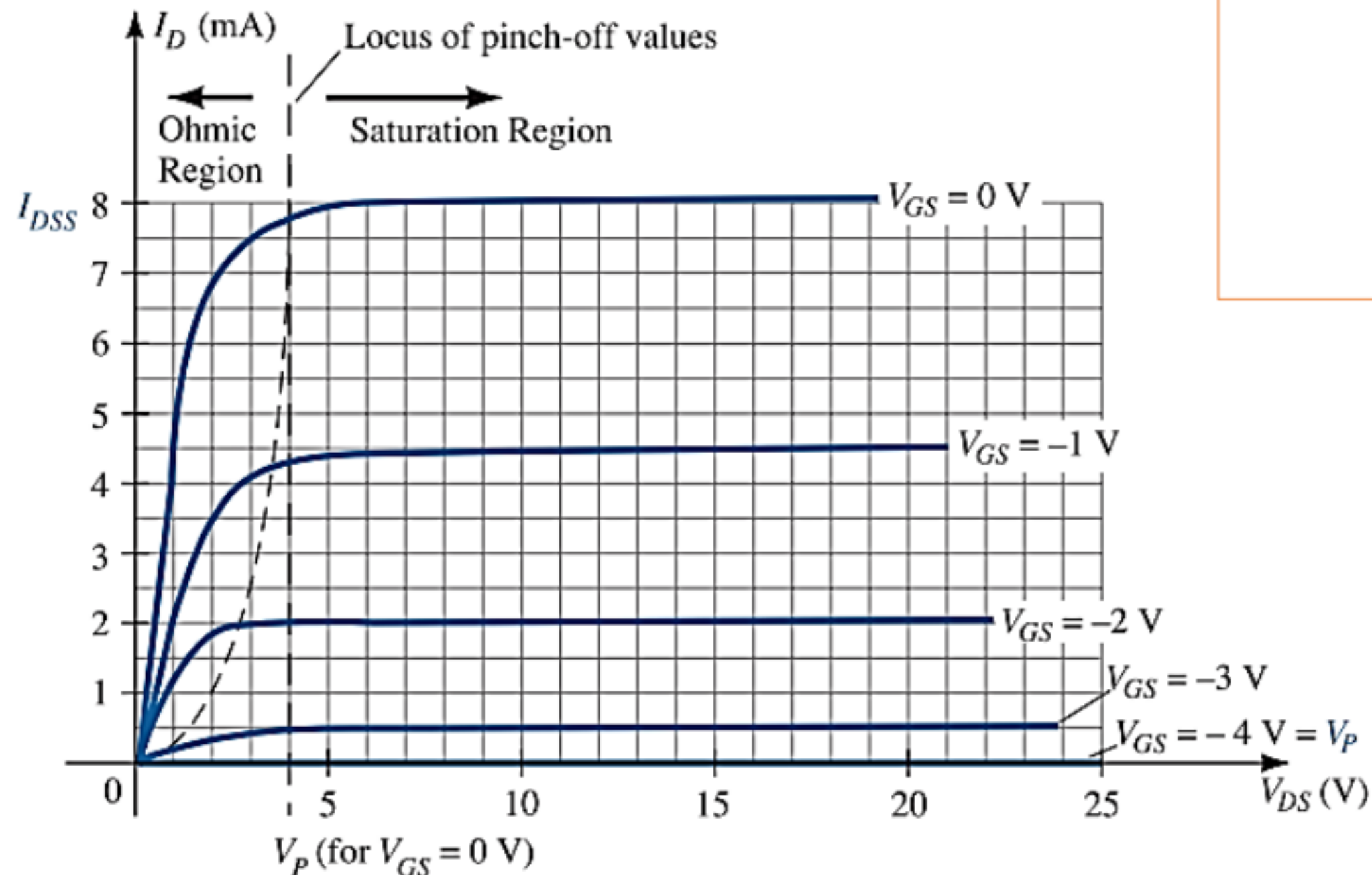


Mode of operation in N-JFET (2) & characteristic curve



Case 02: $V_{GS} < 0$ & $V_{DS} > 0$

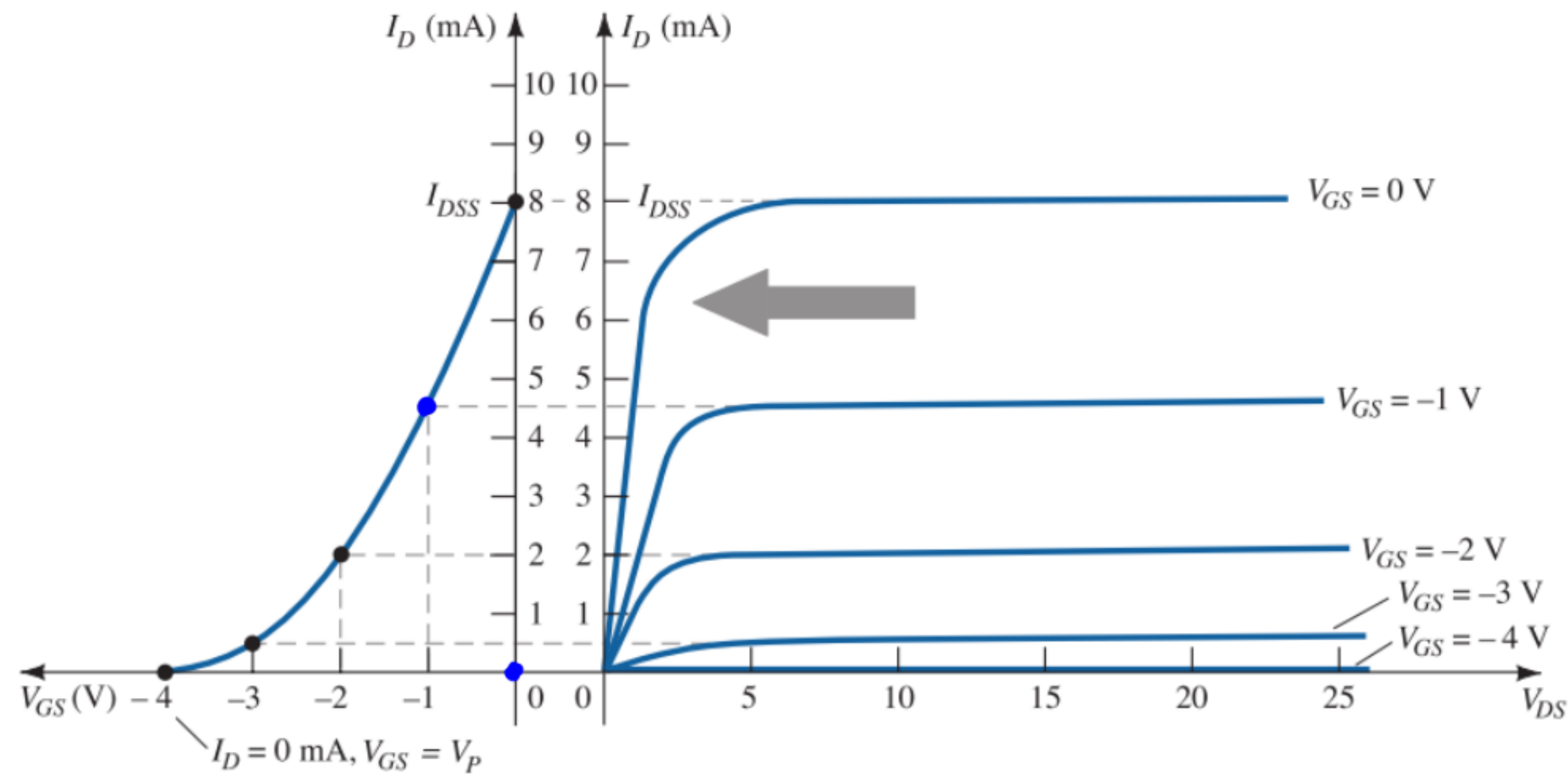
- Due to applying negative V_{GS} , the depletion increases even more, and the JFET will reach pinch-off quicker with lower values of I_{DSS}
- If $V_{GS} = V_p$ is applied, then even with bias voltage, the channel is already pinched-off,
- Thus, $I_D = 0$ at $V_{GS} = V_P$



Transfer Curve from Characteristics Curve of N-JFET



- I_D vs. $V_{GS} \Rightarrow$ transfer curve
- I_D vs. $V_{DS} \Rightarrow$ output characteristic curve



Cut-off & Pinch-off Voltage



- Pinch-off voltage, V_p is the voltage V_{DS} when I_D reaches saturation. ($V_{GS}=0$)
- Cut-off voltage, $V_{GS}(off)$ is the voltage V_{GS} for which $I_D = \text{zero}$.
- Interestingly, both $V_{GS}(off)$ & V_p are equal, but with different signs.

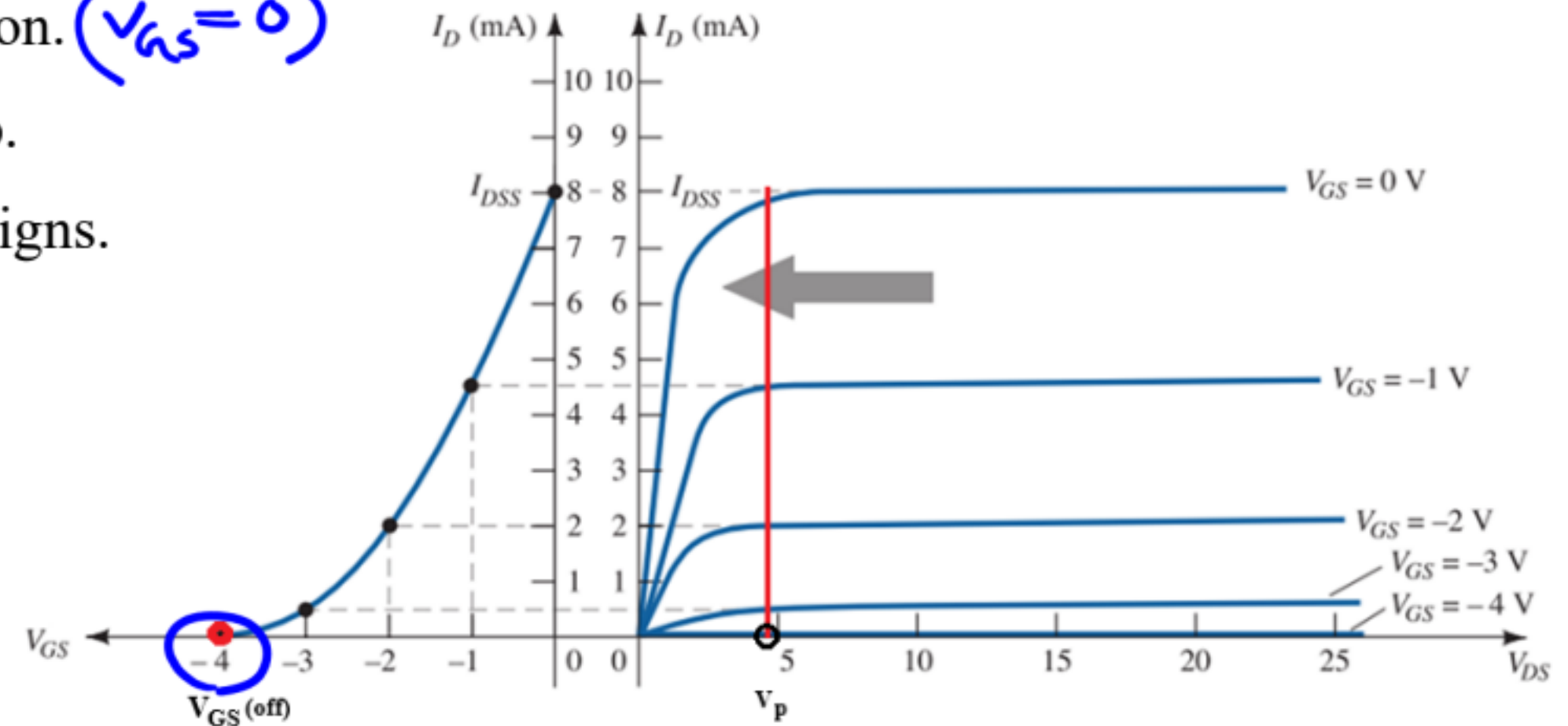
$\therefore V_{GS}(off) = -V_p$ (Source: Floyd – page 393)

$$\therefore I_D = I_{DSS} \left(1 - \underbrace{\frac{V_{GS}}{V_{GS}(off)}}_{-ve} \right)^2 \rightarrow \text{(Floyd)}$$

- According to Floyd, $V_{GS}(off)$ is negative, while V_p is positive.
- But according to Boylestad (see page 393), V_p is negative.

$$\therefore I_D = I_{DSS} \left(1 - \underbrace{\frac{V_{GS}}{V_p}}_{-ve} \right)^2 \rightarrow \text{(Boylestad)}$$

- Whichever you use, make sure to use negative value for V_p or $V_{GS}(off)$ for n-JFET & vice-versa for p-JFET

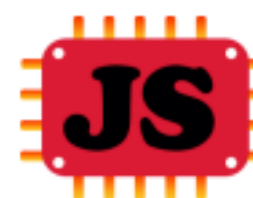
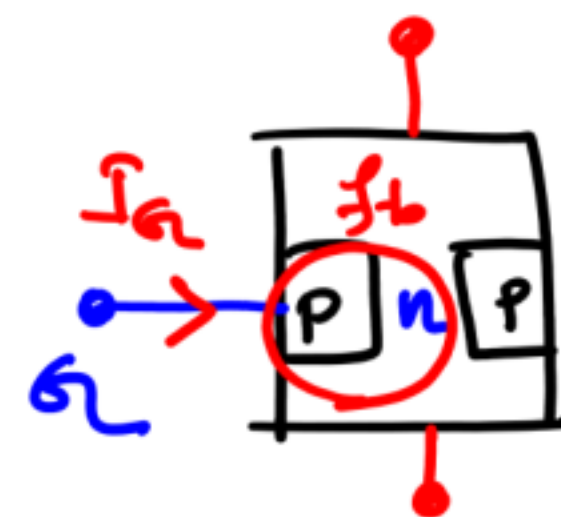


Why is $V_{GS} > 0$ in n-JFET **never** drawn in the characteristic curve?



The characteristic curve typically does not include regions where $V_{GS} > 0$ because: For an N-JFET, the gate is a p-type material, and the source is n-type. **Positive V_{GS} forward-biases the gate-source junction**, leading to a large current through the gate (I_G), which is generally undesirable.

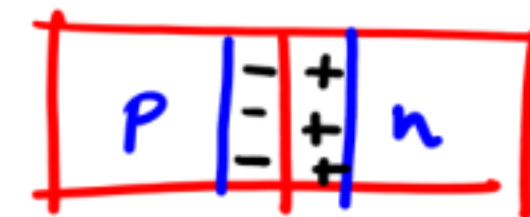
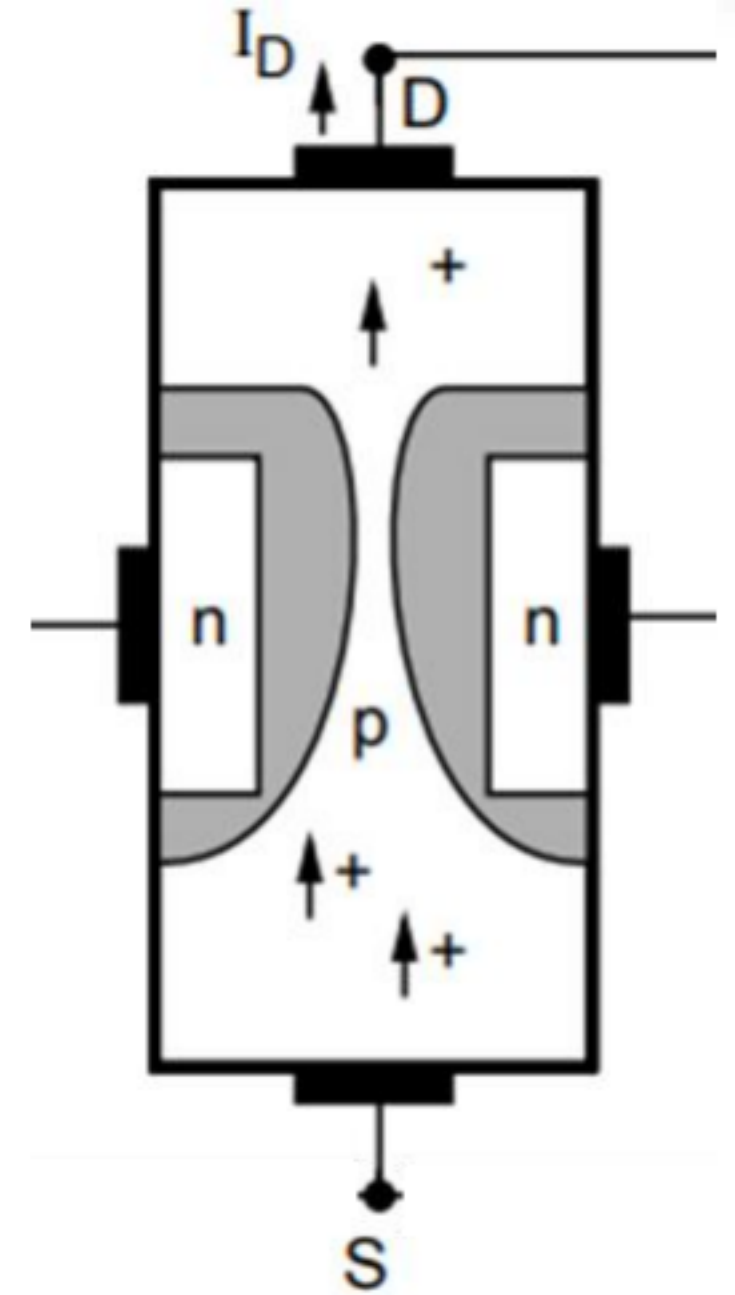
1. **Loss of Control:** The JFET loses its voltage-controlled characteristic because the gate current interferes with the channel operation.
2. **Possible Damage:** Forward biasing beyond the allowed limits can damage the gate-channel junction, as JFETs are not designed to handle significant gate current.
3. **Practical Considerations:** N-JFET is designed to operate in a region where $V_{GS} < 0$. Including $V_{GS} > 0$ in the characteristic curves is not practically useful, as it is outside the intended operating range.



Introduction to P-JFET



- In P-JFET, the channel where the current flow is “p-type”
- The p-channel is surrounded by two n-type layers
- With no bias voltages at any nodes, there are two depletion regions in the JFET. (in between the 02 p-n junctions)
- Depletion regions are created whenever a p-type & an n-type semi-conductor are joined. (Refer to our Diode lecture)

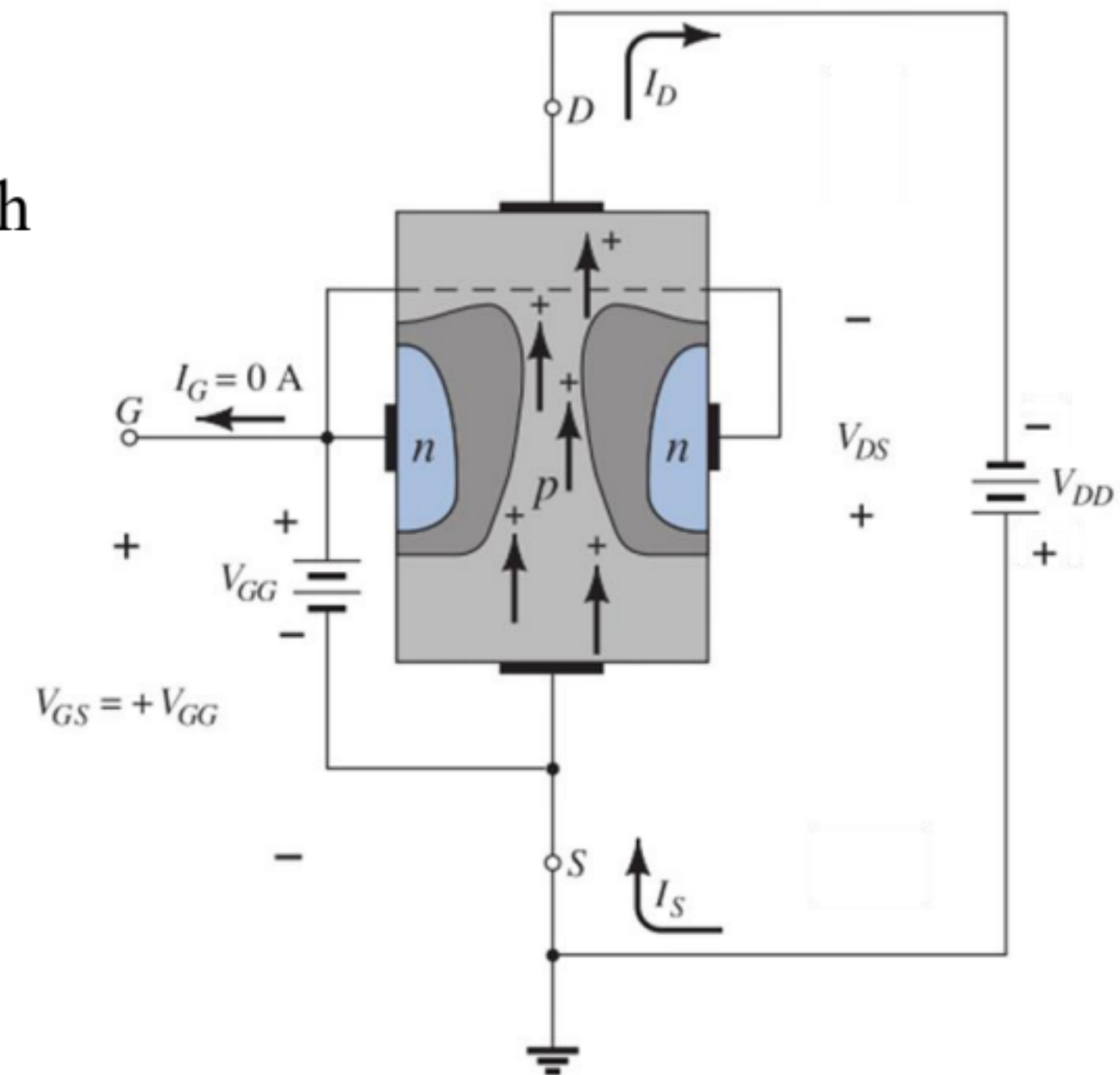


Mode of operation in P-JFET

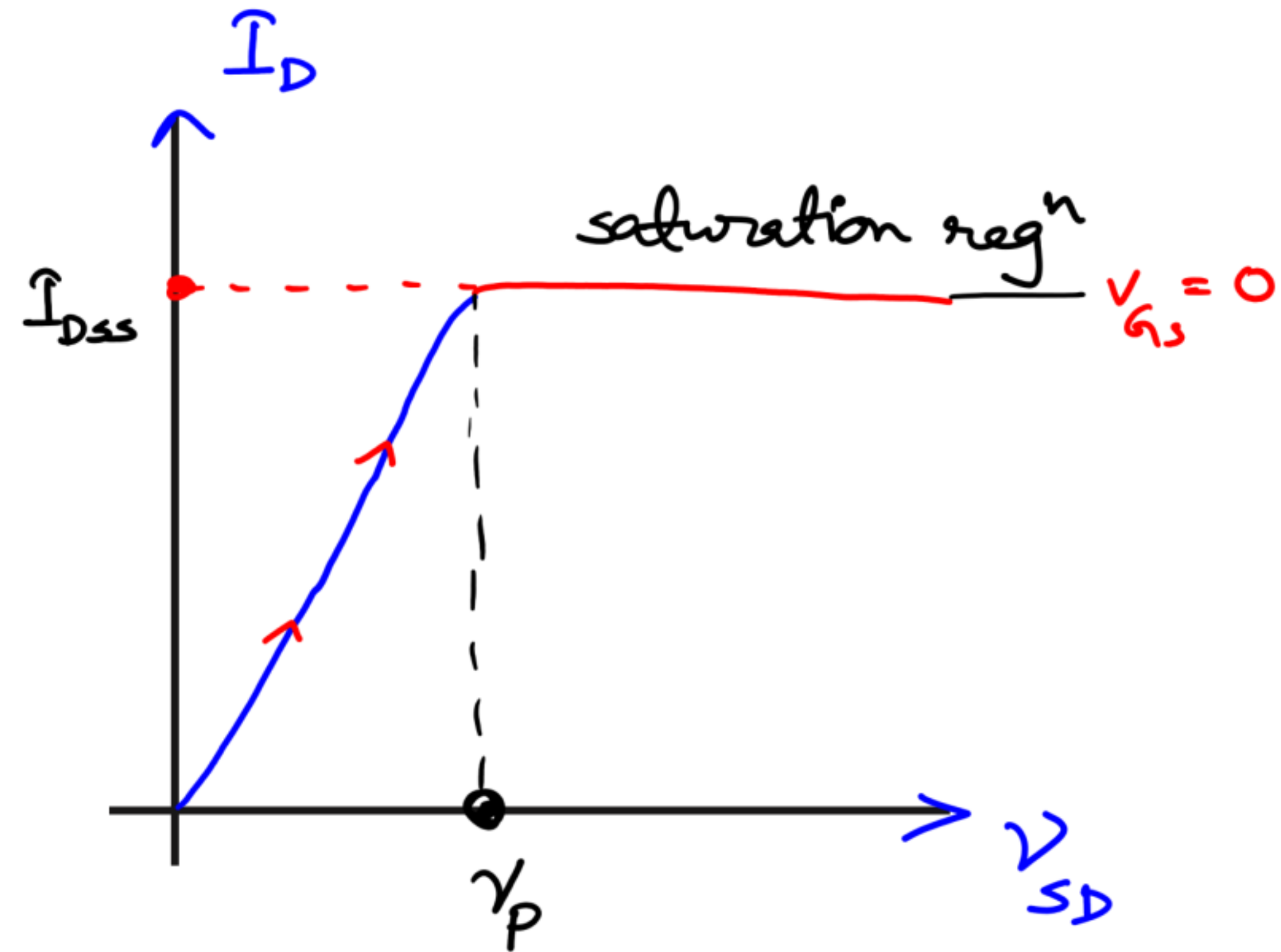
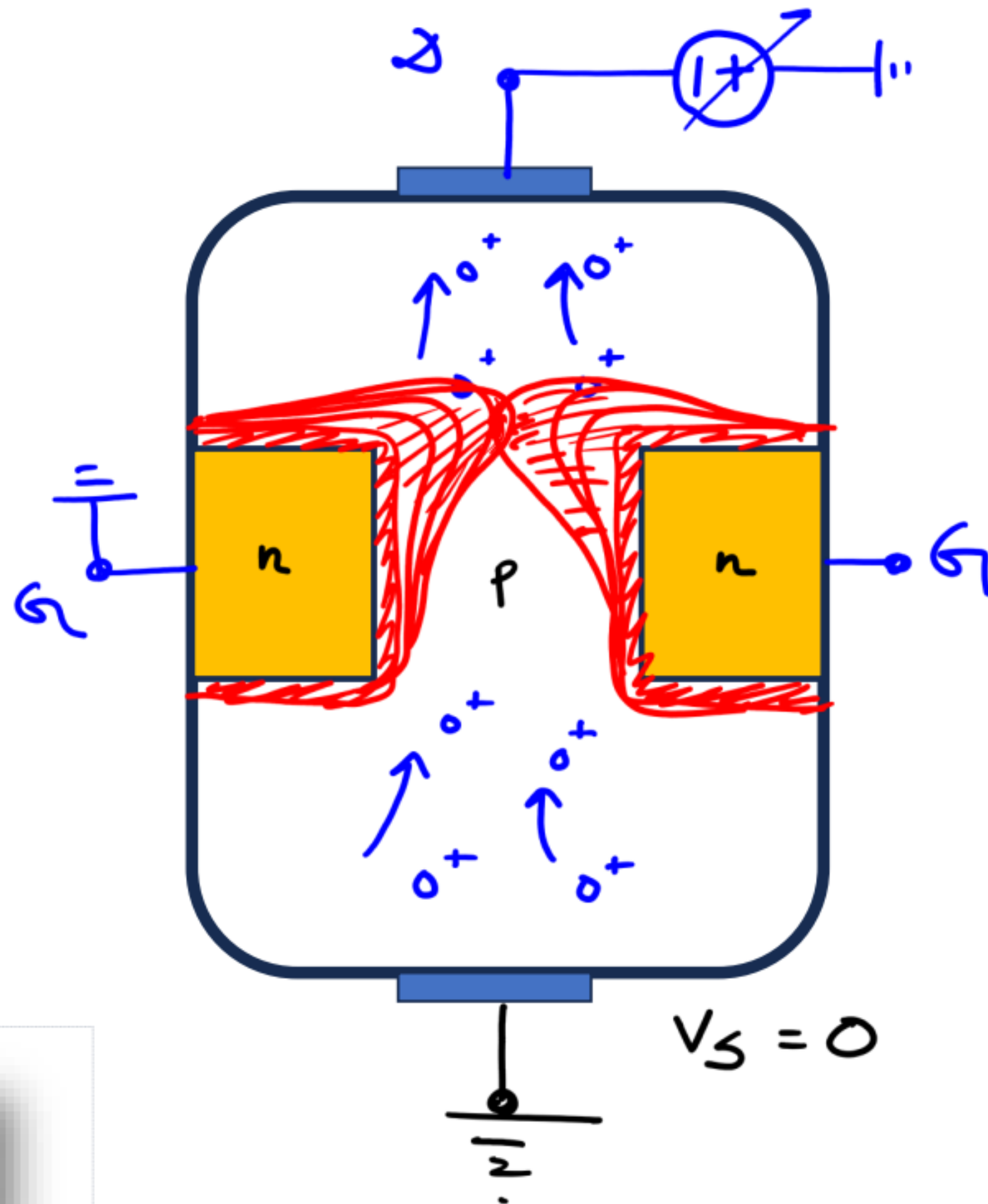


Case 01 & 02: $V_{SD} > 0$ & $V_{GS} \geq 0$

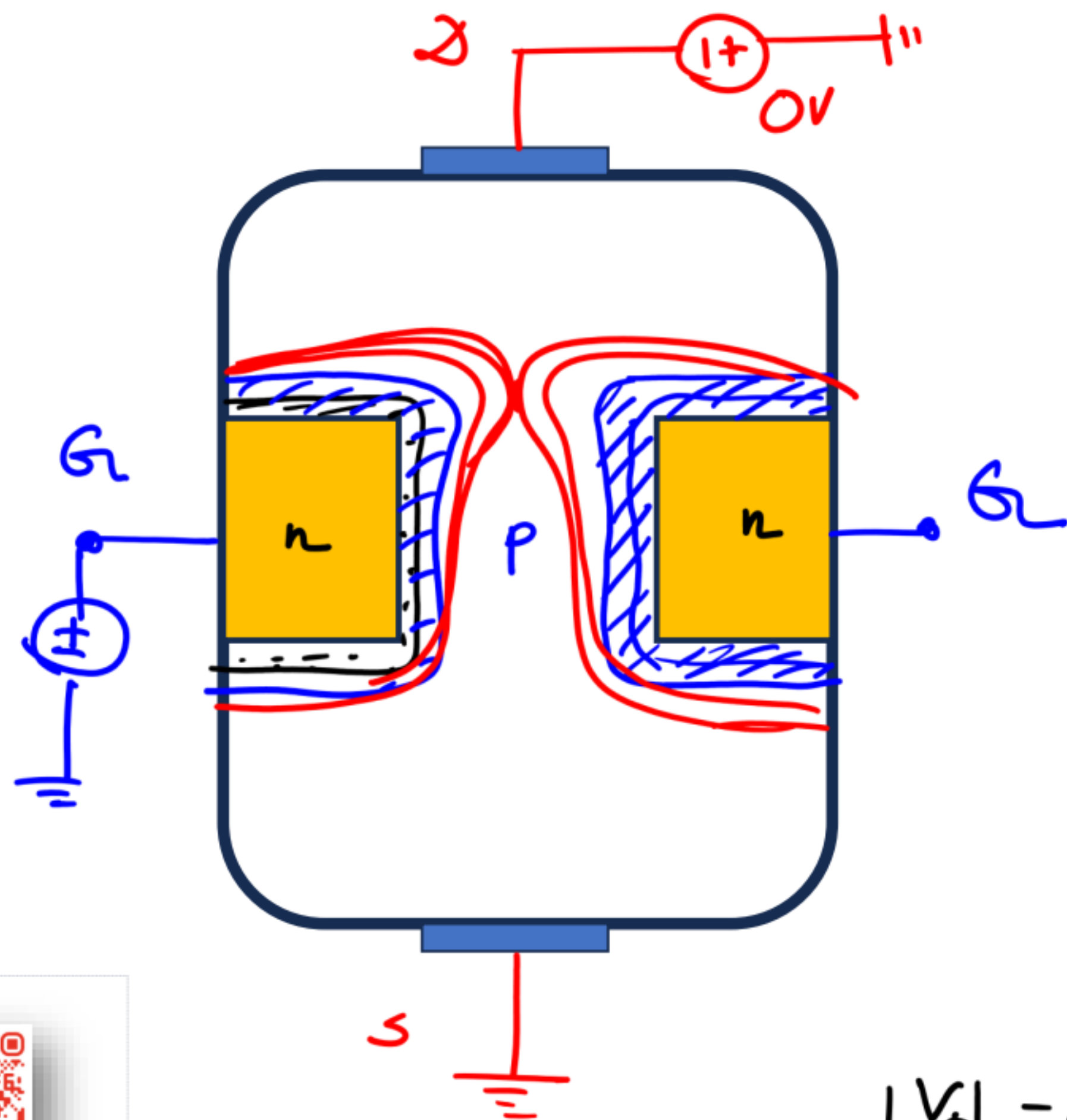
- Due to voltage between source & drain, current starts to flow
- Depletion regions increase in size decreasing the p-channel width
- Current will continue to increase with increasing V_{SD} until the channel is pinched off!
- The maximum value of I_D is termed as I_{DSS}
- Current I_D reaches I_{DSS} when voltage V_{SD} reaches V_p
- The pinch-off condition will be achieved quicker if $V_{GS} > 0$ is also applied!



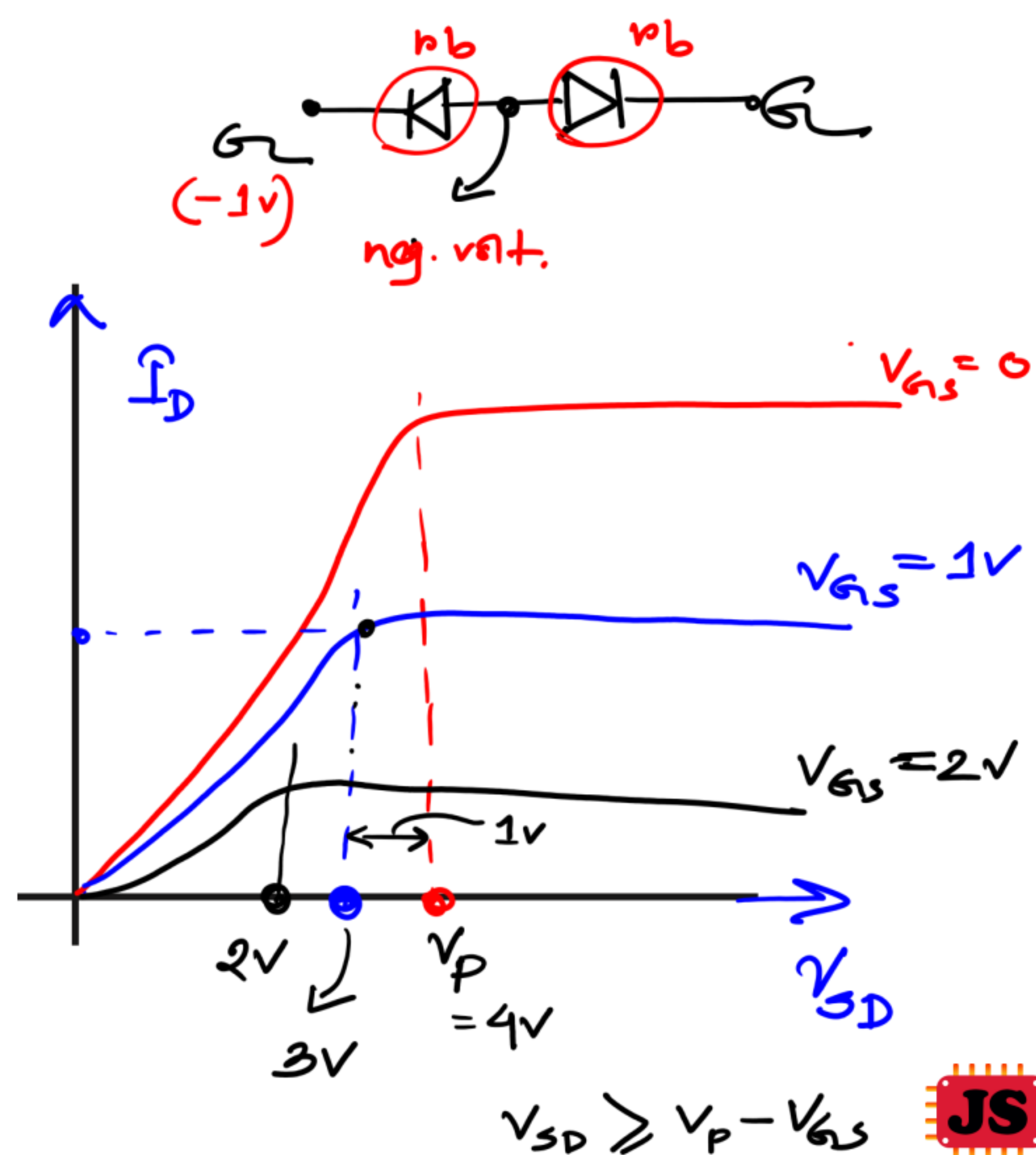
Understanding the mode of operation of p-JFET



Understanding the mode of operation of p-JFET



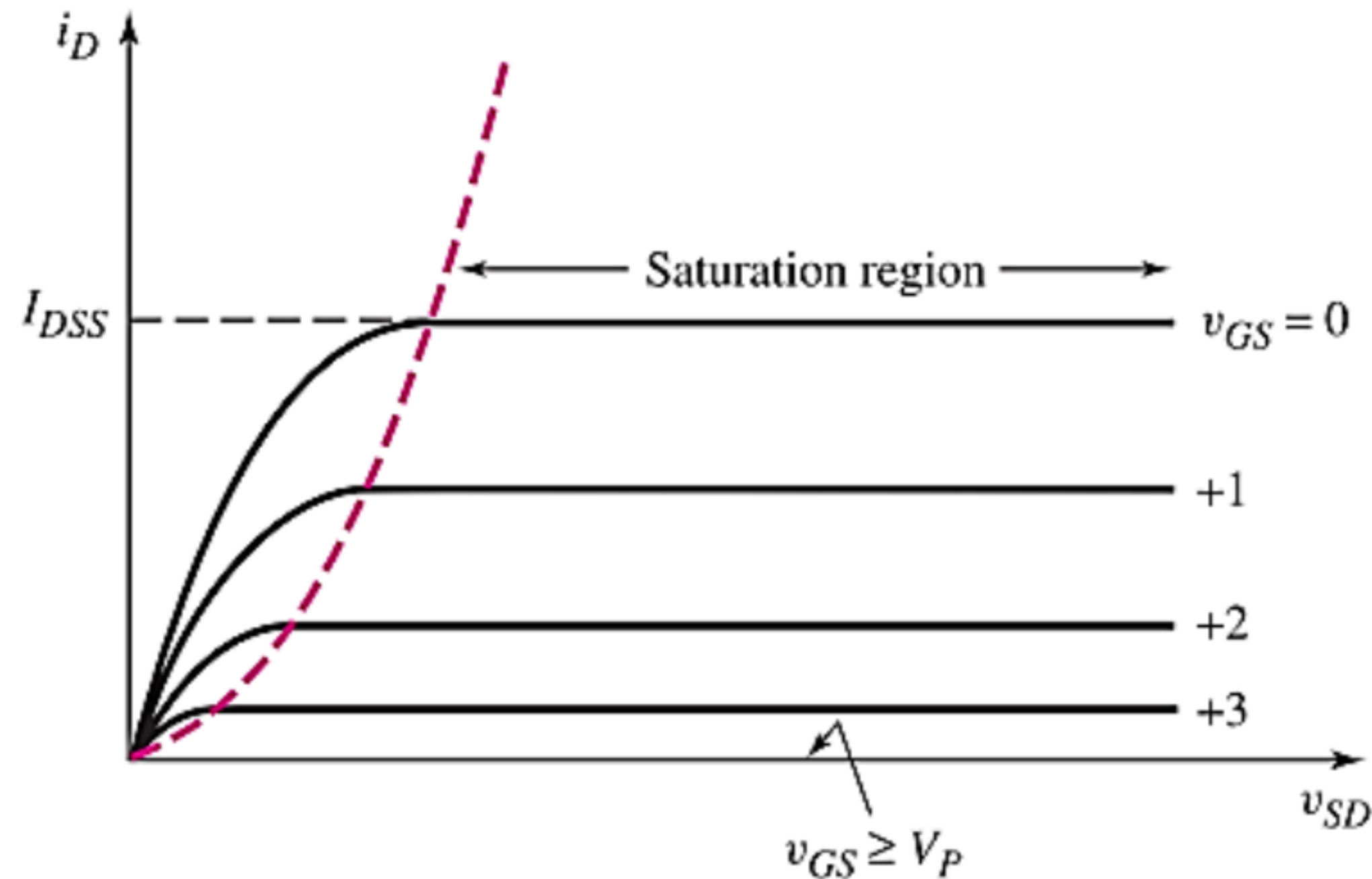
$$|V_p| = 4V$$



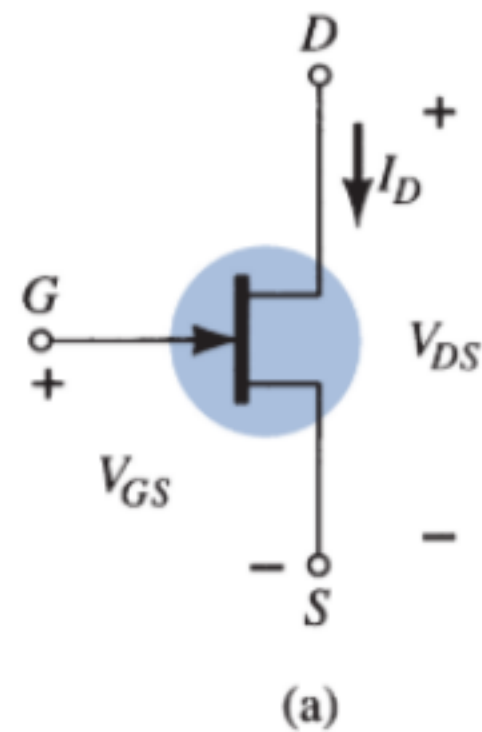
Characteristic curve of P-JFET



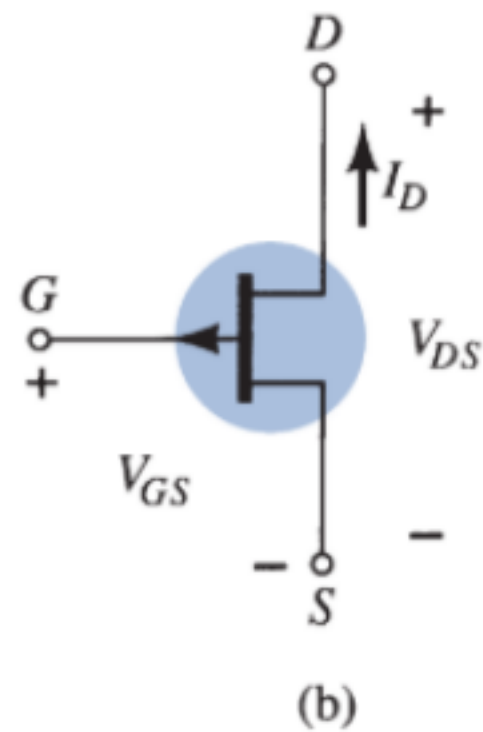
- The characteristic curve is as follows:



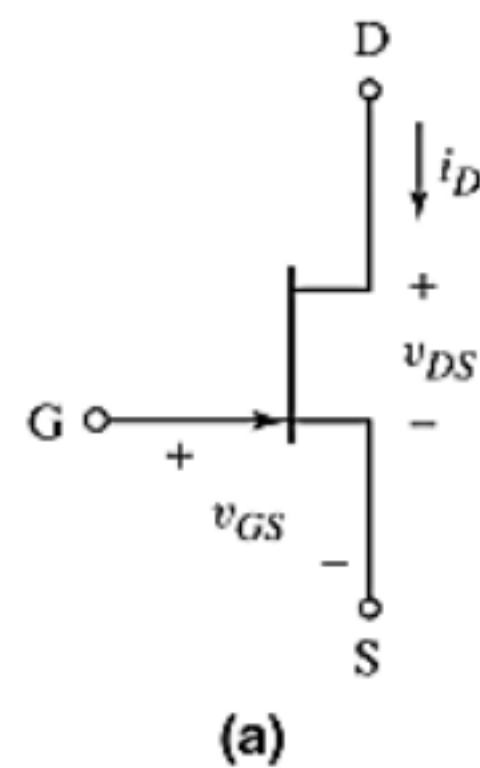
Symbology for JFETs



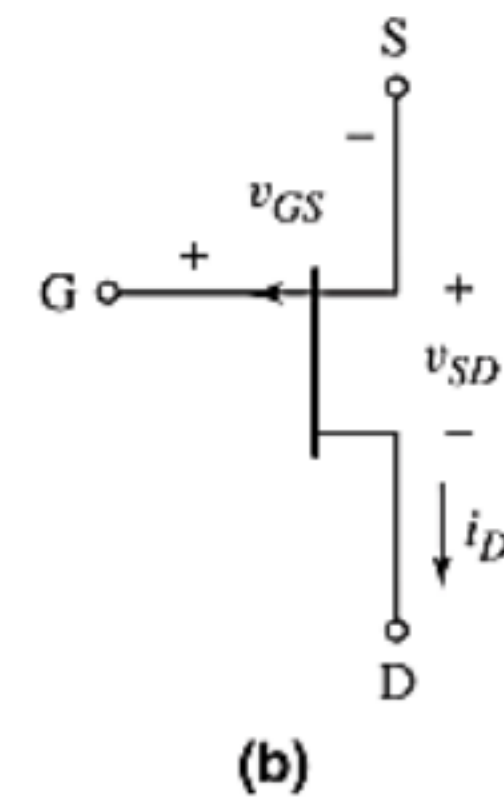
N-JFET



P-JFET



N-JFET



P-JFET

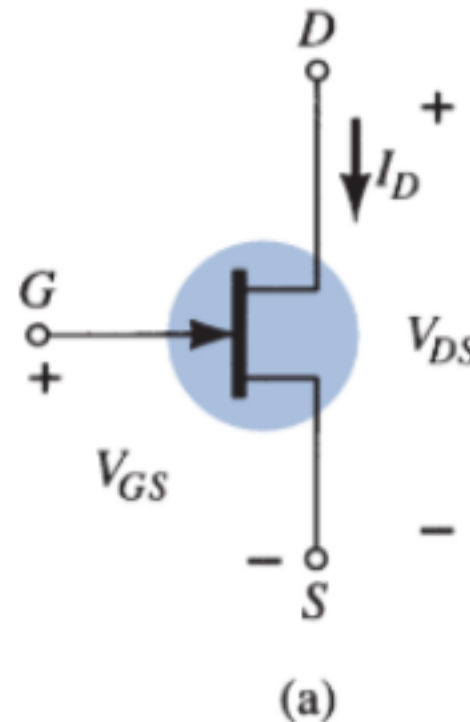
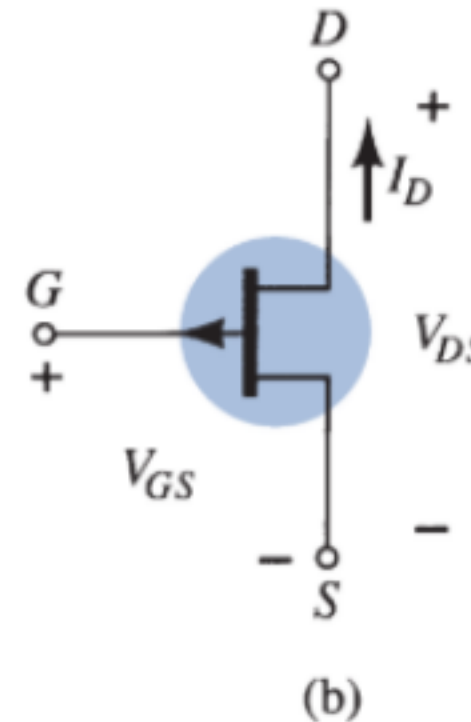
Technique to remember the symbol notation:

1. **N-JFET**: Gate is connected to p-type layer; D-S region is n-type. Arrow is from p $\rightarrow$ n
2. **P-JFET**: Gate is connected to n-type layer; D-S region is p-type. Arrow is from p $\rightarrow$ n



So, to sum things up...



Category	N-JFET	P-JFET
Current flow	From Drain to Source	From Source to Drain
Drain-to-source voltage	$V_{DS} > 0$	$V_{SD} > 0$
Pinch-off Voltage, V_P	$V_P < 0$	$V_P > 0$
Symbology		



Transfer & Characteristic curve of an N-JFET

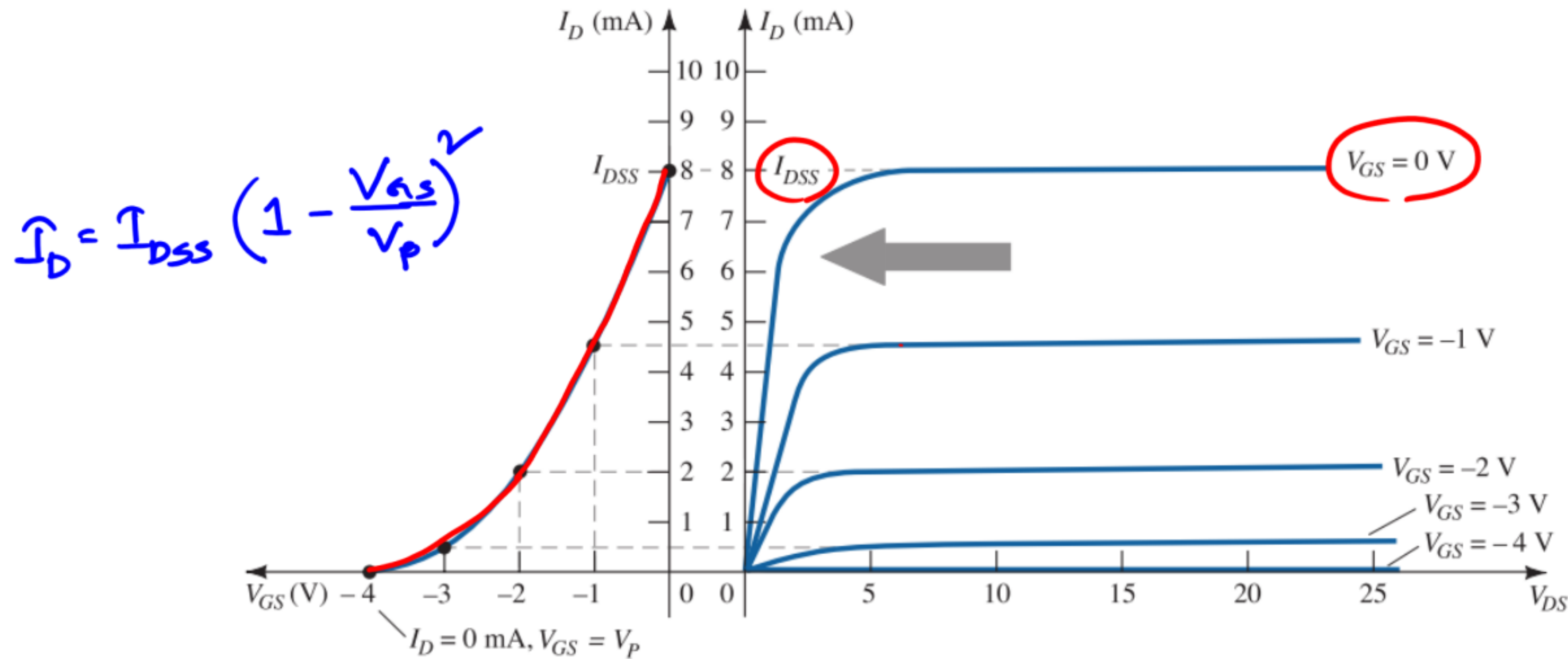


FIG. 17

Obtaining the transfer curve from the drain characteristics.



Equations for N-JFET & P-JFET (1)



- The current-voltage equation (Shockley's equation) for JFET is as follows:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

control variable

constants



V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

- Remember, $V_{GS} = V_P \Rightarrow$ both negative for n-JFET
- Remember, $V_{GS} = V_P \Rightarrow$ both positive for p-JFET

$$\begin{aligned} I_D &= I_{DSS} \left(1 - \frac{0.3V_P}{V_P} \right)^2 \\ &= I_{DSS} (1 - 0.3)^2 \\ &\approx \frac{I_{DSS}}{2} \end{aligned}$$

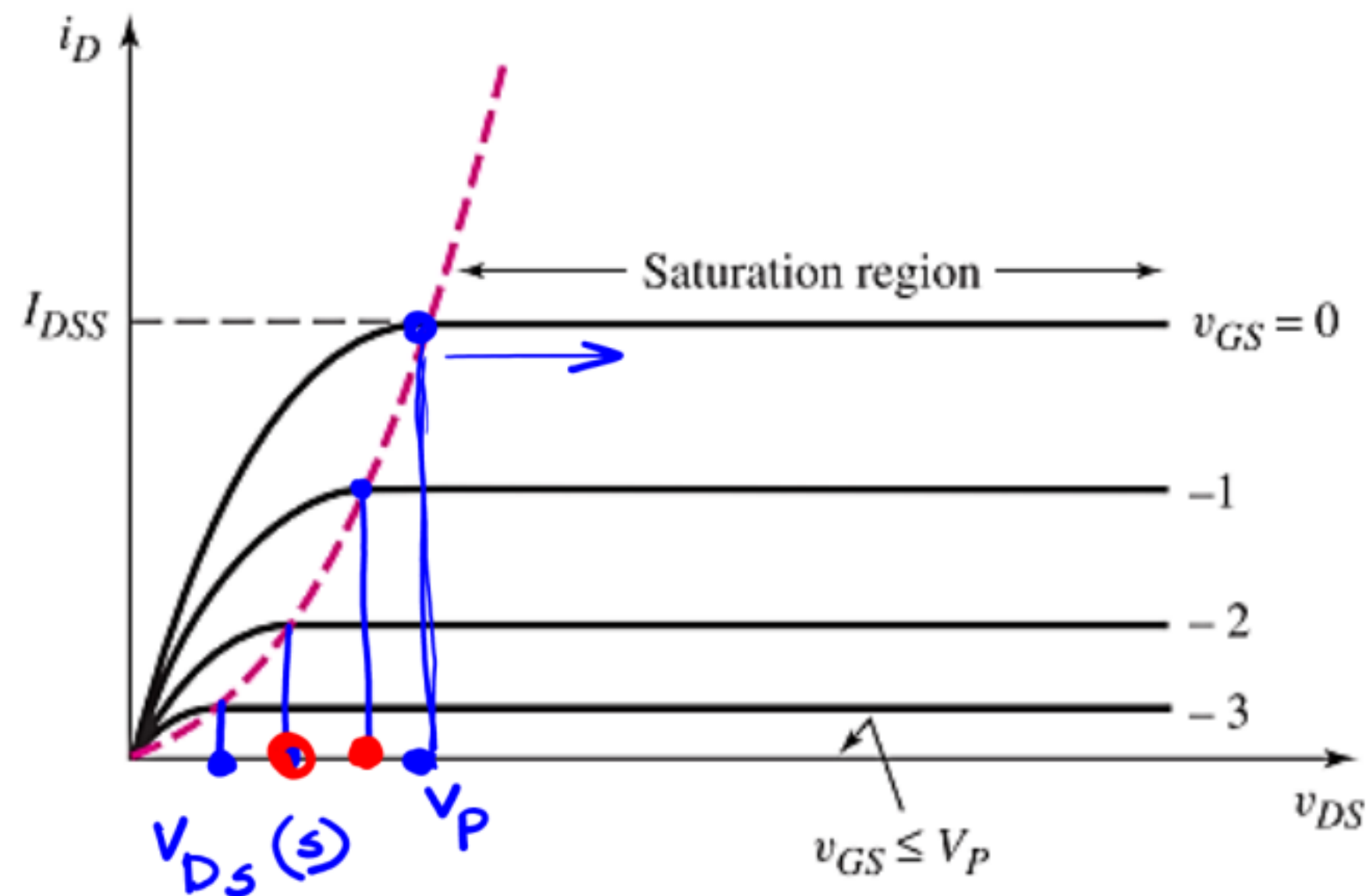


Saturation condition for N-JFET



- A JFET has a fixed pinch-off voltage (V_P)
- The more $|V_{GS}|$ we apply, the quicker the JFET will go into saturation!
- We'll say, an n-JFET will go into saturation if $V_{DS} > V_{DS}(sat) \equiv V_{ov} = V_{GS} - V_P$

$$V_{DS}(sat) \equiv V_{ov} = V_{GS} - V_P$$



$$V_P = -4V$$

$$V_{GS} = -2V$$

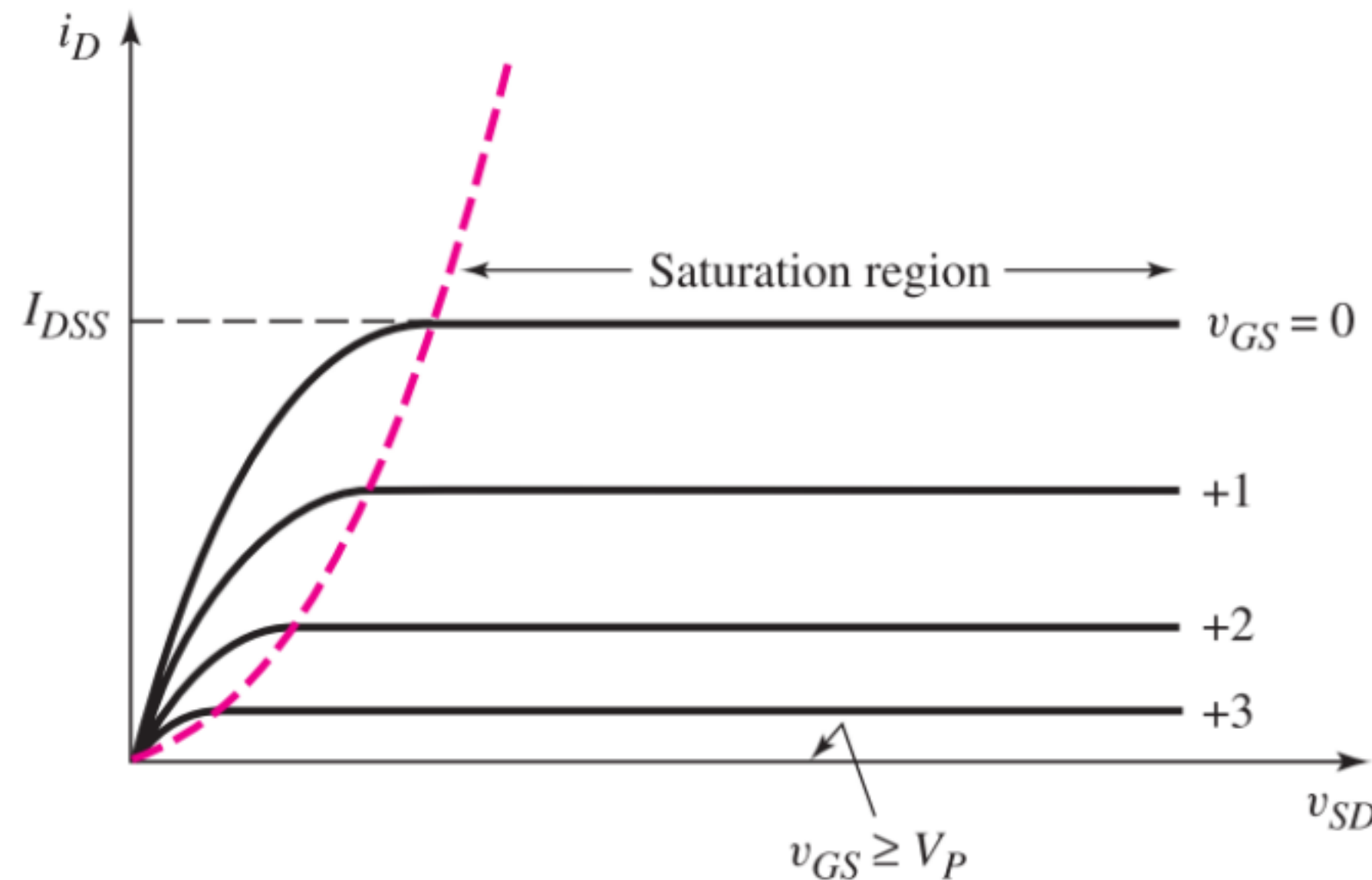
$$\begin{aligned} V_{DS}(s) &= V_{GS} - V_P \\ &= -2 + 4 \\ &= 2V \end{aligned}$$

Saturation condition for P-JFET



- A JFET has a fixed pinch-off voltage (V_P)
- The more $|V_{GS}|$ we apply, the quicker the JFET will go into saturation!
- We'll say, an p-JFET will go into saturation if $V_{SD} > V_{SD}(sat) \equiv V_{ov} = V_P - V_{GS}$

$$V_{SD}(sat) \equiv V_{ov} = V_P - V_{GS}$$



Problem 1: Sketch the transfer curve of a JFET defined by $I_{DSS} = 12 \text{ mA}$ and $V_p = -6\text{V}$

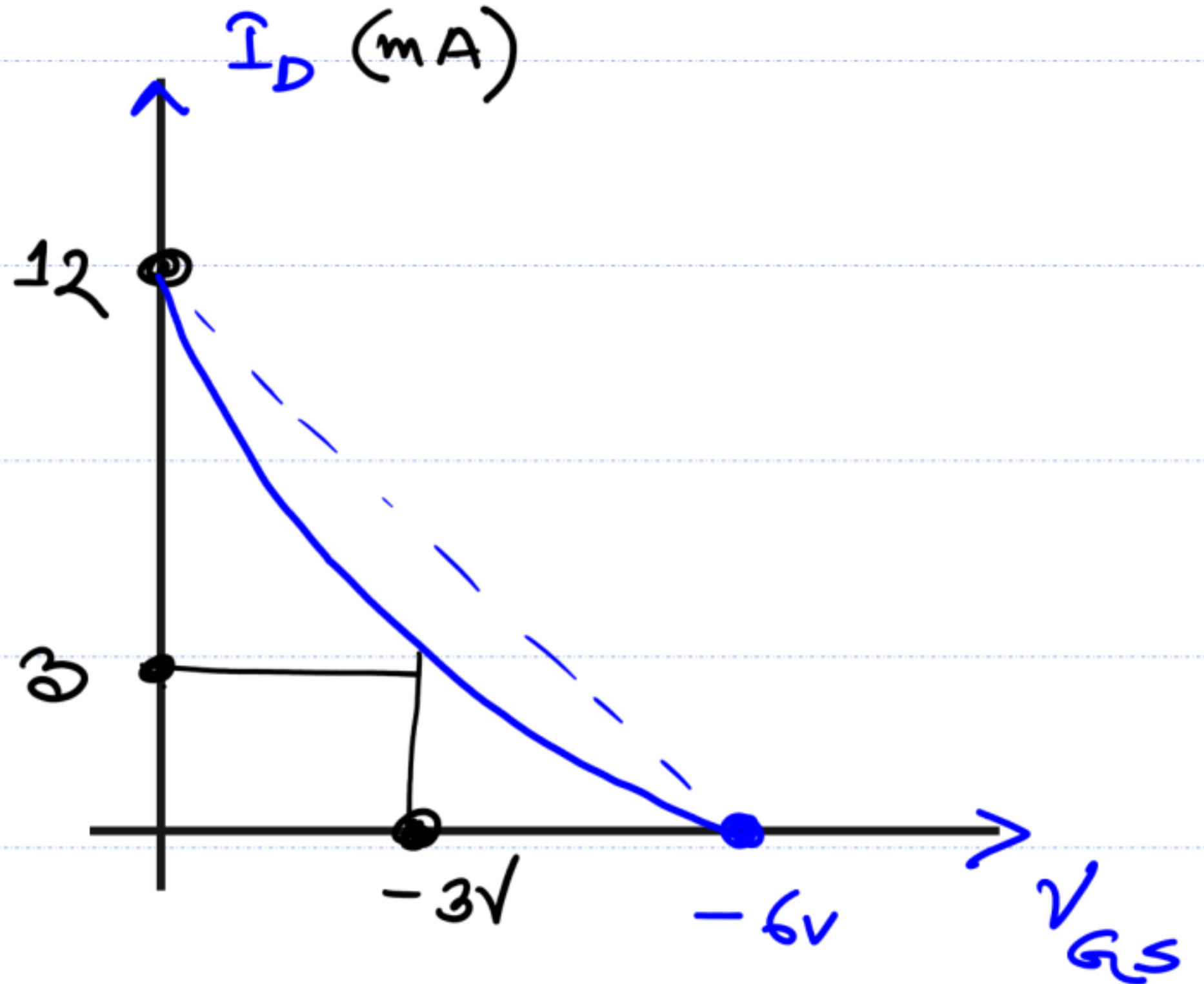
$V_p \rightarrow$ negative $\therefore$ n-JFET

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

• $V_{GS} = 0 \quad \therefore I_D = I_{DSS} = 12 \text{ mA}$

• $V_{GS} = V_p \quad \therefore I_D = 0 \text{ mA}$

• $V_{GS} = -3 \quad \therefore I_D = 12 \left(1 - \frac{-3}{-6} \right)^2$
 $= 3 \text{ mA}$



Problem 2: Sketch the transfer curve of a JFET defined by $I_{DSS} = 4 \text{ mA}$ and $V_p = 3 \text{ V}$

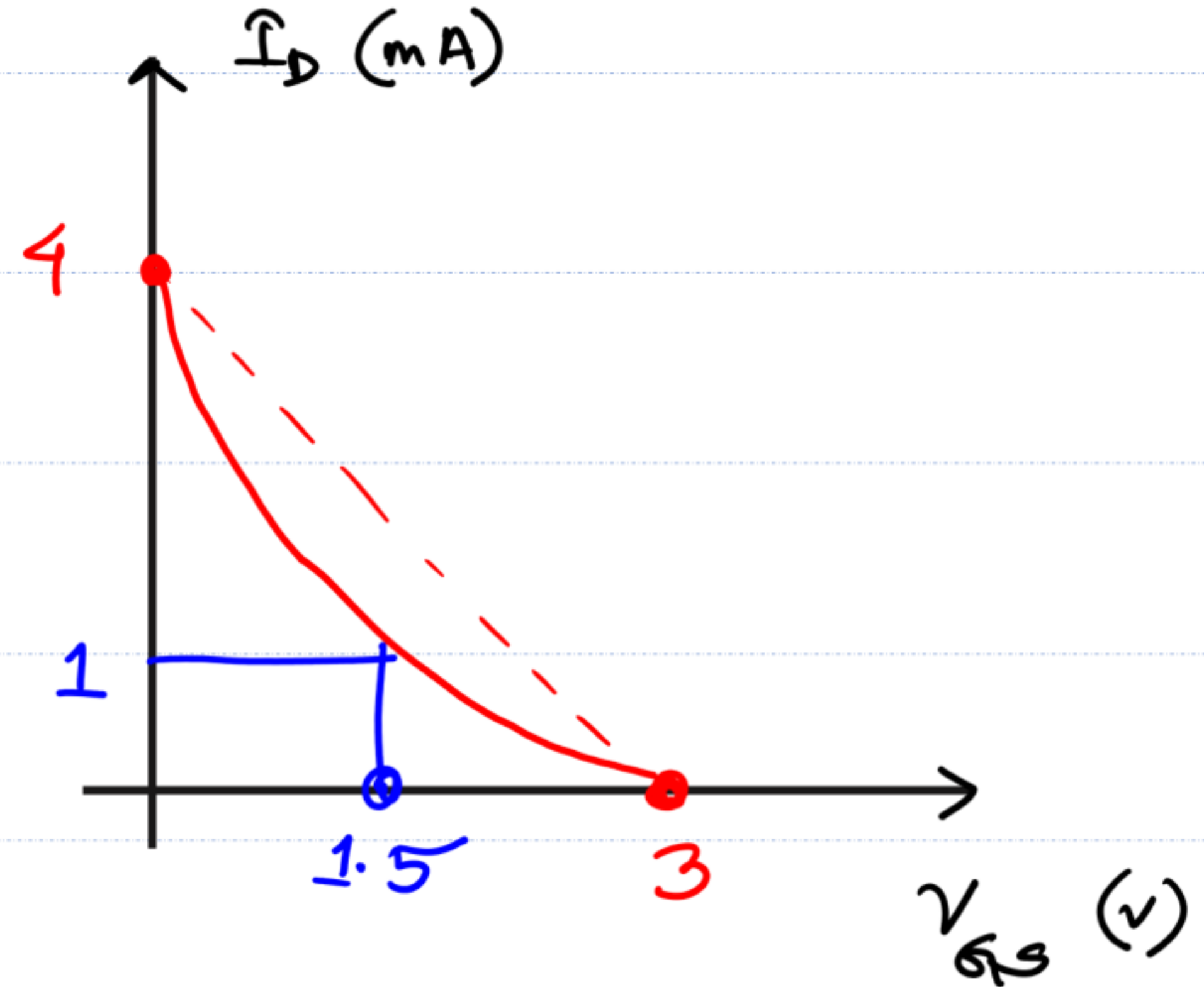
$V_p > 0 \text{ V}$ (positive) $\therefore$ p-JFET

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

• $V_{GS} = 0 \rightarrow I_D = I_{DSS} = 4 \text{ mA}$

• $V_{GS} = V_p \rightarrow I_D = 0 \text{ mA}$

• $V_{GS} = 1.5 \rightarrow I_D = 1 \text{ mA}$



Problem 3:

Objective: Calculate i_D and $v_{DS}(\text{sat})$ in an n-channel pn JFET.

Assume the saturation current is $I_{DSS} = 2 \text{ mA}$ and the pinchoff voltage is $V_P = -3.5 \text{ V}$. Calculate i_D and $v_{DS}(\text{sat})$ for $v_{GS} = 0, V_P/4$, and $V_P/2$.

$$v_{DS} > \underbrace{v_{GS} - V_P}_{V_{DS}(\text{sat})} \equiv V_{DS}(\text{sat})$$

$$I_D = 2 \left(1 - \frac{v_{GS}}{V_P} \right)^2$$

- $I_D = 2 \left(1 - \frac{0}{V_P} \right)^2 = 2 \text{ mA} \longrightarrow V_{DS}(s) = 0 - (-3.5) = 3.5 \text{ V}$
- $I_D = 2 \left(1 - \frac{1}{4} \right)^2 = 1.125 \text{ mA} \longrightarrow V_{DS}(s) = \frac{-3.5}{4} - (-3.5) = 2.625 \text{ V}$
- $I_D = 2 \left(1 - \frac{1}{2} \right)^2 = 0.5 \text{ mA} \longrightarrow V_{DS}(s) = \frac{-3.5}{2} - (-3.5) = 1.75 \text{ V}$

Problem 4:

Objective: Design a JFET circuit with a voltage divider biasing circuit.

Consider the circuit shown in Figure 3.61(a) with transistor parameters $I_{DSS} = 12 \text{ mA}$, $V_P = -3.5 \text{ V}$, and $\lambda = 0$. Let $R_1 + R_2 = 100 \text{ k}\Omega$. Design the circuit such that the dc drain current is $I_D = 5 \text{ mA}$ and the dc drain-to-source voltage is $V_{DS} = 5 \text{ V}$.

$$V_S = -5 + I_D R_S = -2.5 \text{ V}$$

$$V_{DD} - V_S = 5 - (-2.5) = 7.5 = I_D R_D + V_{DS} = 5 R_D + 5$$

$$\therefore R_D = \frac{7.5 - 5}{5} = 0.5 \text{ k}\Omega$$

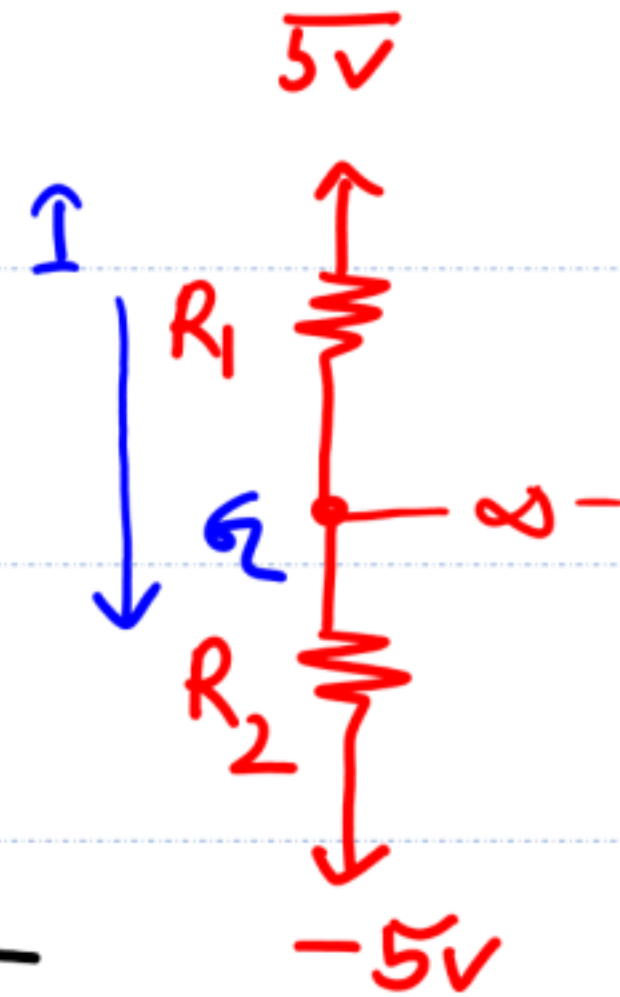
$$\therefore V_G = V_{GS} + V_S = -3.74 \text{ V}$$

$$V_G = -5 + I R_2$$

$$\begin{cases} = -5 + \frac{10}{R_1 + R_2} \times R_2 \\ = -3.74 \end{cases}$$

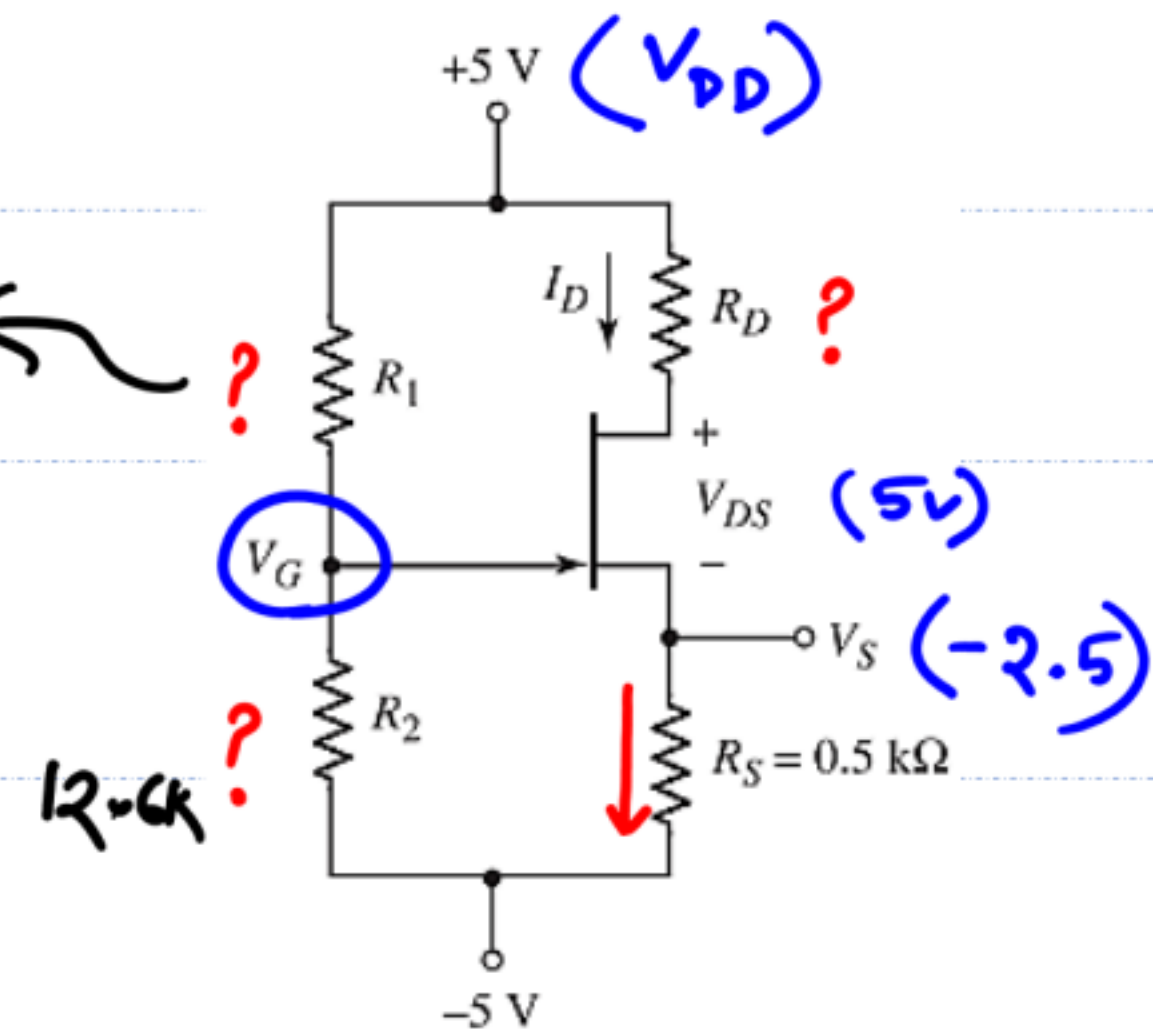
$$I = \frac{10}{R_1 + R_2}$$

$$\therefore R_2 = 12.6 \text{ k}\Omega$$



$$I_D = 12 \left(1 + \frac{V_{GS}}{3.5} \right)^2 = 5$$

$$\therefore V_{GS} = \left(\sqrt{\frac{5}{12}} - 1 \right) 3.5 = -1.24 \text{ V}$$



Problem 5: Find the Quiescent values of current and voltage for the circuit shown here.

Given: $I_{DSS} = 2.5\text{mA}$ & $V_P = 2.5\text{V}$

$$I_D = I_Q = 0.8 = \frac{V_D - (-9)}{4} \rightarrow \underline{V_D = -5.8\text{V}}$$

$$I_D = 0.8 = 2.5 \left(1 - \frac{V_{GS}}{2.5} \right)^2$$

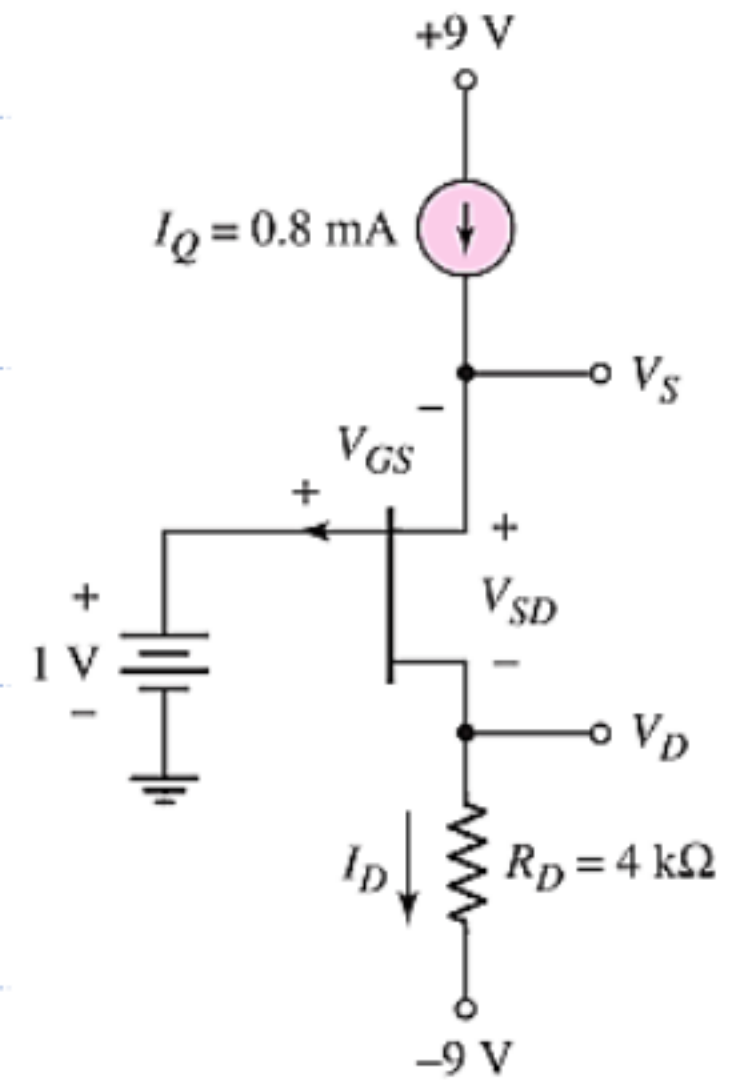
$$\Rightarrow V_{GS} = \left(\sqrt{\frac{0.8}{2.5}} - 1 \right) \times 2.5 = 1.086\text{V}$$

$$V_G = 1\text{V}$$

$$\therefore V_S = V_G - V_{GS} = 1 - 1.086 = -0.086\text{V}$$

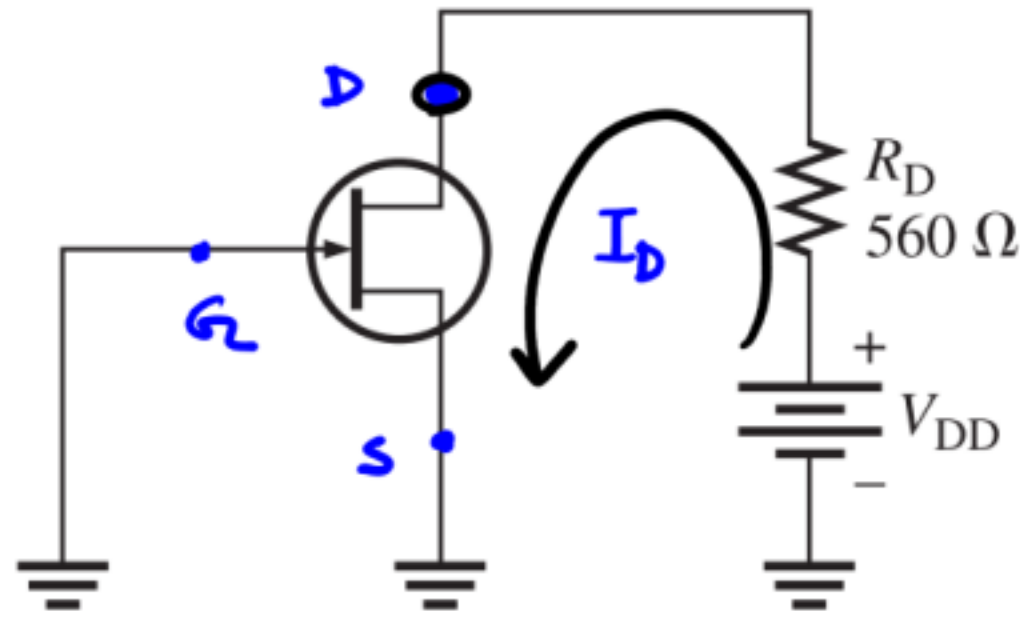
$$V_P - V_{GS} = 2.5 - 1.086 = 1.41\text{V}$$

$$V_{SD} = 5.714\text{V} > V_P - V_{GS} \quad \therefore \text{saturation}$$



Ignore the calc.
error in the
video

Problem 6: For the following circuit, $V_{GS}(off) = -4V$ & $I_{DSS} = 12mA$. Determine the minimum value of V_{DD} required to put the device in the constant-current region of operation when $V_{GS} = 0V$



$$V_{DD} = V_D + I_D R_D$$

$$V_{DD} = V_{DS} + I_D R_D$$

$$\therefore V_{DD}(\min) = V_{DS}(\min \text{ at sat}) + I_D R_D$$

$$= V_{DS}(\text{sat}) + I_D R_D$$

$$= V_{GS} - V_{GS}(\text{off}) + I_D R_D$$

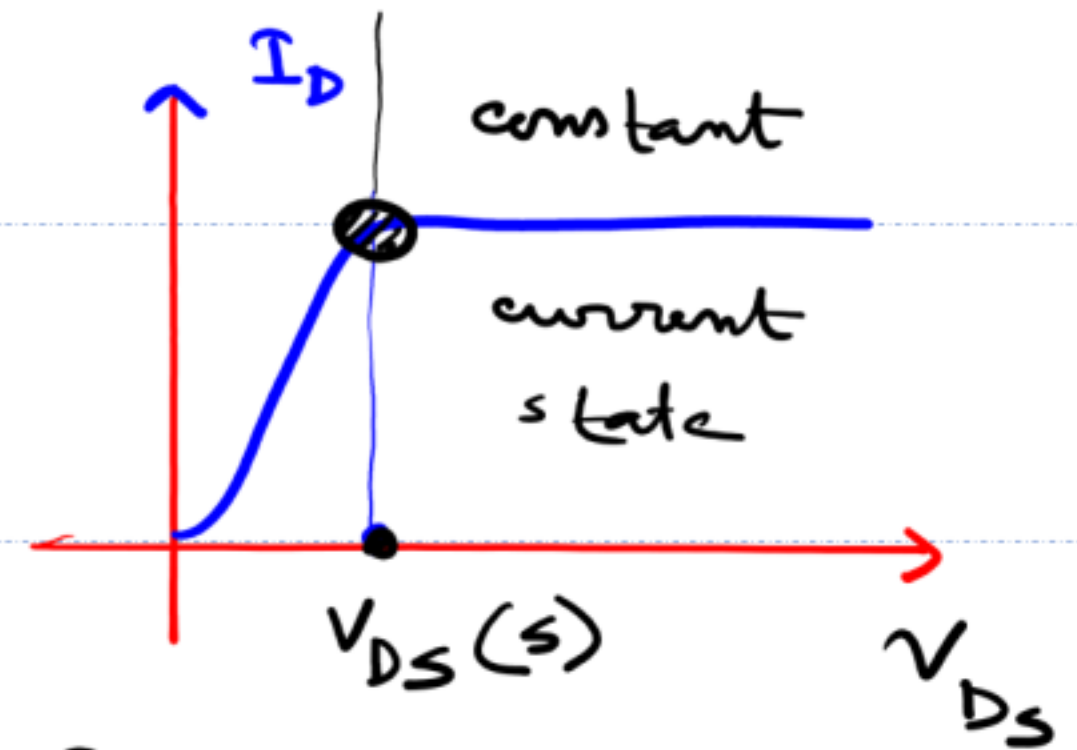
$$= 0 - (-4) + I_D R_D$$

$$= 4 + 12m \times 560 = 10.72V$$

$$V_{GS}(\text{off}) = -4V$$

$$I_D = I_{DSS} \left(1 - \frac{0}{V_{GS}(\text{off})}\right)^2$$

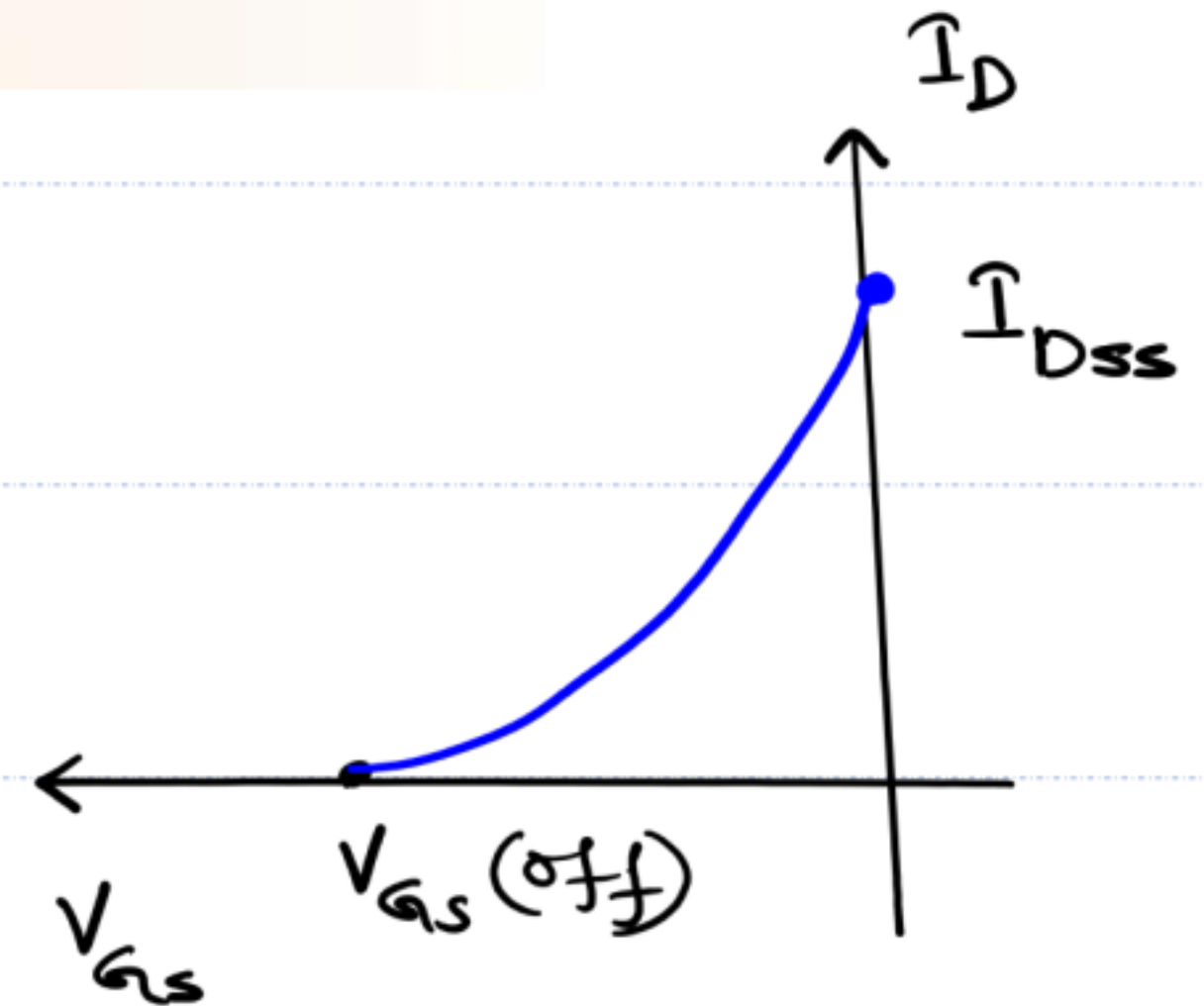
$$= 12mA$$



Problem 7: A particular p-channel JFET has a $V_{GS}(\text{off}) = +4 \text{ V}$. What is I_D when $V_{GS} = +6 \text{ V}$?

Solution:

The p-channel JFET requires a positive gate-to-source voltage. The more positive the voltage, the less the drain current. When $V_{GS} = 4 \text{ V}$, $I_D = 0$. Any further increase in V_{GS} keeps the JFET cut off, so I_D remains 0.



Biasing of a JFET



- **Meaning of Biasing:** Biasing is the process of applying external DC voltages (or currents) to a transistor so that it operates at a desired point on its characteristics, called the Q-point (quiescent point).
- **In the case of JFET:** Biasing means setting the Gate-Source Voltage (V_{GS}) in such a way that the JFET works properly in its active (or ohmic/saturation) region depending on the application.
- **Why we bias:**
 - To control the drain current (I_D) for a given application.
 - To ensure the JFET operates in the correct region (typically the saturation region for amplification).
 - To avoid distortion by keeping the signal swing within limits.



Types of Biasing for JFET



1. Fixed-Bias Configuration
2. Self-Bias Configuration
3. Voltage-Divider Biasing
4. Common-Gate Configuration
5. Common-Drain Configuration (Source Follower)

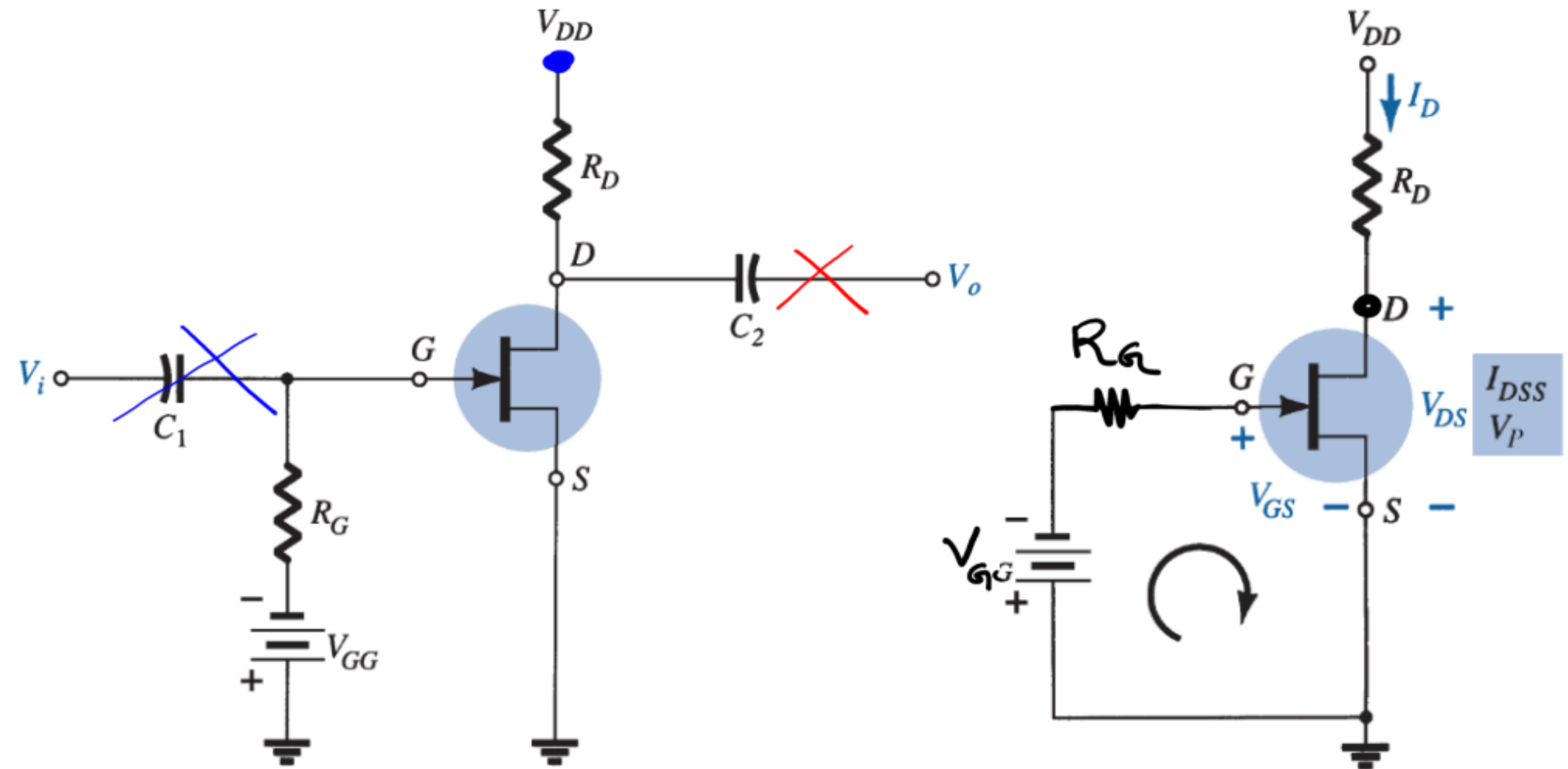


Fixed-Bias Configuration

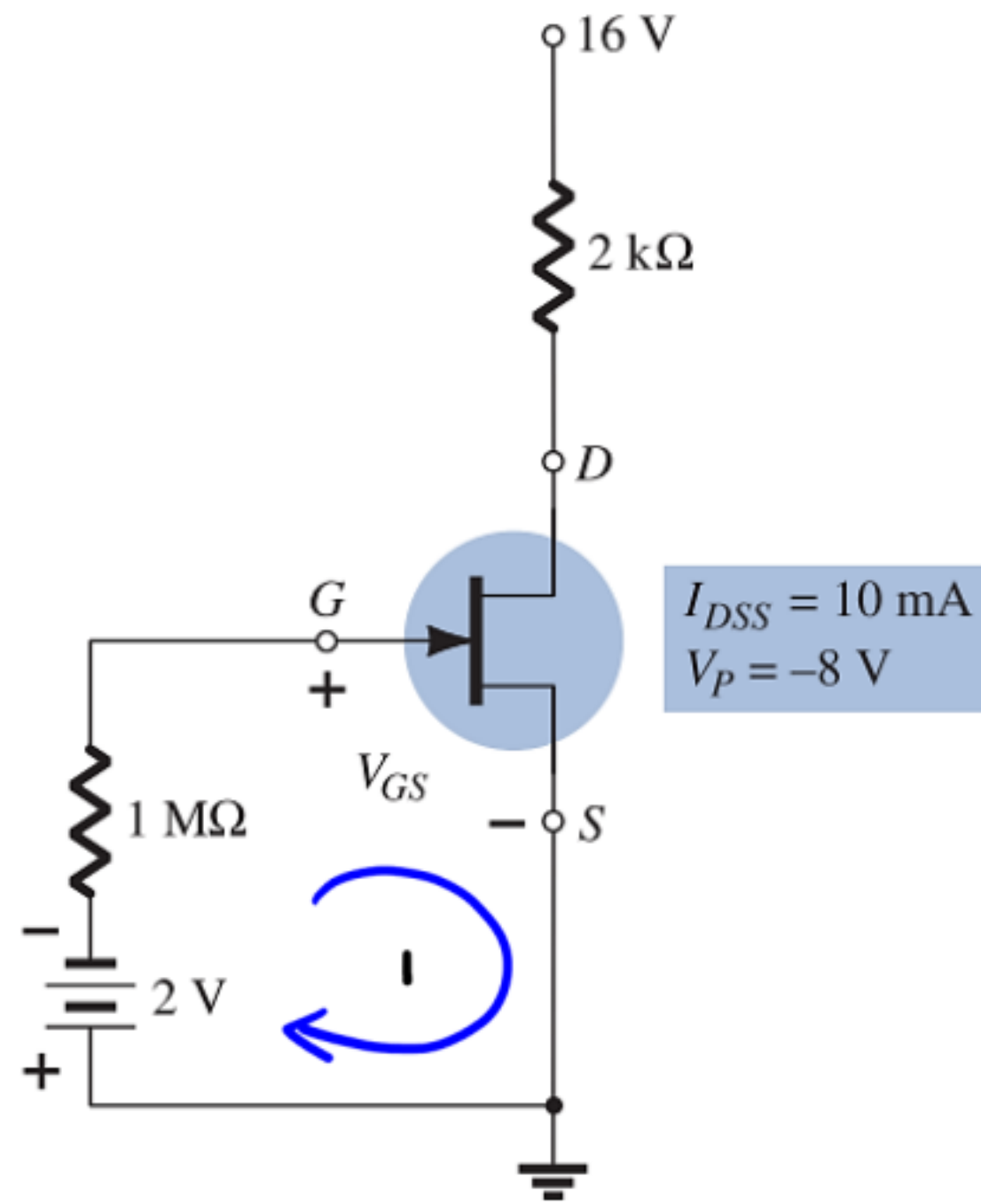


- In this method, a DC voltage is directly applied between gate and source through a resistor, keeping the gate at a fixed negative potential (for n-channel JFET). It is simple but suffers from poor stability against parameter variations.

- $V_{GS} = -V_{GG}$
- $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$
- $V_{DS} = V_{DD} - I_D R_D$
- $V_S = 0$
- $V_D = V_{DS}$
- $V_G = V_{GS}$



Problem 8: Determine V_{GSQ} , I_{DQ} , V_{DS} , V_D , V_G , V_S



$$\text{KVL at } \textcircled{1} : +2 + V_{GS} = 0 \quad \therefore V_{GSQ} = -2 \text{ V}$$

$$V_S = 0 \text{ V} \quad \therefore V_G - V_S = -2$$

$$\therefore V_G = -2 \text{ V}$$

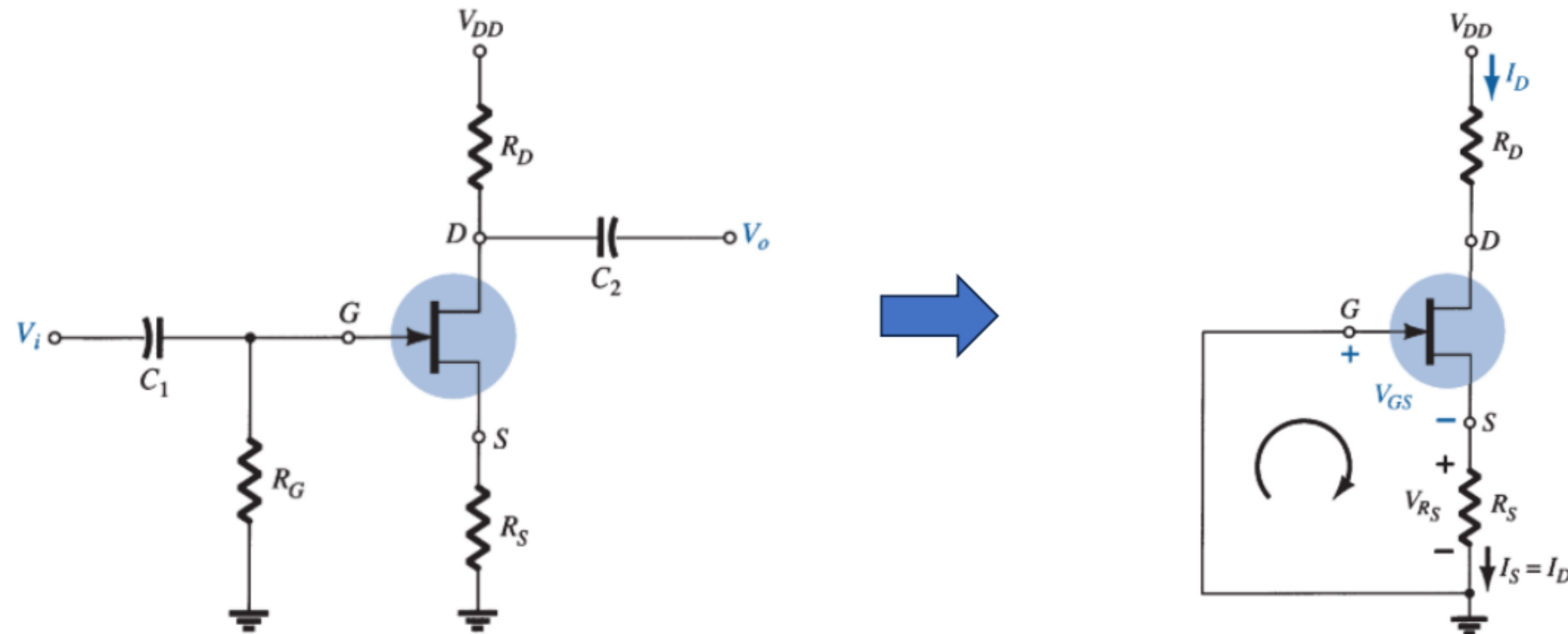
$$I_{DQ} = 10 \text{ mA} \left(1 - \frac{-2}{-8} \right)^2 = 5.625 \text{ mA}$$

$$\therefore V_{DS} = V_D = 16 - 5.625 \times 2 = 4.75 \text{ V}$$

Self-Bias Configuration (1)



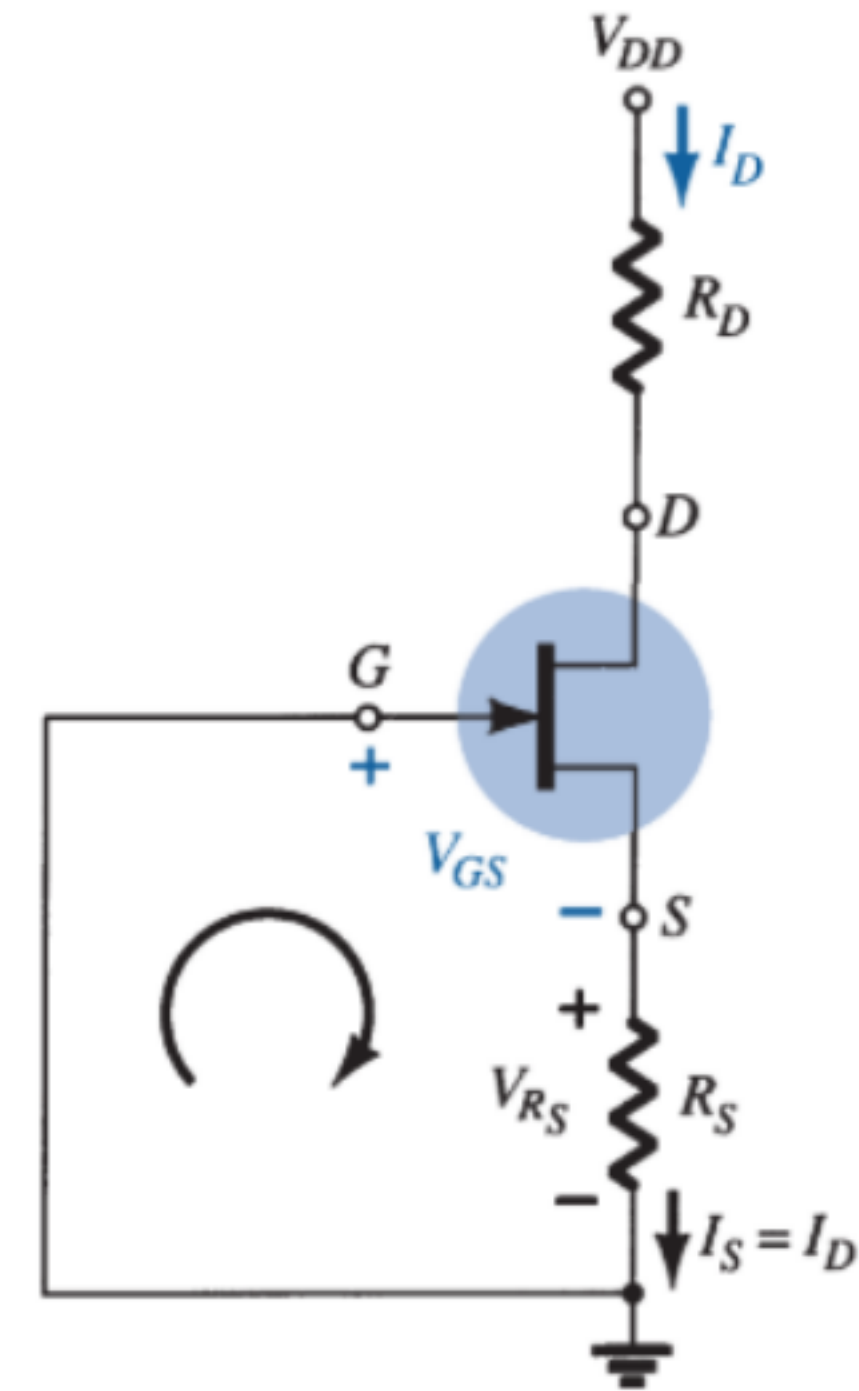
- The self-bias configuration eliminates the need for two dc supplies.
- The controlling gate-to-source voltage is now determined by the voltage across a resistor R_S introduced in the source leg of the configuration as shown in the following figure.



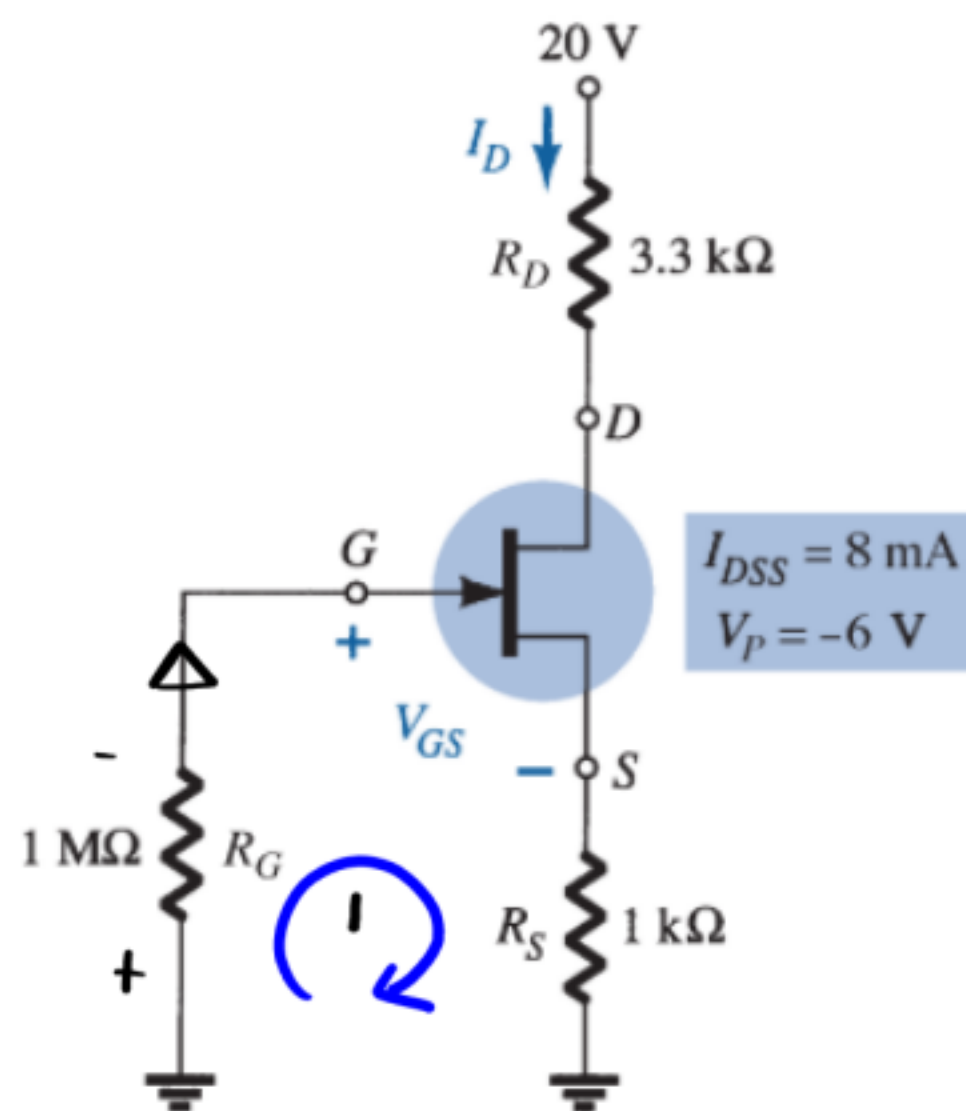
Self-Bias Configuration (2)



- $V_{GS} + V_{R_S} = 0$
- $V_{GS} = -V_{R_S} = -I_D R_S$
- $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$
- $I_D = I_{DSS} \left(1 - \frac{-I_D R_S}{V_p} \right)^2 \Rightarrow \text{quadratic equation}$
- Solve the quadratic equation to get I_{DQ}



Problem 9: Find V_{GSQ} , I_{DQ} , V_{DS} , V_D , V_G , V_S



$$\text{KVL at ① : } +0 \times R_G + V_{GS} + I_D R_S = 0$$

$$\therefore V_{GS} = -I_D \times 1k \text{ V} \quad \text{--- ①}$$

$$I_D = 8 \text{ mA} \left(1 - \frac{-1k I_D}{-6} \right)^2 = 8 \text{ m} (1 - 166.67 I_D)^2$$

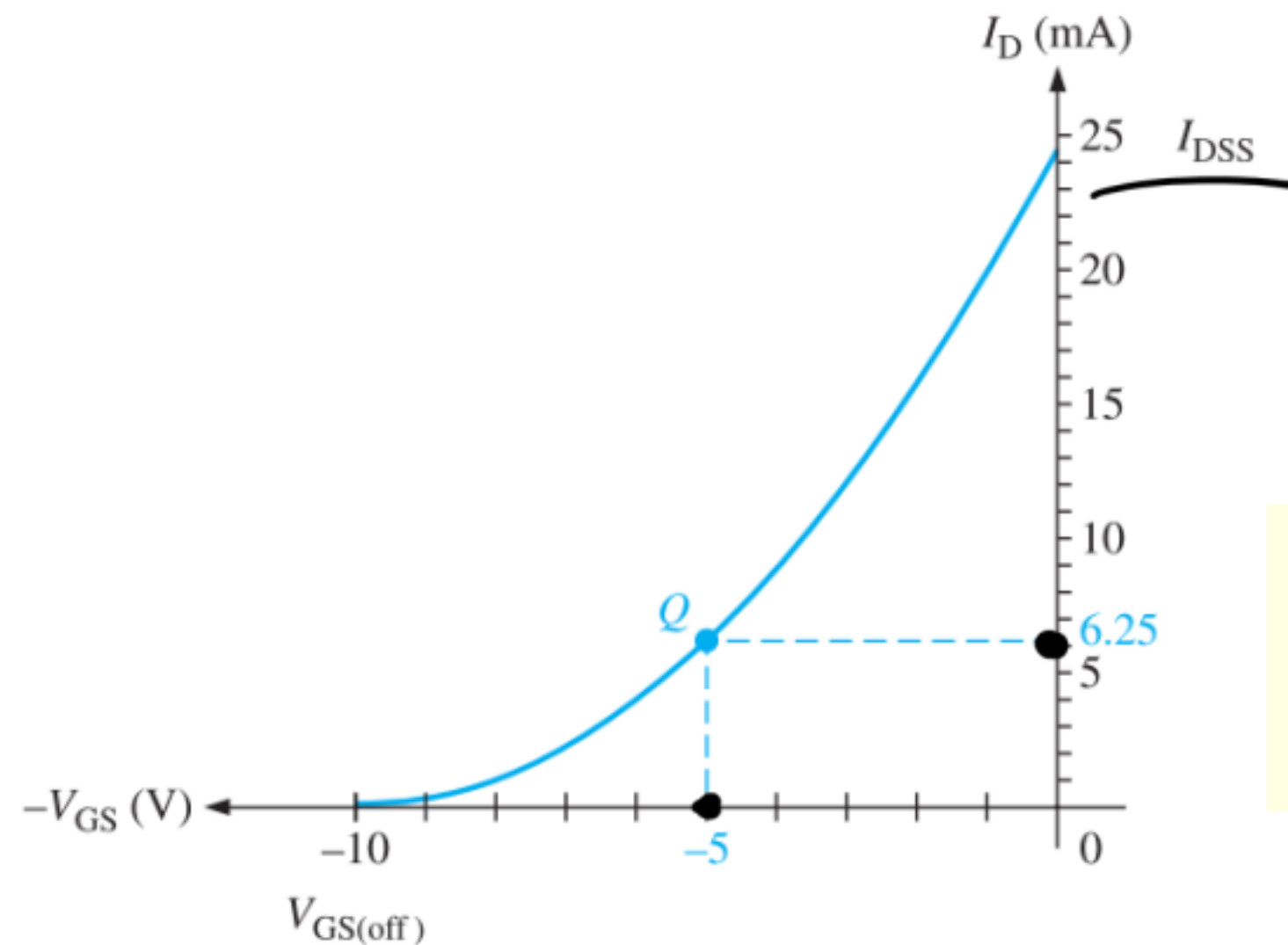
$$(222.23) I_D^2 + (-3.67) I_D + (8 \times 10^{-3}) = 0$$

$$\therefore I_D = 0.014, 2.58 \times 10^{-3} \longrightarrow 2.58 \text{ mA} \quad (\because I_D \leq I_{DSS})$$

$$\therefore V_{GS} = -2.6 \text{ V} \quad ; \quad V_S = I_D \times 1k = 2.6 \text{ V} \quad ; \quad V_D = 20 - I_D R_D = 11.5 \text{ V}$$

Problem 10:

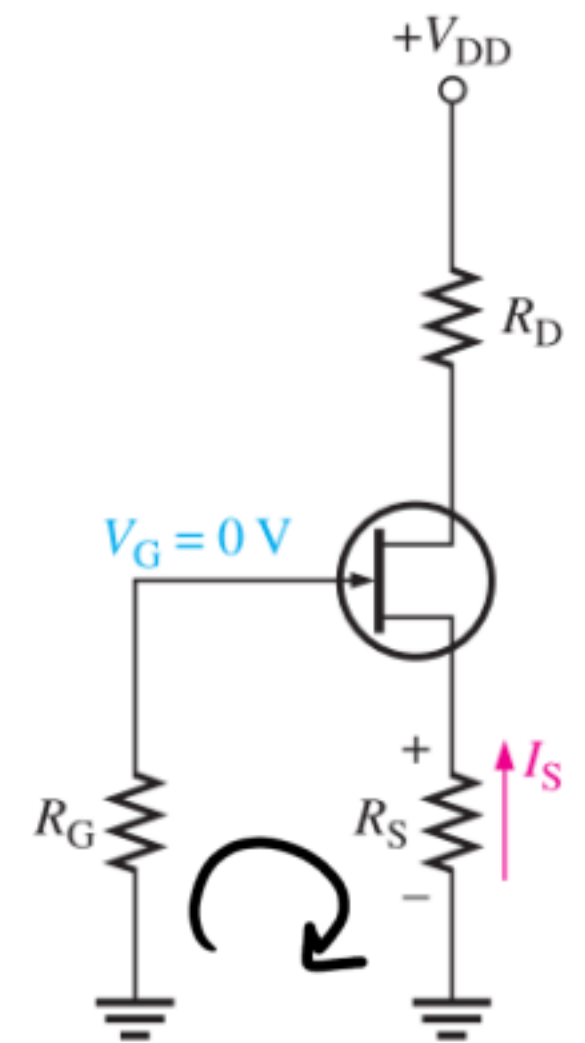
Determine the value of R_S required to self-bias an n -channel JFET that has the transfer characteristic curve shown in Figure 8–18 at $V_{GS} = -5$ V.



$$V_{GS} = -I_D R_S$$
$$\therefore R_S = \left| \frac{V_{GS}}{I_D} \right|$$

From the graph, $I_D = 6.25$ mA when $V_{GS} = -5$ V. Calculate R_S .

$$R_S = \left| \frac{V_{GS}}{I_D} \right| = \frac{5 \text{ V}}{6.25 \text{ mA}} = 800 \, \Omega$$



$$I_D = 25 \left(1 - \frac{-5}{-10} \right)^2 = 6.25 \text{ mA}$$



Problem 11:

Determine the value of R_S required to self-bias a p -channel JFET with datasheet values of $I_{DSS} = 25 \text{ mA}$ and $V_{GS(off)} = 15 \text{ V}$. V_{GS} is to be 5 V .

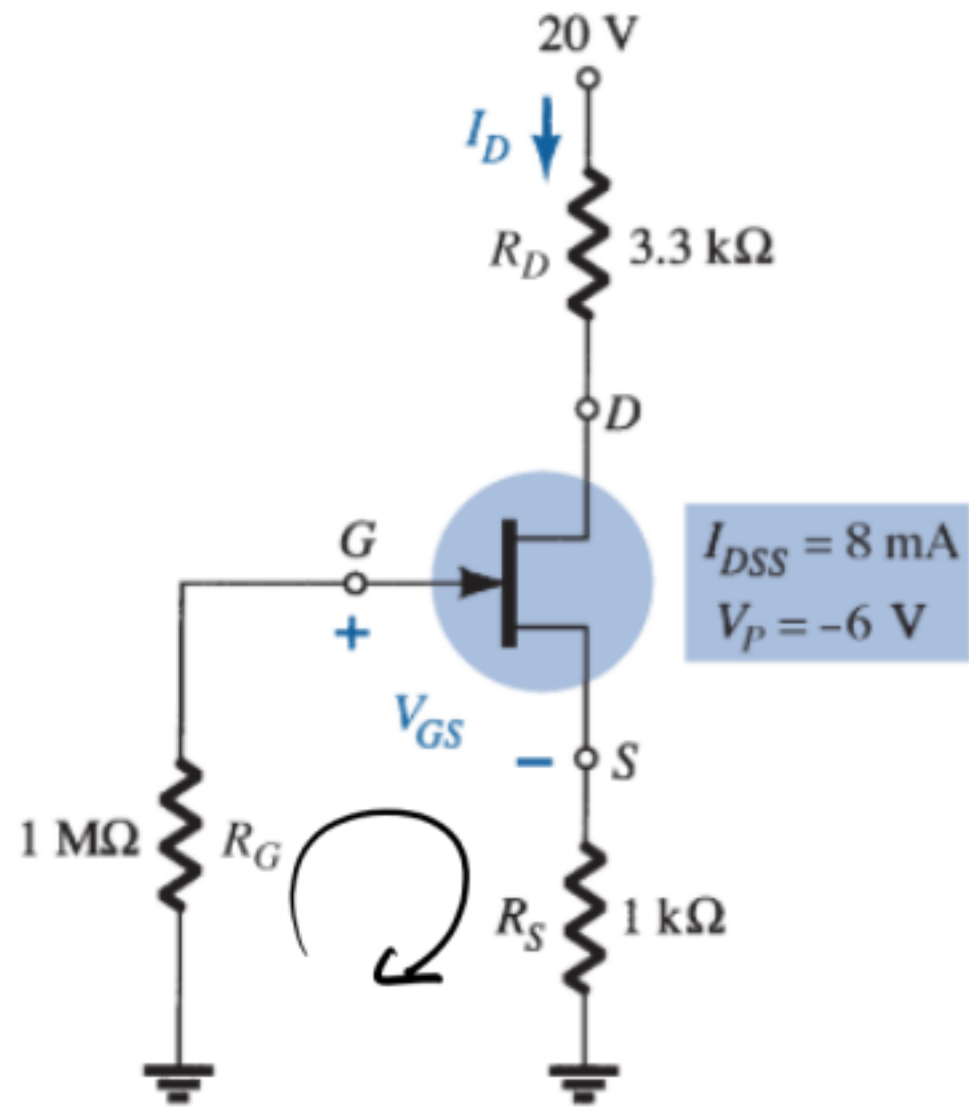
$$\begin{aligned} I_D &\cong I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 = (25 \text{ mA}) \left(1 - \frac{5 \text{ V}}{15 \text{ V}} \right)^2 \\ &= (25 \text{ mA})(1 - 0.333)^2 = 11.1 \text{ mA} \end{aligned}$$

Now, determine R_S .

$$R_S = \left| \frac{V_{GS}}{I_D} \right| = \frac{5 \text{ V}}{11.1 \text{ mA}} = \mathbf{450 \, \Omega}$$

Problem 12: Show the Q-point in the characteristic curve of the JFET in the following circuit.

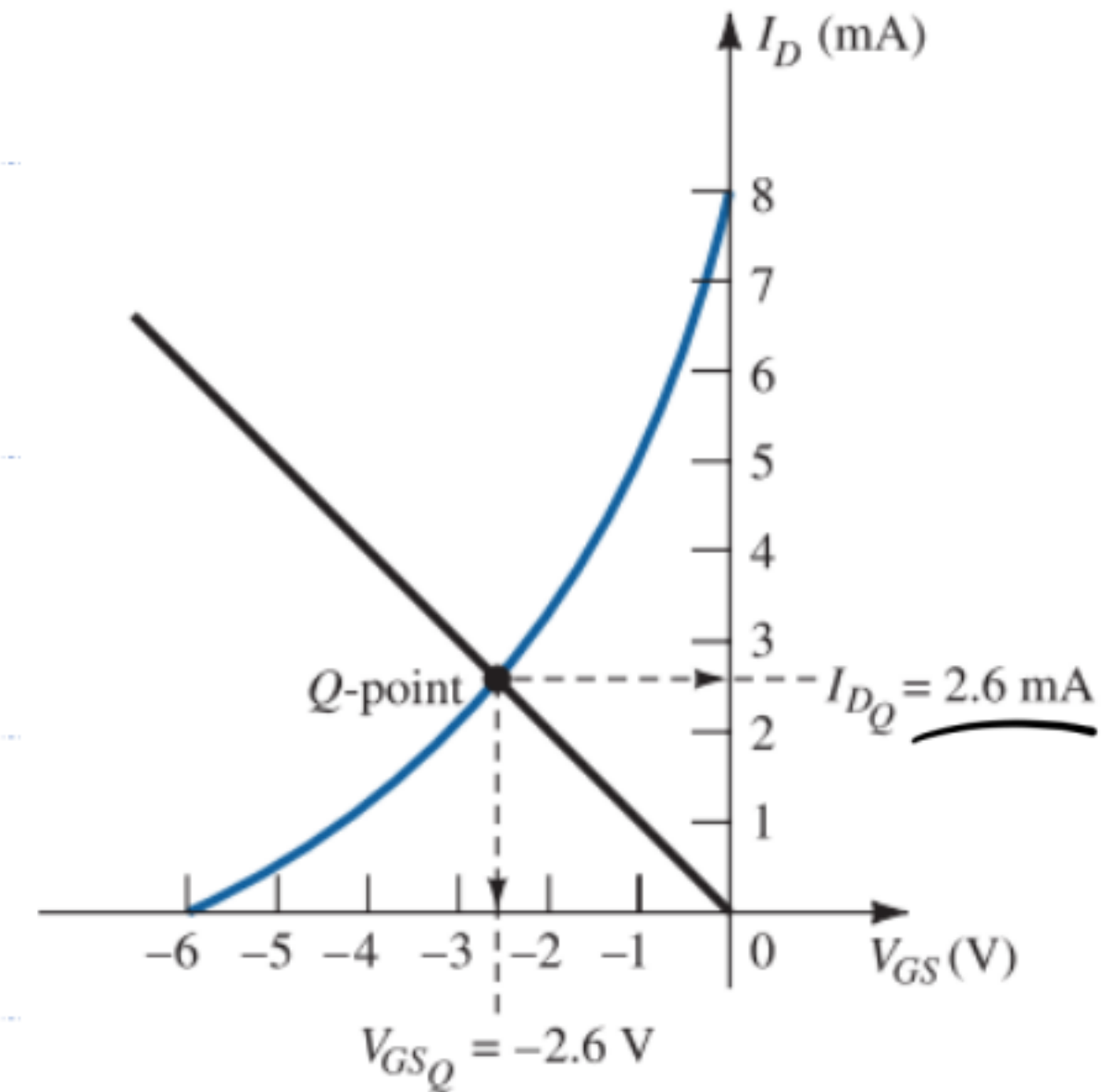
I_D, V_{GS}



$$\hat{I}_D = 8 \text{ mA} \left(1 - \frac{-1 \text{ k} \hat{I}_D}{-6} \right)^2$$

$$\therefore \hat{I}_D = 2.6 \text{ mA} \quad (\text{Problem 9})$$

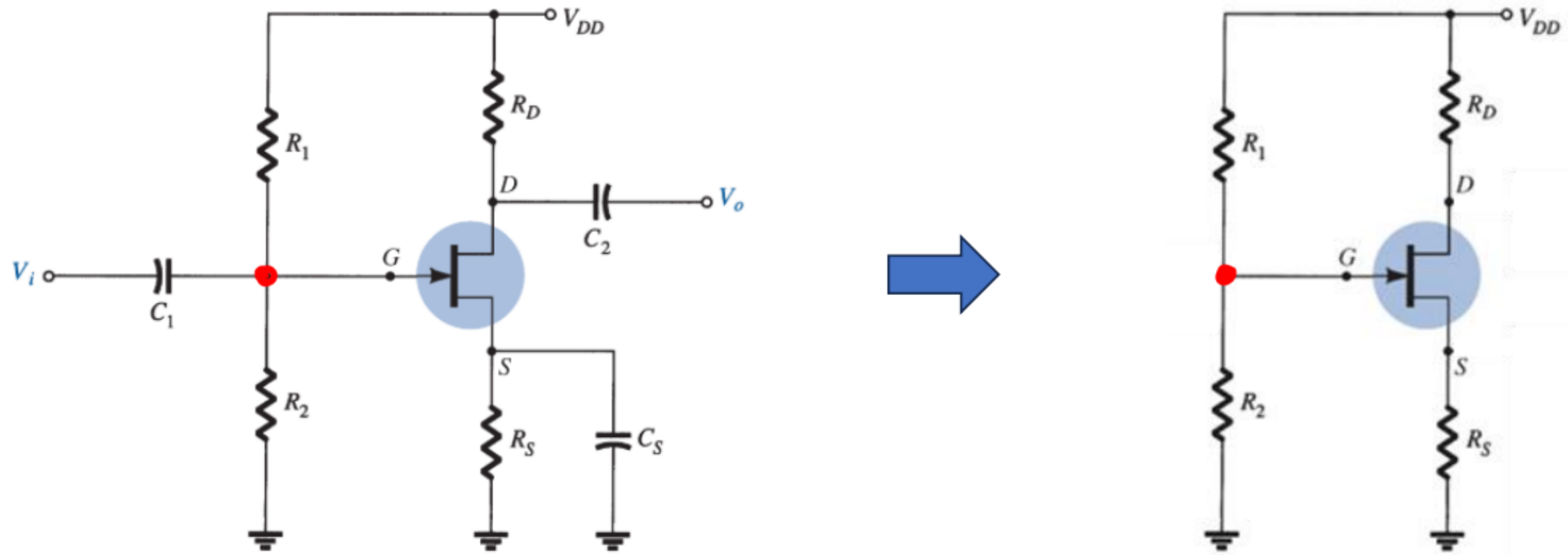
$$\therefore V_{GS} = -\hat{I}_D R_S = -2.6 \text{ V}$$



Voltage-Divider Bias Configuration (1)



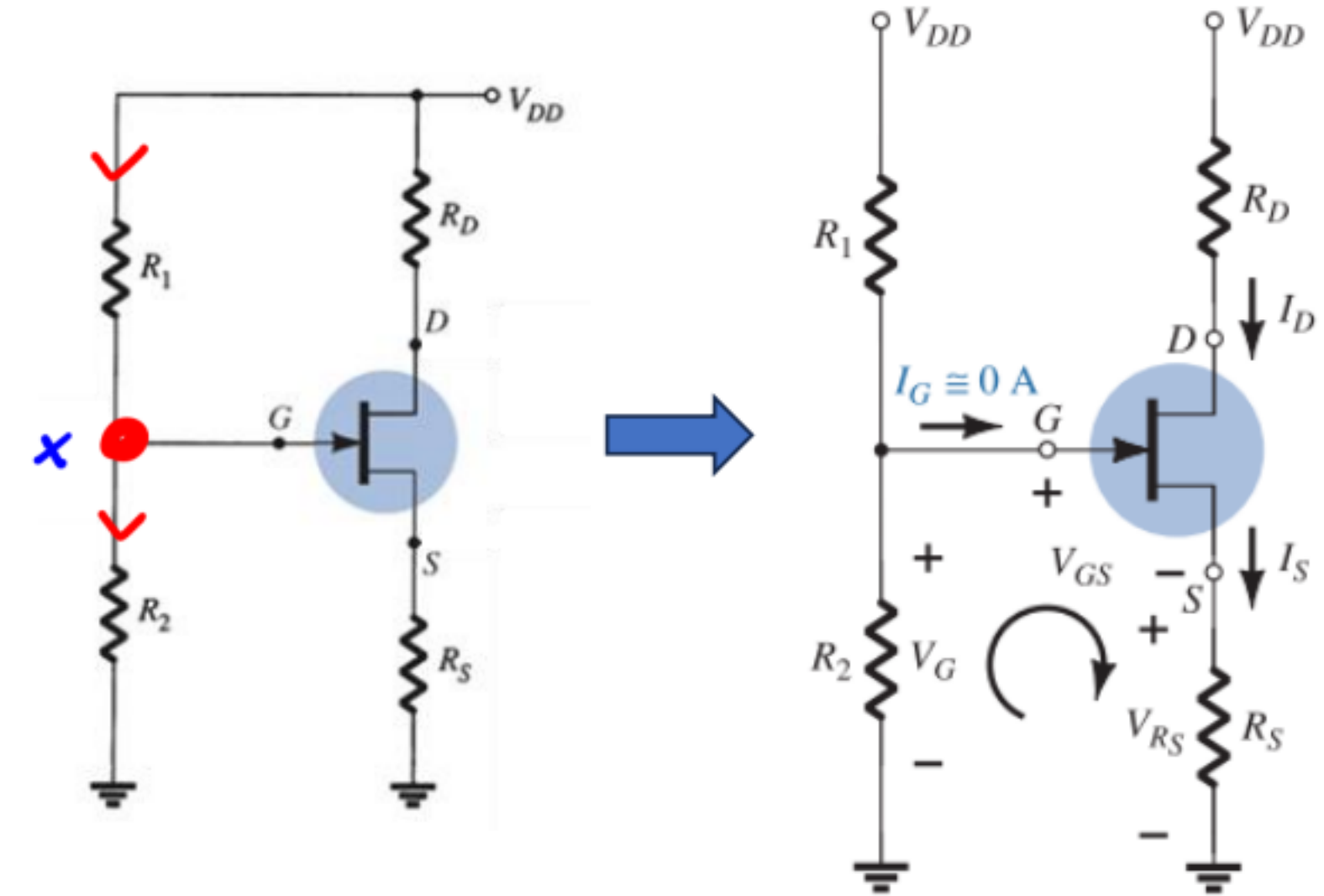
- The voltage-divider bias arrangement applied to BJT transistor amplifiers is also applied to FET amplifiers.



Voltage-Divider Bias Configuration (2)



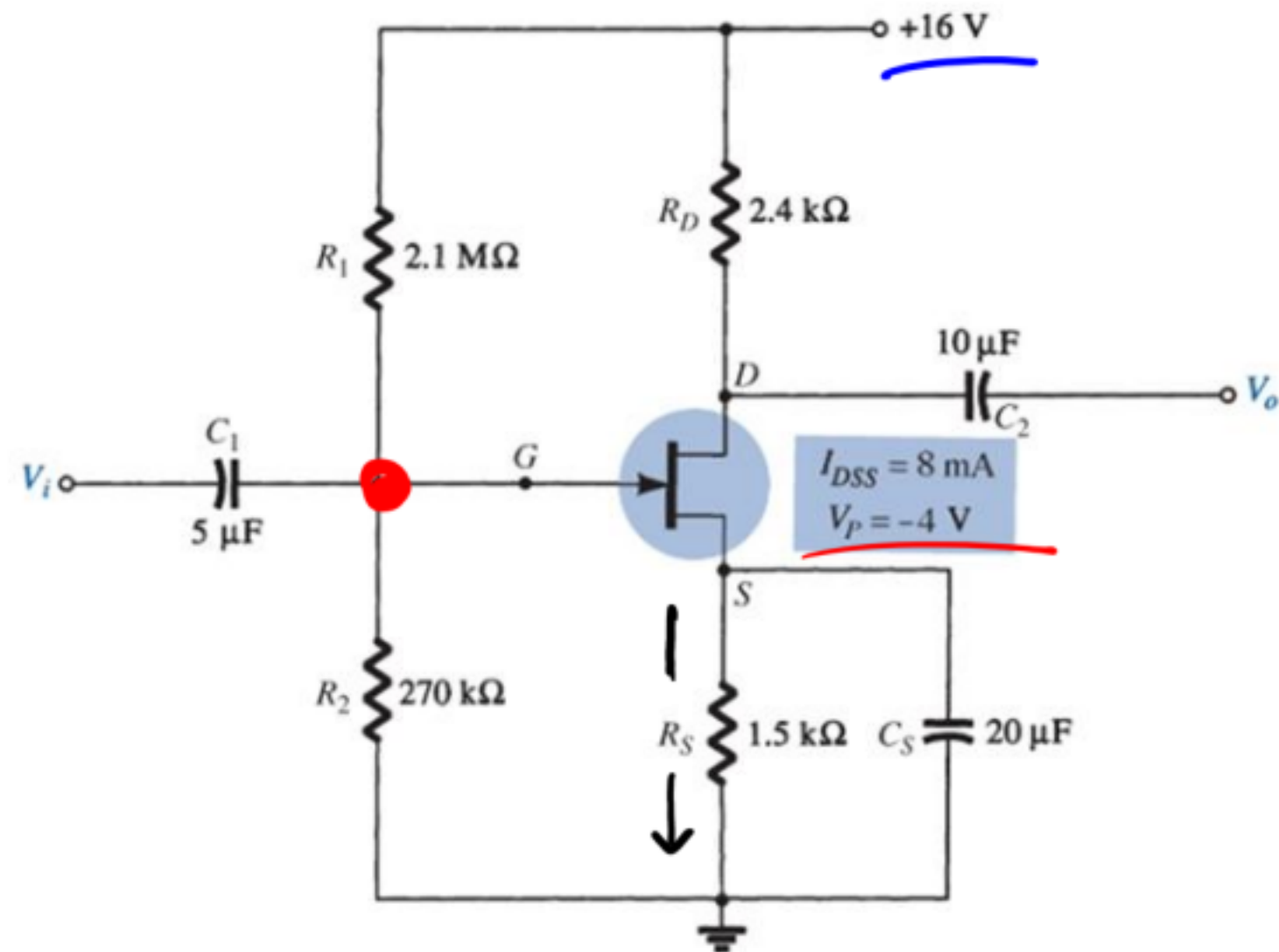
- $V_G = V_{DD} \frac{R_2}{R_1 + R_2}$
- $V_S = I_D R_S$
- $V_{GS} = V_G - V_S = V_G - I_D R_S$ — ①
- $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$ ←
- $I_D = I_{DSS} \left(1 - \frac{V_G - I_D R_S}{V_p} \right)^2 \Rightarrow$ Quadratic equation
- Solve the quadratic equation to get I_{DQ}



$$V_x = V_{DD} \times \frac{R_2}{R_1 + R_2}$$



Problem 13: Find V_{GSQ} , I_{DQ} , V_{DS} , V_D , V_G , V_S



Solving only for DC : $V_G = 16 \times \frac{270k}{270k + 2.1M} = 1.82V$

$$V_{GS} = 1.82 - I_D (1.5k) \quad \text{--- ①}$$

$$I_D = 8m \left(1 - \frac{1.82 - 1.5k I_D}{-4} \right)^2$$

$$I_D = 8m \left(1 + 0.455 - 375 I_D \right)^2$$

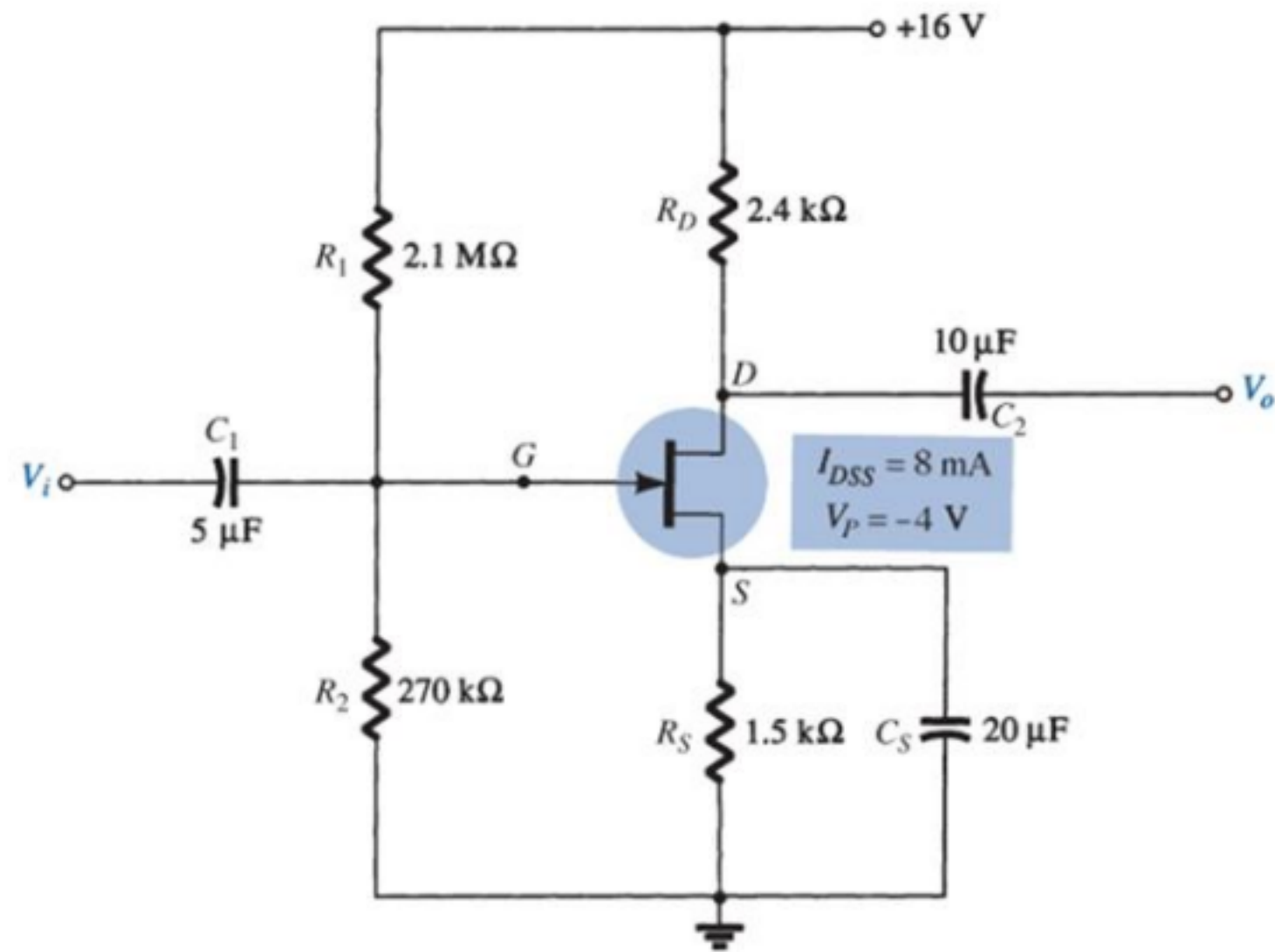
$$\therefore (1125) I_D^2 + (-9.73) I_D + (0.017) = 0$$

$$I_D = 2.4mA / 6.23mA$$

$$\therefore V_{GS} = 1.82 - 6.23 \times 1.5 \\ = -7.5V \rightarrow \text{X}$$

$$\therefore V_{GS} = -1.78V$$

Problem 13: Find V_{GSQ} , I_{DQ} , V_{DS} , V_D , V_G , V_S



$$I_D = 2.4 \text{ mA} \quad V_{GS} = -1.78 \text{ V}$$

$$V_D = V_{DD} - I_D R_D = 10.24 \text{ V}$$

$$V_S = I_D R_S = 3.6 \text{ V}$$

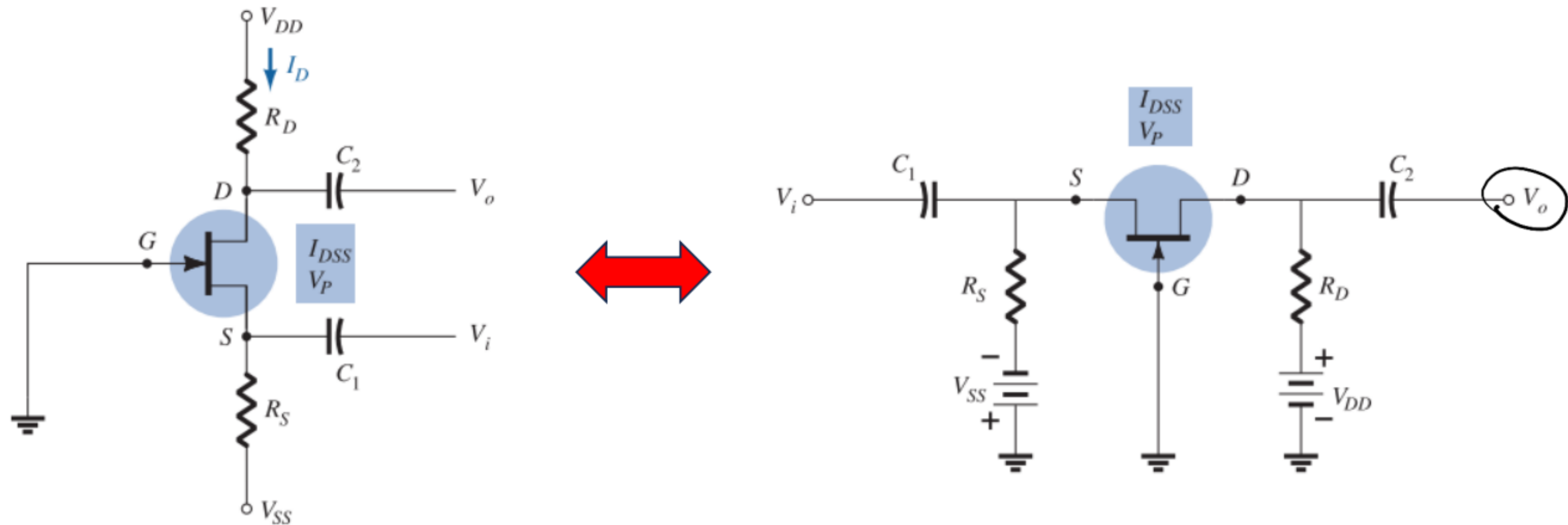
$$\therefore V_{DS} = V_D - V_S = 6.64 \text{ V}$$

$$V_G = 1.82 \text{ V}$$

Common-Gate Configuration (1)



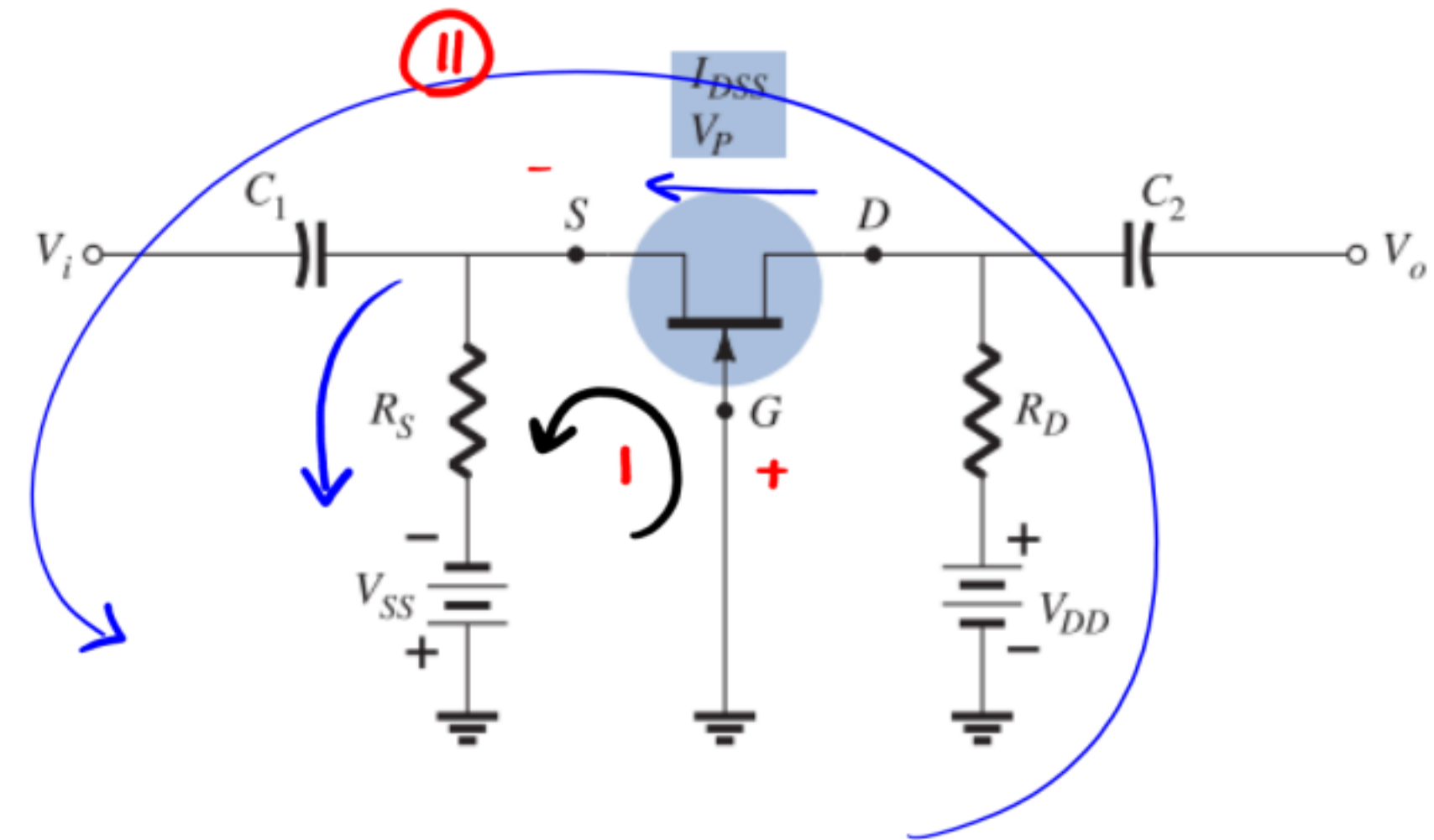
- Here, the gate terminal is grounded and the input signal typically applied to the source terminal and the output signal obtained at the drain terminal as shown in the circuits below.
- The common-gate configuration circuit can be drawn in the following two ways:



Common-Gate Configuration (2)



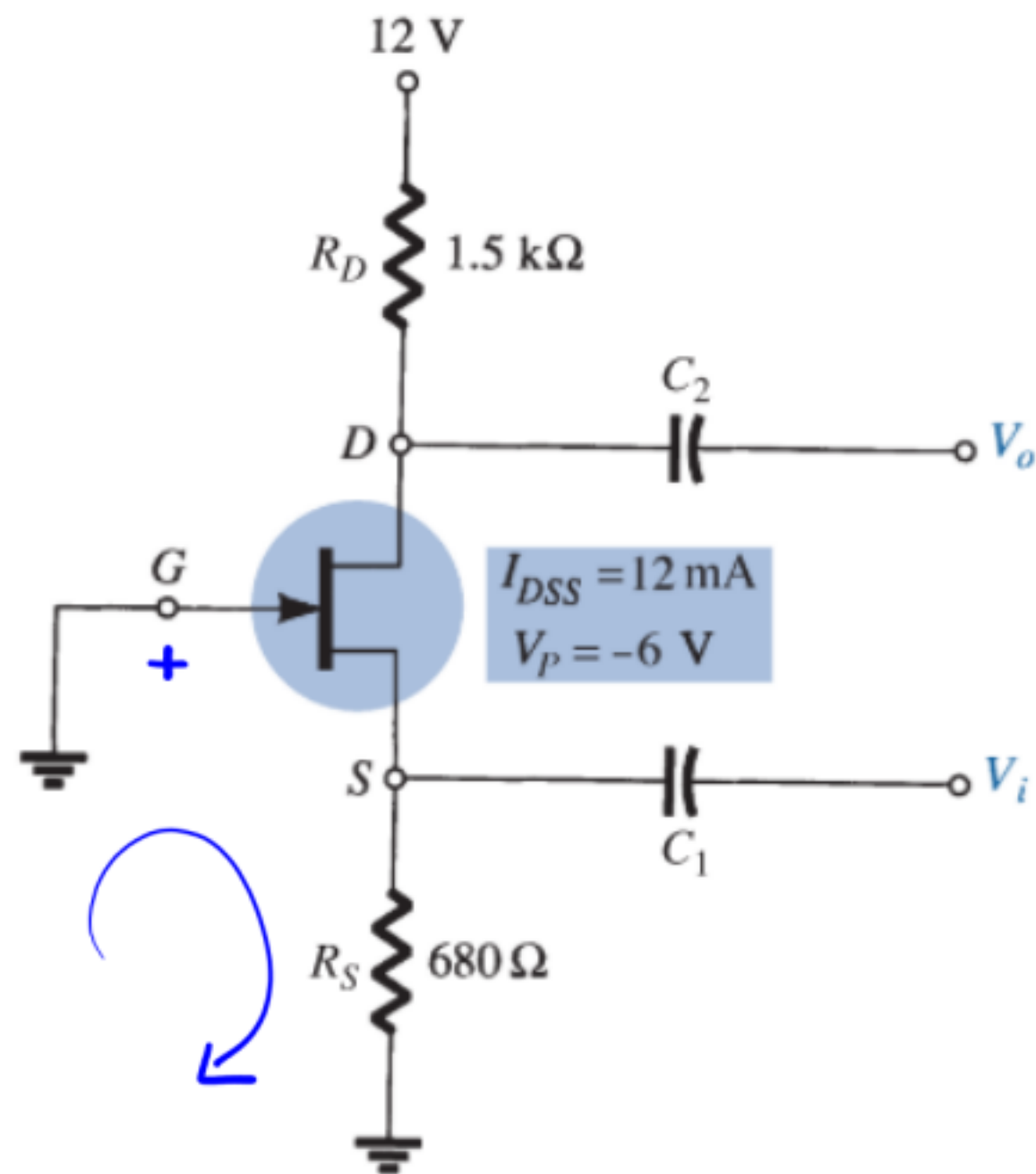
- $V_{GS} = V_{SS} - I_D R_S$
- $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$
- $I_D = I_{DSS} \left(1 - \frac{V_{SS} - I_D R_S}{V_P} \right)^2 \Rightarrow$ Quadratic equation
- Solve the quadratic equation to get I_{DQ}
- $-V_{DD} + I_D R_D + V_{DS} + I_D R_S - V_{SS} = 0$
- $V_{DS} = V_{DD} + V_{SS} - I_D (R_D + R_S)$



$$V_{GS} + I_D R_S - V_{SS} = 0$$



Problem 14: Find V_{GSQ} , I_{DQ} , V_{DS} , V_D , V_G , V_S



$$V_{GS} + I_D R_S = 0 \quad \therefore V_{GS} = -680 I_D \quad \text{--- ①}$$

$$I_D = 12 \text{ m} \left(1 - \frac{-680 I_D}{-6} \right)^2 = 12 \text{ m} (1 - 113.33 I_D)^2$$

$$\therefore (154.13) I_D^2 + (-3.72) I_D + (12 \times 10^{-3}) = 0$$

$$I_{DQ} = \underline{0.02 \text{ A}}, \underline{3.83 \text{ mA}}$$

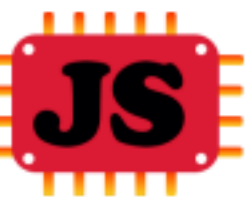
$$\therefore V_{GSQ} = -2.6 \text{ V}$$

$$\therefore V_{DS} = 12 - I_D (R_S + R_D) = 3.65 \text{ V}$$

$$V_D = 12 - I_D R_D = 6.3 \text{ V}$$

$$\therefore V_S = 2.58 \text{ V}$$

JFET AC ANALYSIS



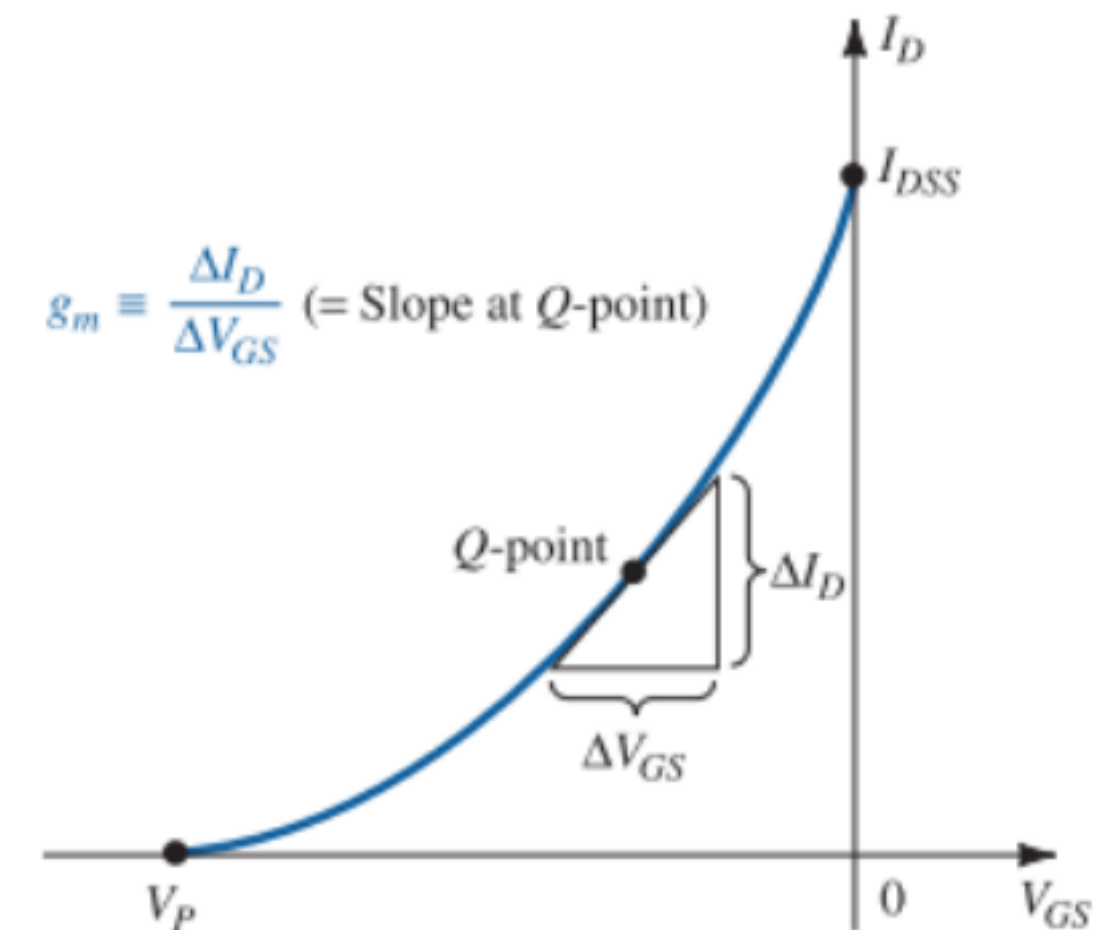
Najmus Sakib, BUET (EEE-14) ; WhatsApp: 01892777317

JFET small-signal model



- DC gate-to-source voltage controls the level of DC drain current through the JFET
- V_{GS} and I_D are related by Shockley's equation: $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$
- The change in drain current that will result from a change in V_{GS} can be determined using the **trans-conductance factor g_m**

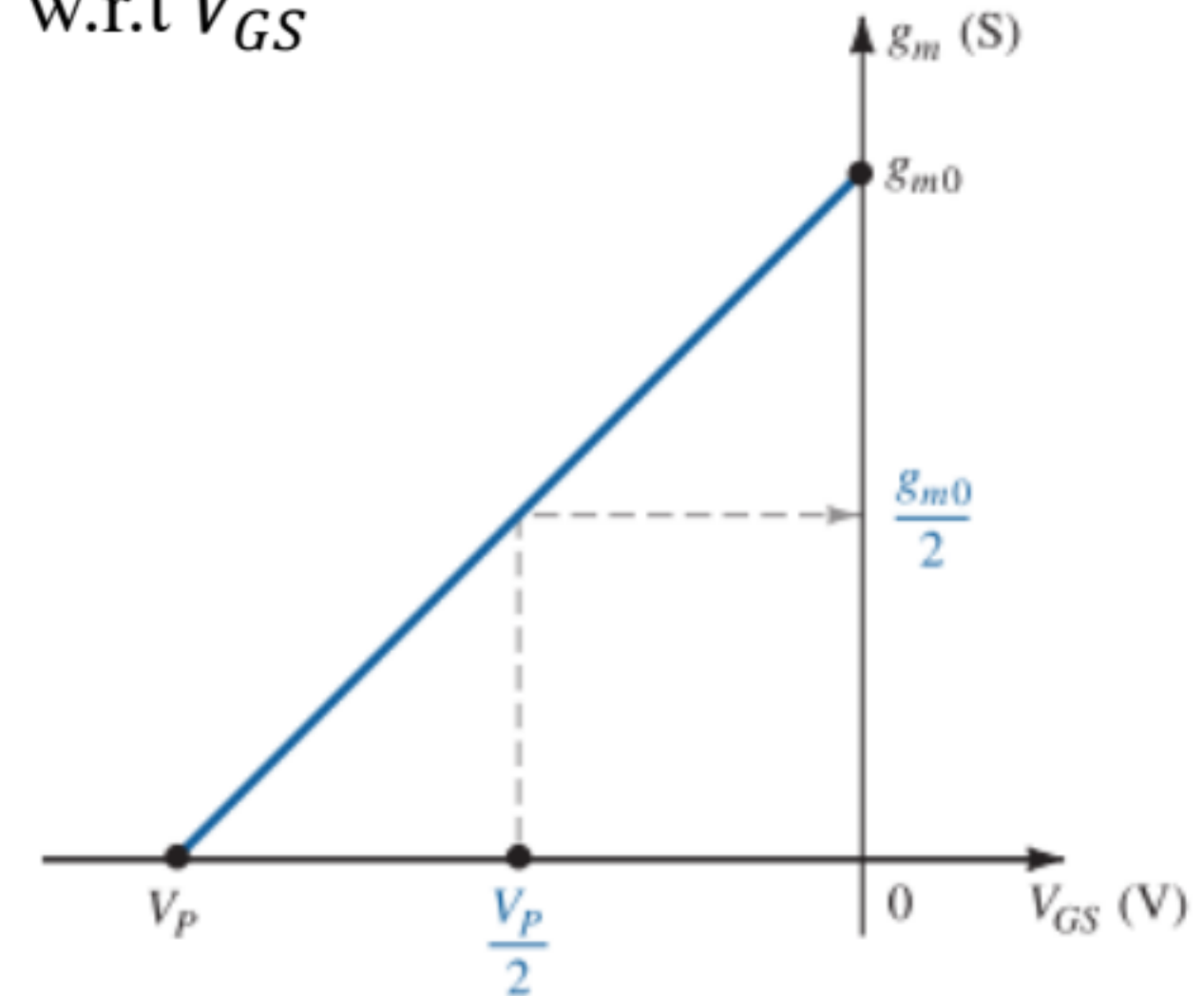
- $\Delta I_D = g_m \Delta V_{GS} \Rightarrow g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{d(I_D)}{d(V_{GS})}$
- g_m is also written as g_{fs} or y_{fs} in spec. sheets.



Trans-conductance factor g_m



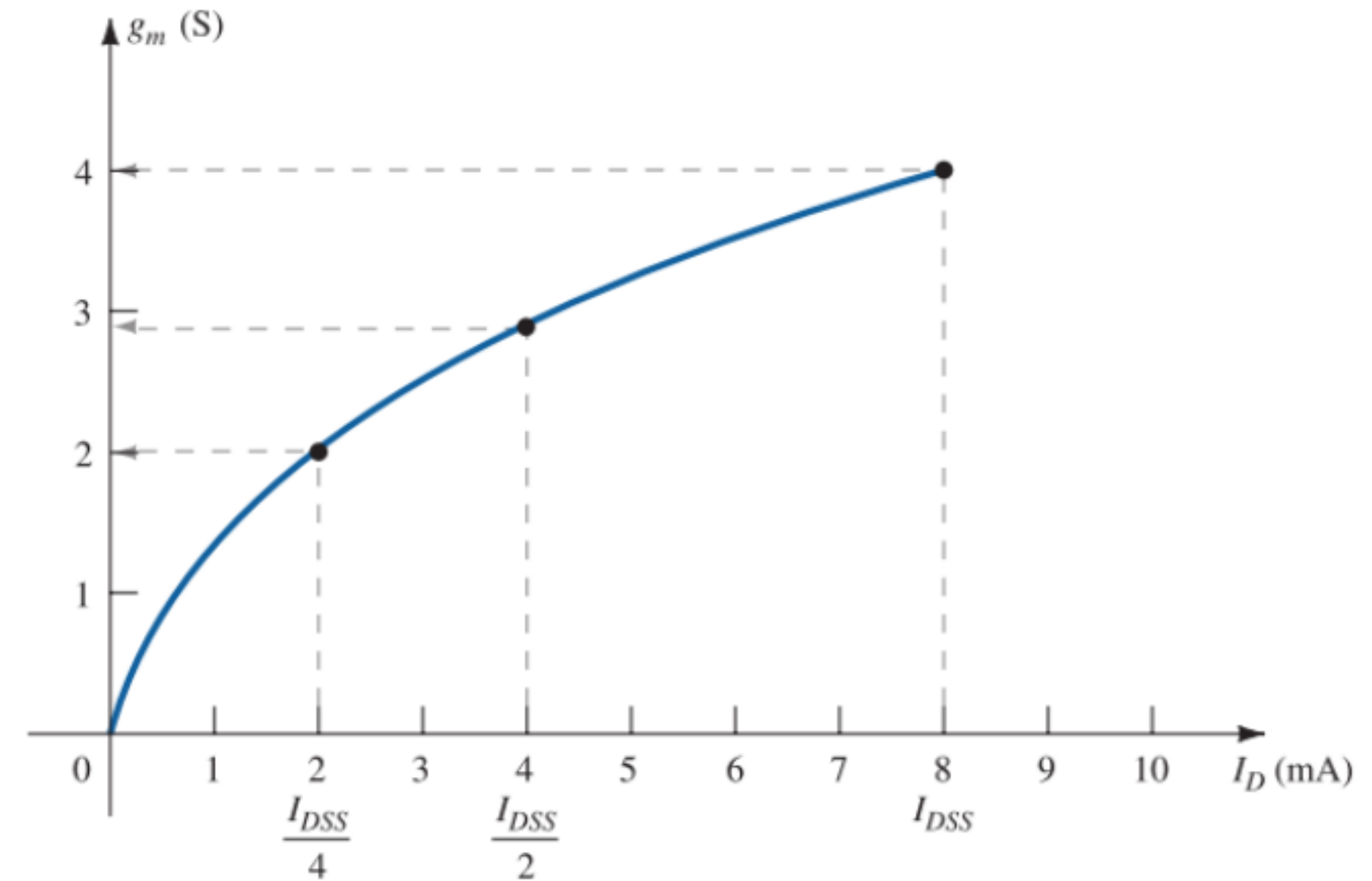
- An accurate formula can be derived for g_m by taking the derivative of I_D w.r.t V_{GS}
- $g_m = \frac{2 I_{DSS}}{|V_P|} \left[1 - \frac{V_{GS}}{V_P} \right]$ —✓
- At $V_{GS} = 0$, $g_m(max) = g_{m0} = \frac{2 I_{DSS}}{|V_P|} \left[1 - \frac{0}{V_P} \right] = \frac{2 I_{DSS}}{|V_P|}$
- $g_m = g_{m0} \left[1 - \frac{V_{GS}}{V_P} \right]$ —✓
- The unit of g_m is **Siemens (S)**
- If $V_{GS(off)}$ is given instead, use $g_m = g_{m0} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]$ ($V_{GS(off)} \rightarrow -ve$ for N-JFET)



Effect of I_D on g_m



- $g_m = g_{m0} \left[1 - \frac{V_{GS}}{V_P} \right]$
- $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$
- Combining these two equations, we get:
- $I_D = I_{DSS} \left(\frac{g_m}{g_{m0}} \right)^2$
- $g_m = g_{m0} \sqrt{\frac{I_D}{I_{DSS}}}$



Problem 15:

The following information is included on the datasheet in Figure 8–14 for a 2N5457 JFET: typically, $I_{DSS} = 3.0 \text{ mA}$, $V_{GS(off)} = -6 \text{ V}$ maximum, and $g_{fs(max)} = 5000 \mu\text{S}$. Using these values, determine the forward transconductance for $V_{GS} = -4 \text{ V}$, and find I_D at this point.

$g_{m0} = g_{fs} = 5000 \mu\text{S}$. Use Equation 8–2 to calculate g_m .

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right) = (5000 \mu\text{S}) \left(1 - \frac{-4 \text{ V}}{-6 \text{ V}} \right) = \mathbf{1667 \mu\text{S}}$$

Next, use Equation 8–1 to calculate I_D at $V_{GS} = -4 \text{ V}$.

$$I_D \cong I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 = (3.0 \text{ mA}) \left(1 - \frac{-4 \text{ V}}{-6 \text{ V}} \right)^2 = \mathbf{333 \mu\text{A}}$$

$$g_{fs} = g_m$$

$$g_{fs(max)} = g_{m(max)} \\ = g_{m0}$$

$$\hat{I}_D = 3\text{m} \left(\frac{1667}{5000} \right)^2 = 333.5 \mu\text{A}$$

Input & Output Impedance of JFET



- Input impedance of all commercial JFETs is sufficiently large enough to approximate an open circuit.

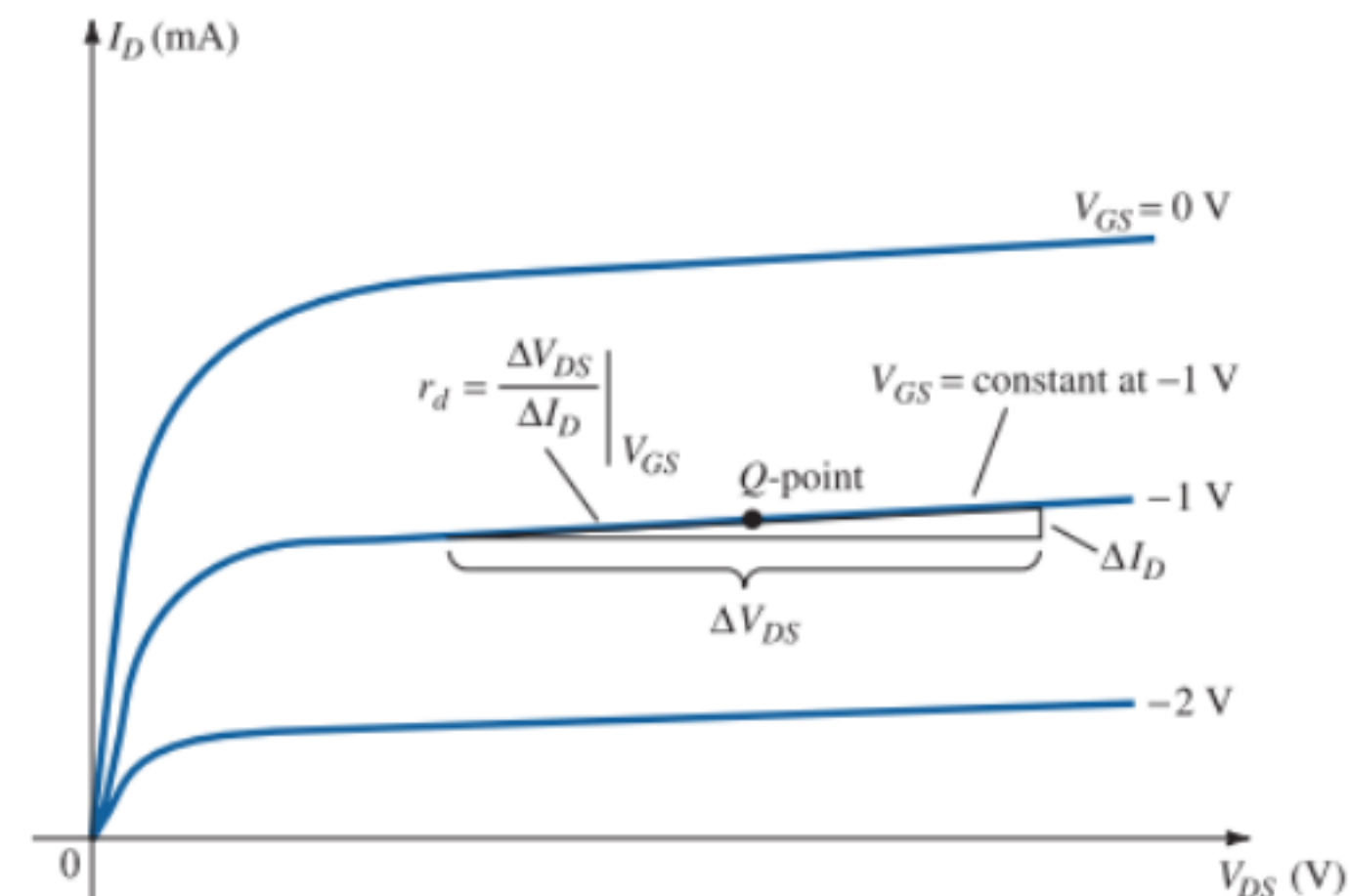
$$Z_i(\text{JFET}) = \infty$$

- The output impedance of JFETs is similar in magnitude to that of conventional BJTs.
- It is the effective resistance appearing across drain-to-source as current flows through it.
- On JFET specification sheets, Z_o will typically appear as g_{os} or y_{os} with the units of μS .

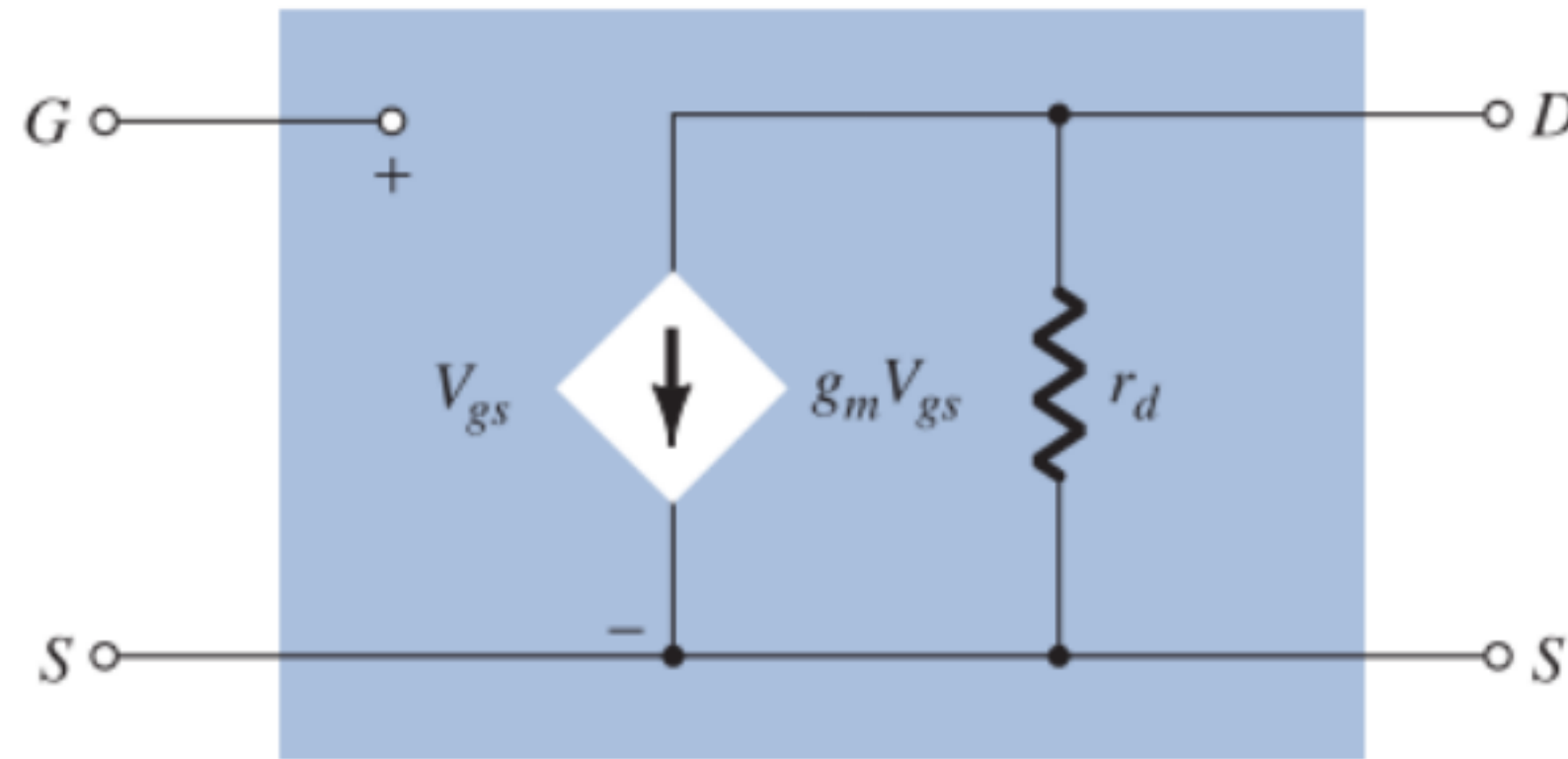
- $Z_o = \frac{V_o}{I_o}$ (when $V_i = 0$)

- $Z_o(\text{JFET}) = r_d = \frac{1}{g_{os}} = \frac{\Delta V_{DS}}{\Delta I_D}$

$g_{os} \rightarrow$ slope of characteristic curve for JFET



JFET AC Equivalent Circuit



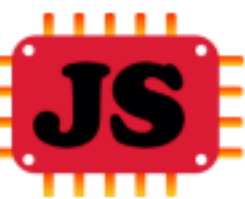
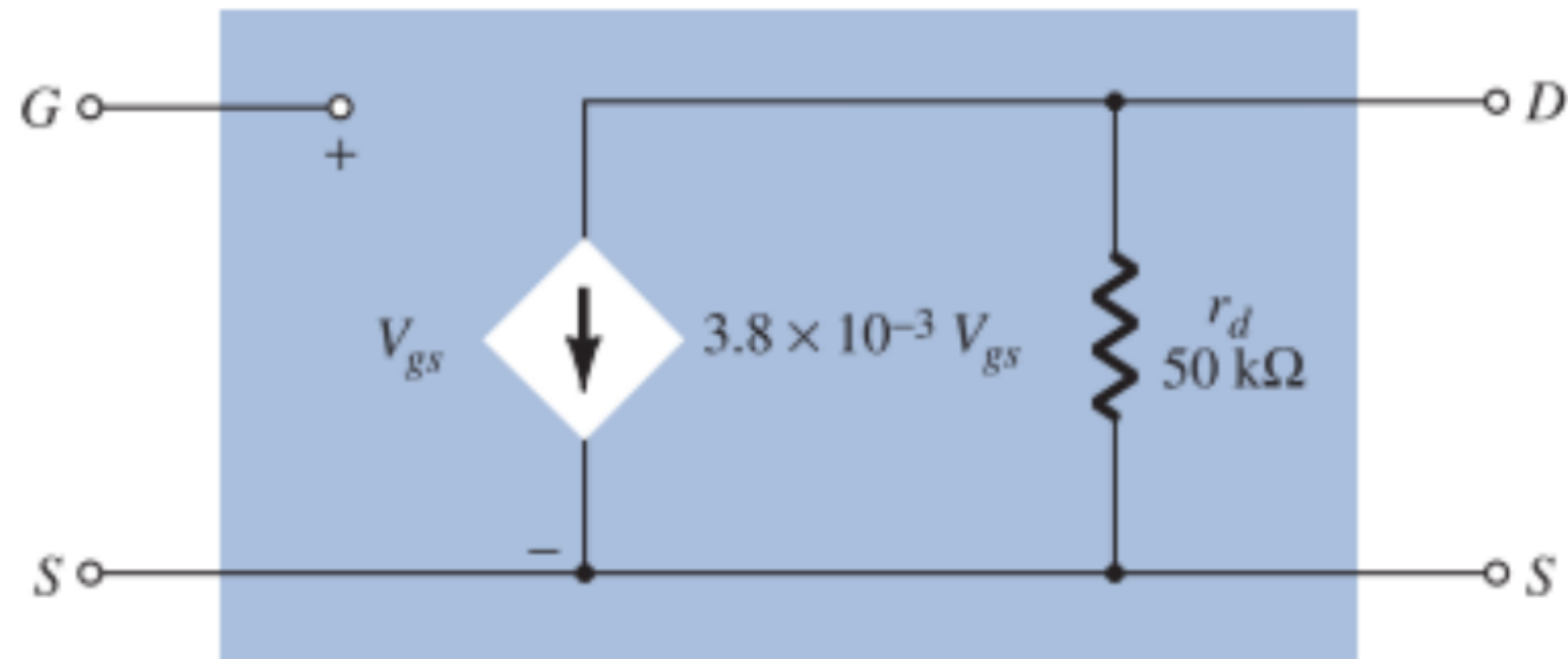
- Notes : Gate-to-Source voltage is V_{gs} , not V_{GS} ; to distinguish it from DC.



Problem 16: Given $g_{fs} = 3.8 \text{ mS}$ & $g_{os} = 20 \text{ } \mu\text{S}$, sketch the FET ac equivalent model.

$$g_m = g_{fs} = 3.8 \times 10^{-3} \text{ S}$$

$$r_d = \frac{1}{g_{os}} = \frac{1}{20 \text{ } \mu\text{S}} = 50 \text{ k}\Omega$$



Procedures in AC analysis



Pre-requisite: Do DC Analysis of the Circuit first and find out g_m & r_d

AC Analysis Steps:

1. Short the coupling capacitors.
2. Short the DC voltages.
3. Replace the JFET with its equivalent AC small-signal model.

Redraw the circuit using the above 03 rules.

$$\begin{aligned} \bullet \quad g_m &= \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) \\ &= g_{m0} \left(1 - \frac{V_{GS}}{V_P}\right) \end{aligned}$$

$$\bullet \quad r_d = \frac{1}{g_o} = \frac{\Delta V_{DS}}{\Delta I_D}$$

$$X_c = \frac{1}{2\pi fC}$$



Before we begin with our analysis...



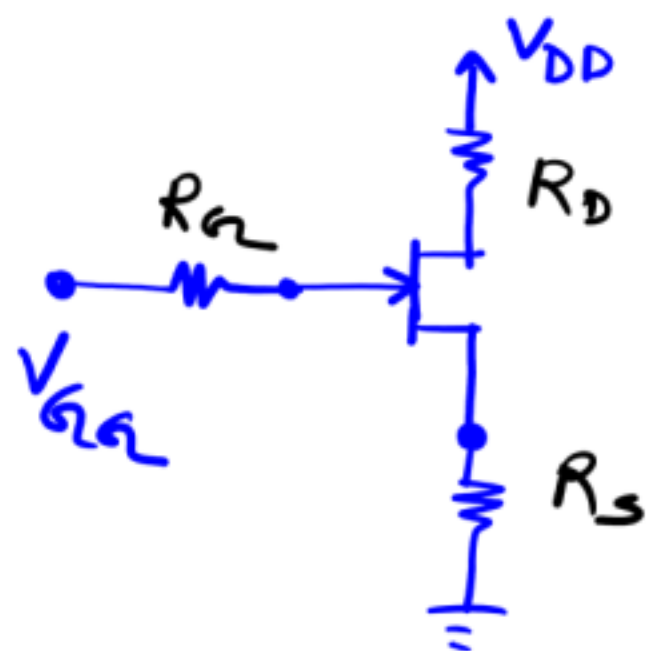
AC analysis for the following biasing configurations will be discussed here:

1. Fixed-Bias Configuration (Source $\rightarrow$ grounded, Gate $\rightarrow$ Input, Drain $\rightarrow$ Output)
2. Self-Bias Configuration (Source $\rightarrow$ grounded, Gate $\rightarrow$ Input, Drain $\rightarrow$ Output)
3. Voltage-Divider Biasing (Gate $\rightarrow$ Input, Drain $\rightarrow$ Output)
4. Common-Gate Configuration (Gate $\rightarrow$ grounded, Source $\rightarrow$ Input, Drain $\rightarrow$ Output)
5. Common-Drain Configuration (Source Follower) (Drain $\rightarrow$ grounded, Gate $\rightarrow$ Input, Source $\rightarrow$ Output)

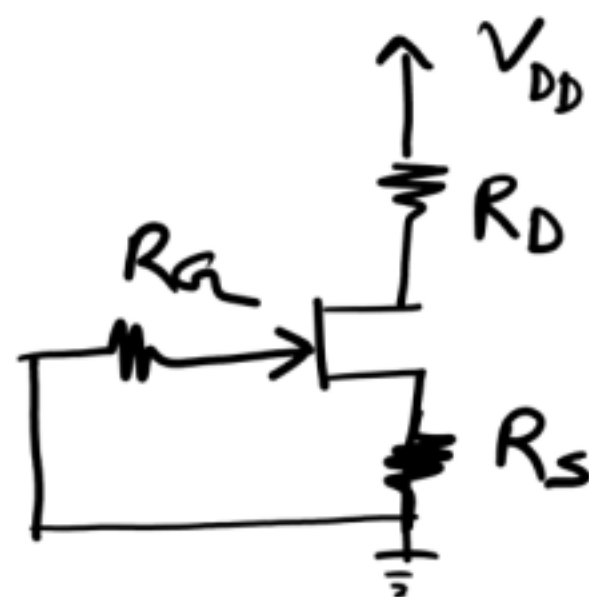
G (*gate*) $\rightarrow$ S (*source*) $\rightarrow$ D (*drain*)



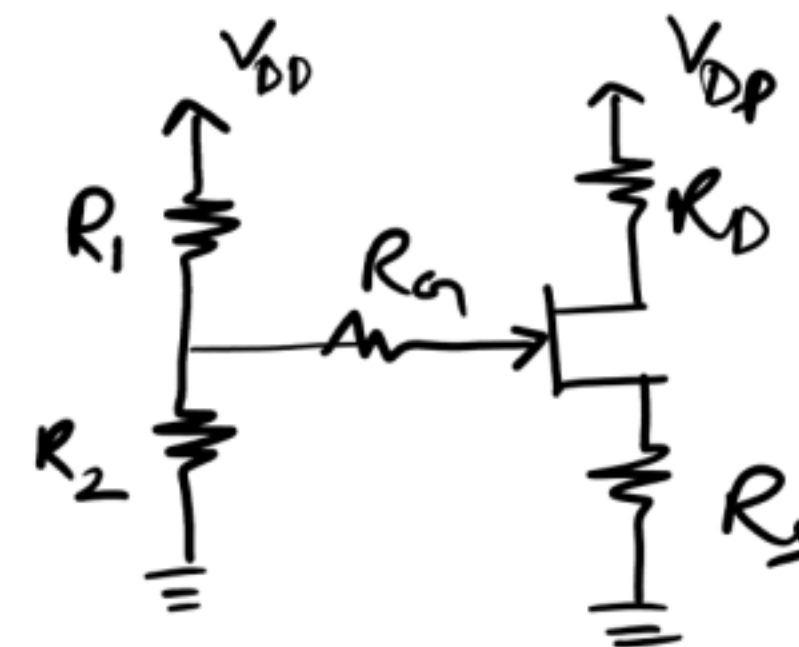
Fixed-Bias Configuration



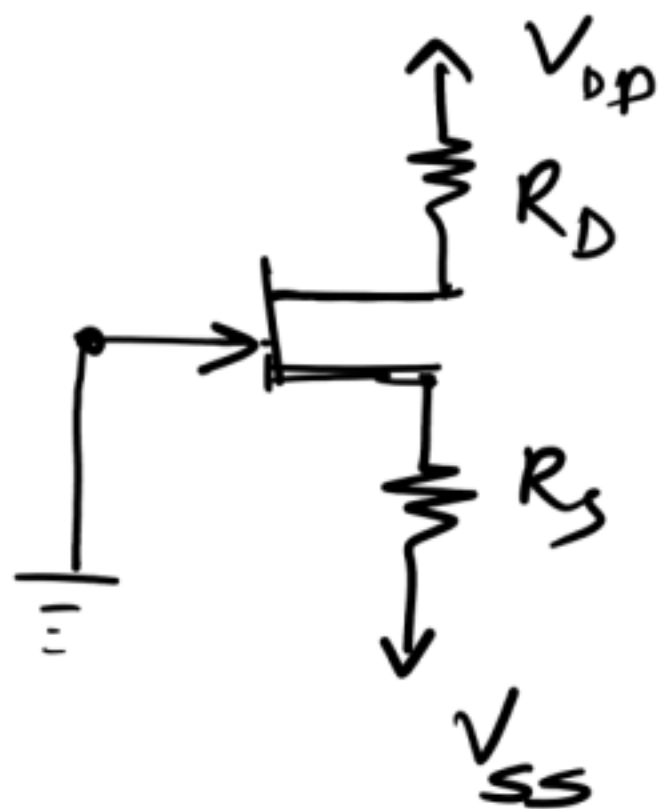
Self-Bias Configuration



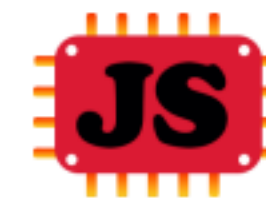
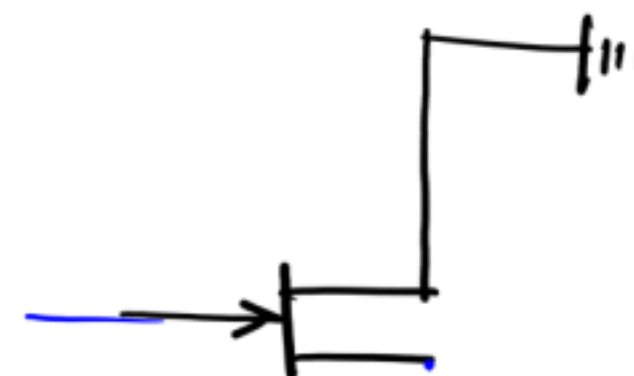
Voltage-Divider Biasing



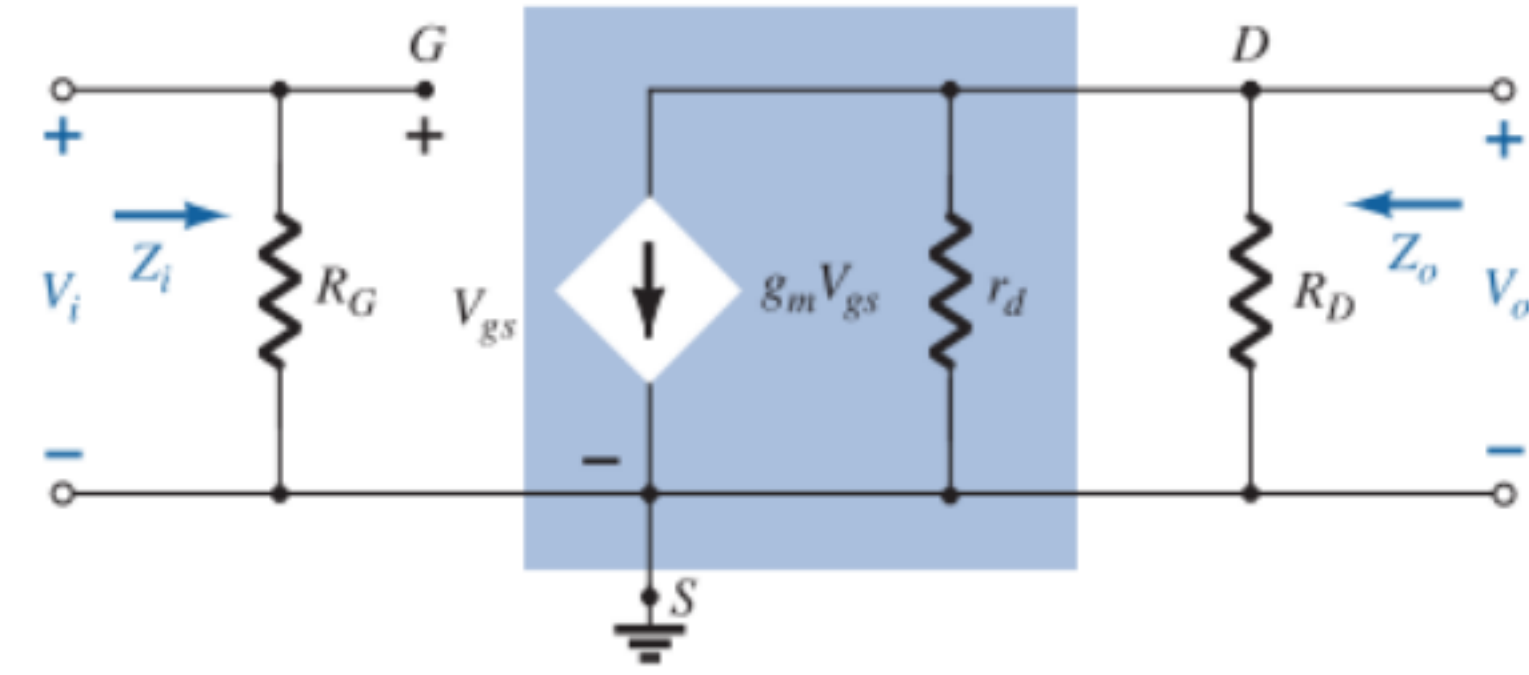
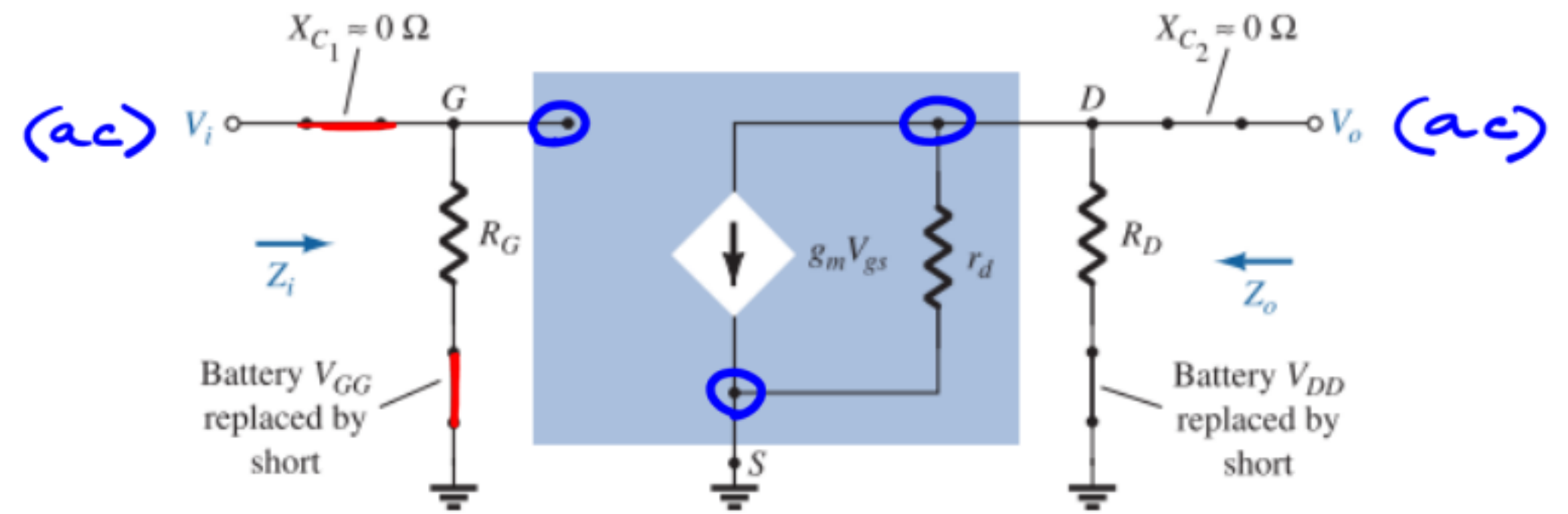
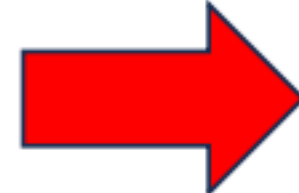
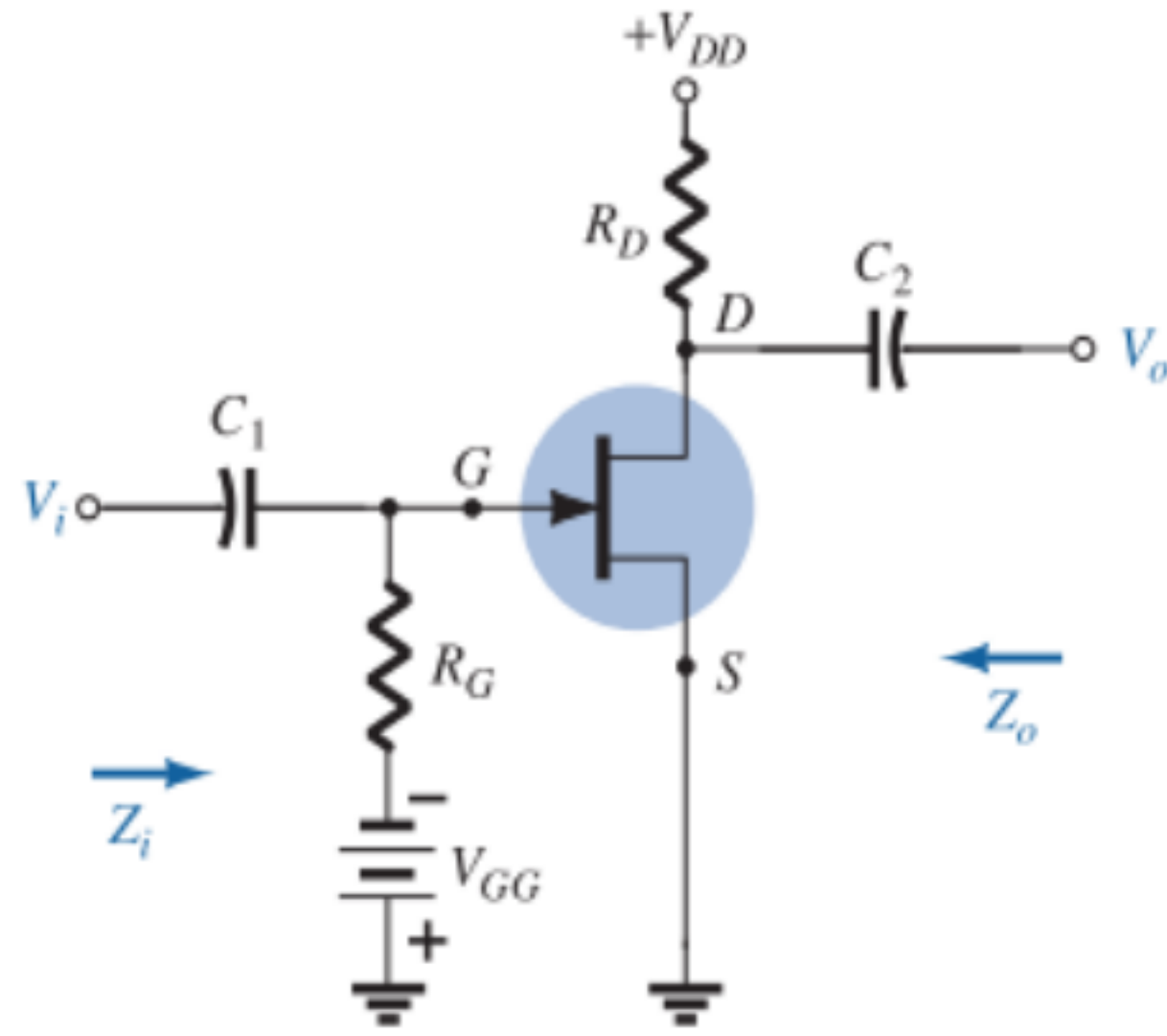
Common-Gate Configuration



Common-Drain Configuration



AC Analysis for Fixed-Bias Configuration (1)



- Input impedance, $Z_i = R_G$

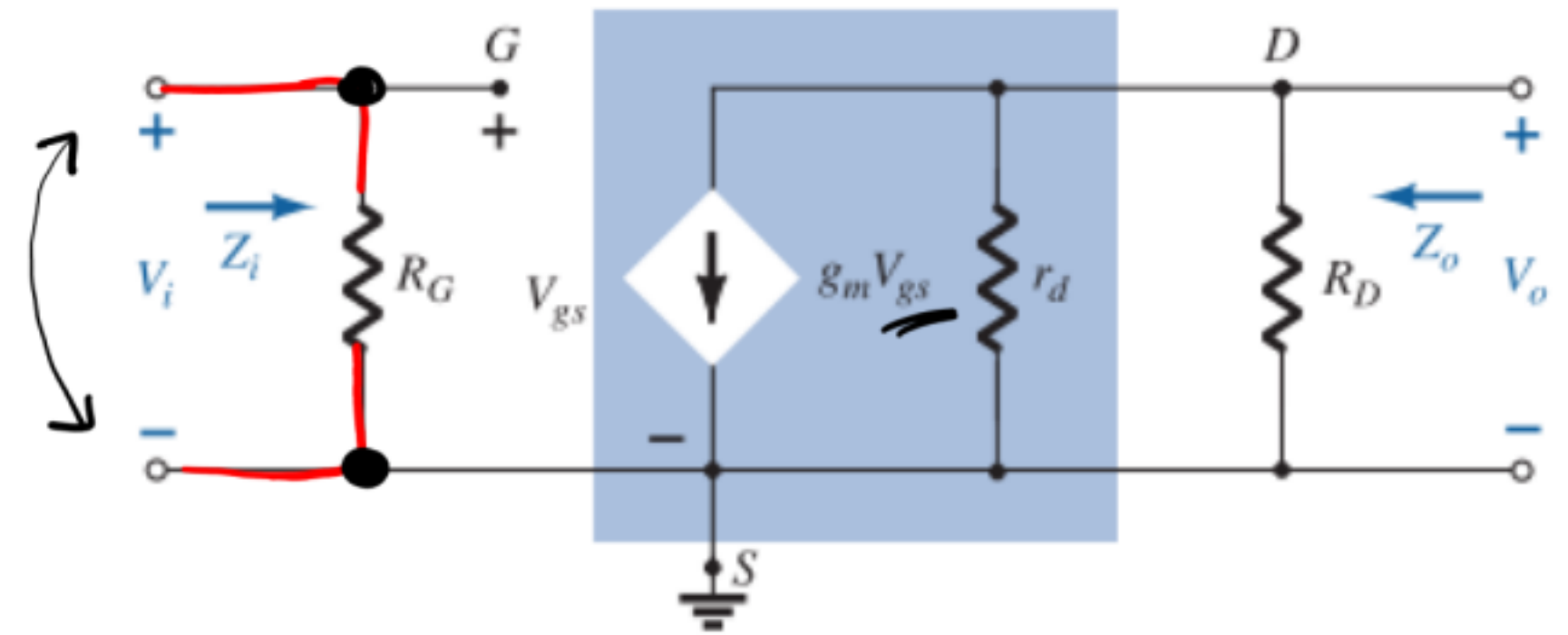
- Output impedance, $Z_o = \frac{V_o}{I_o}$ (when $V_i = 0$)

$$= r_d \parallel R_D = \frac{R_D r_d}{R_D + r_d}$$

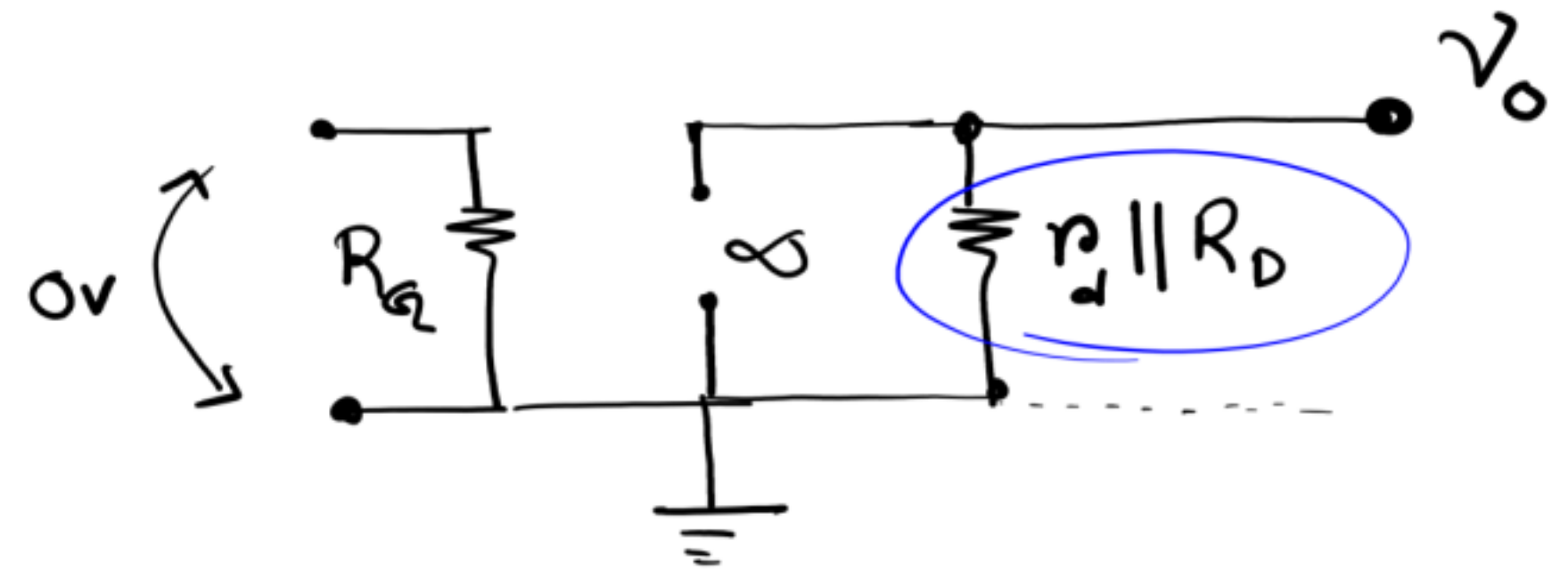
- When $r_d \geq 10R_D$, $Z_o \cong R_D$

$$Z_o = \frac{R_D}{\left(\frac{R_D}{r_d} + 1\right)} \cong R_D$$

→ very small



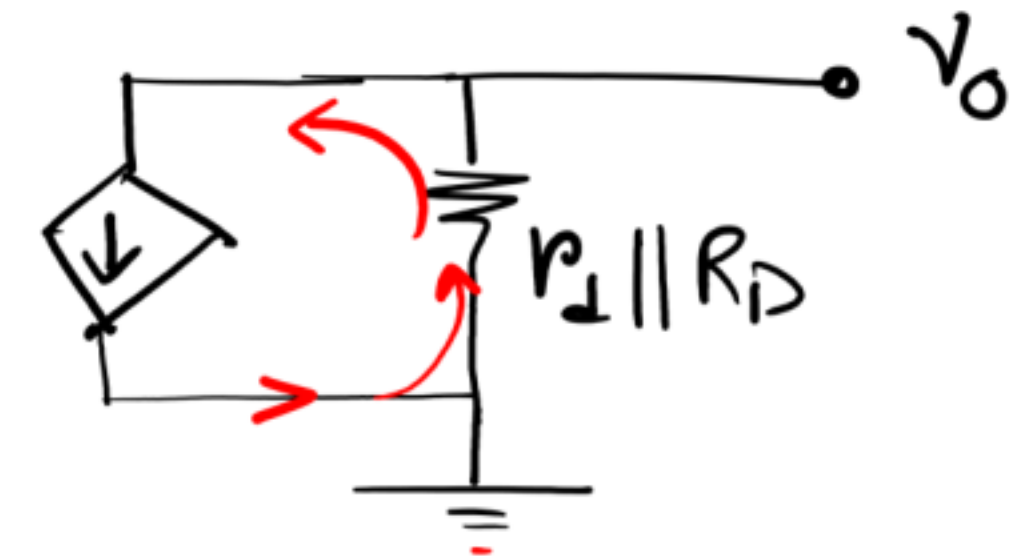
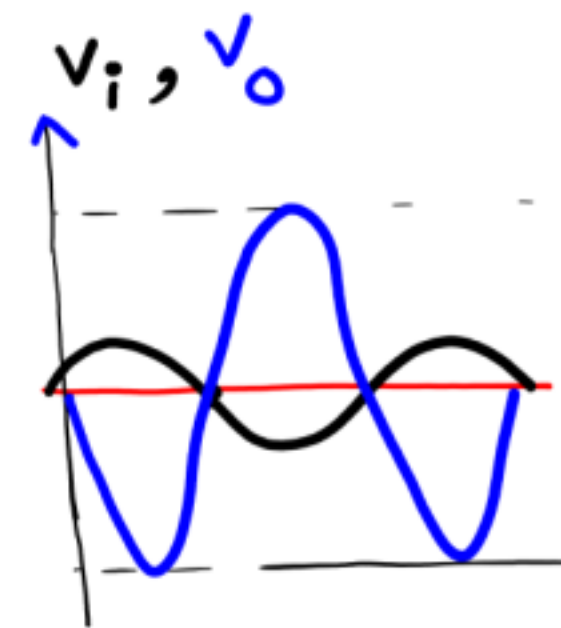
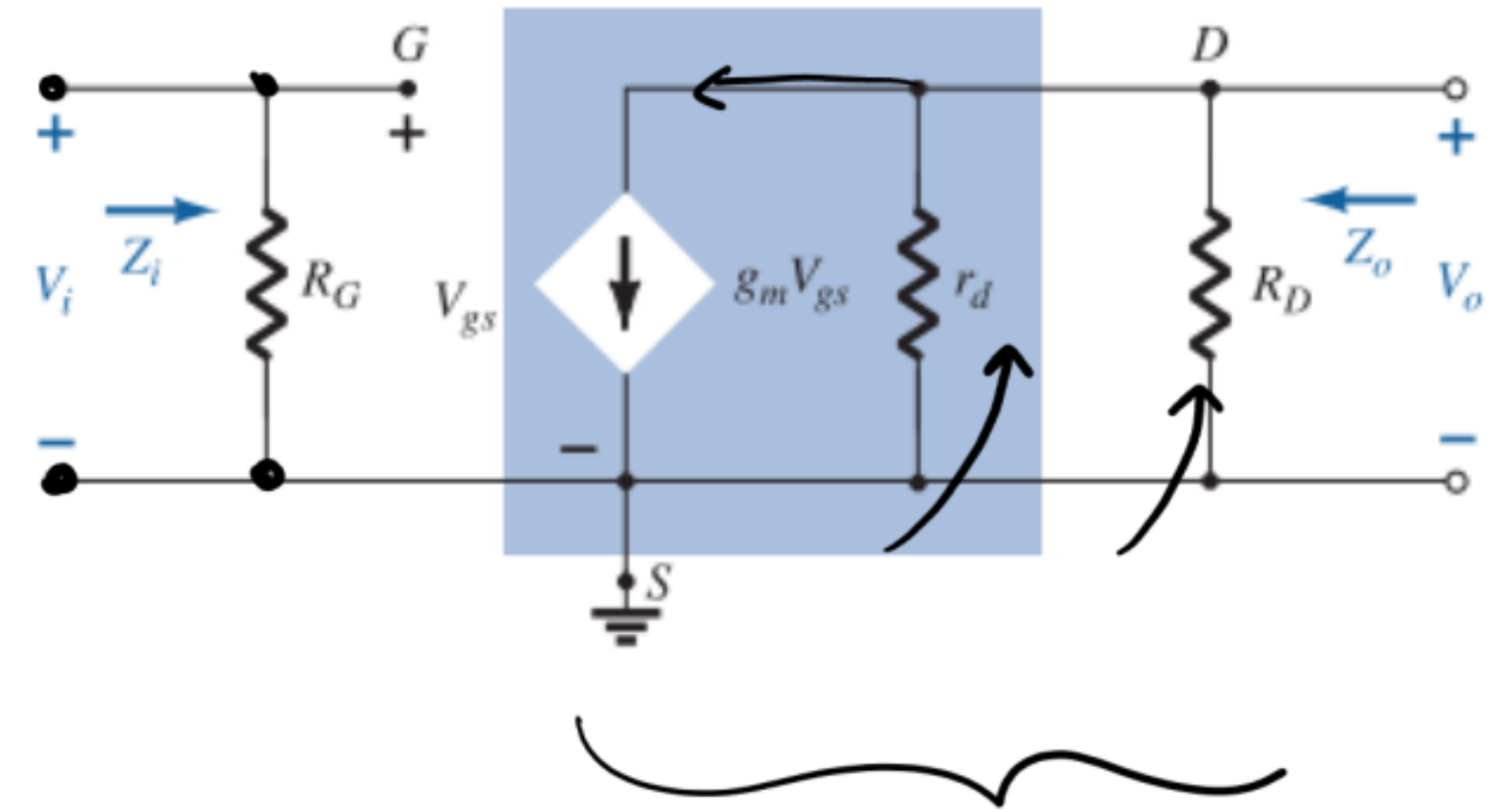
$$V_{gs} = V_i$$



AC Analysis for Fixed-Bias Configuration (2)



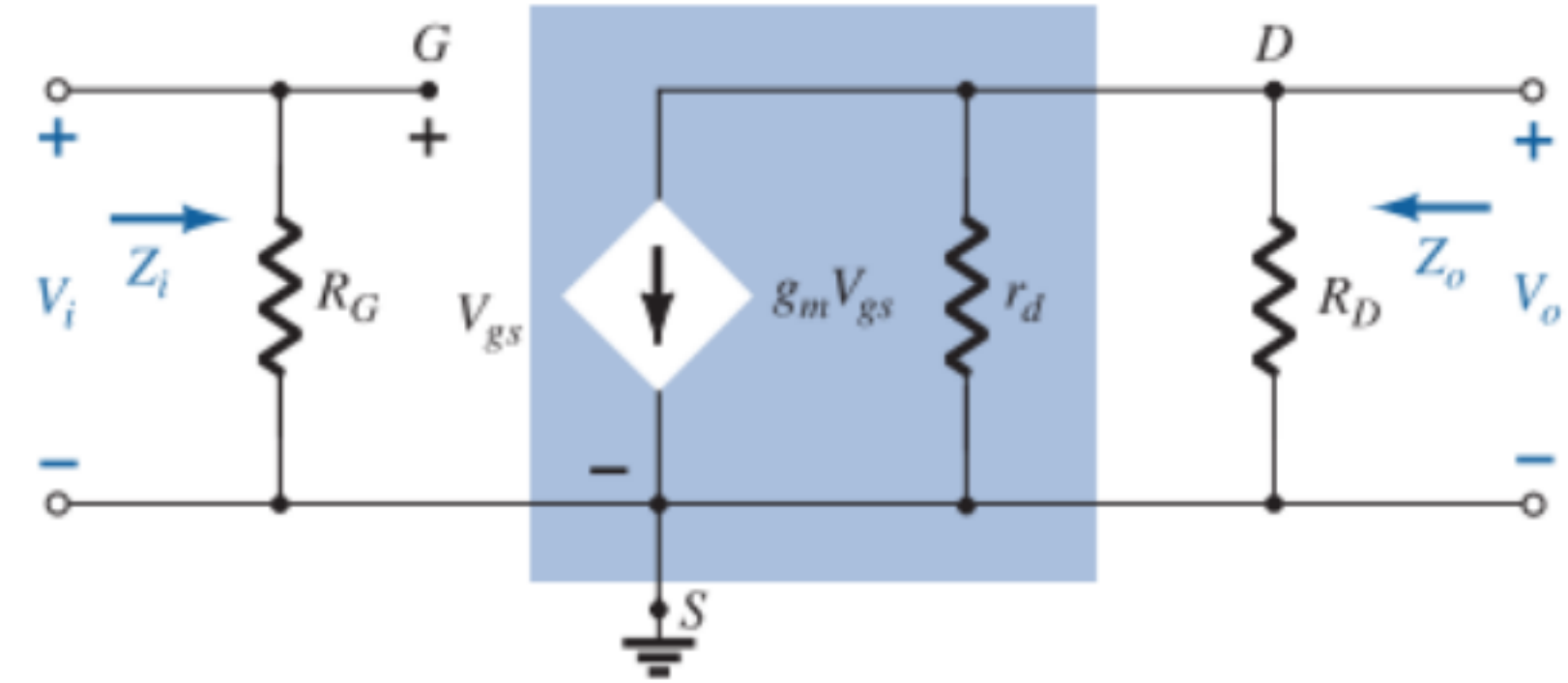
- Input Voltage, $v_i = V_{gs}$
- Output Voltage, $v_o = -g_m V_{gs} (R_D \parallel r_d)$
 $= -g_m v_i (R_D \parallel r_d)$
- Voltage gain, $A_v = \frac{v_o}{v_i} = -g_m (R_D \parallel r_d)$
- The negative sign in A_v clearly reveals a phase shift of 180° between input and output voltages.



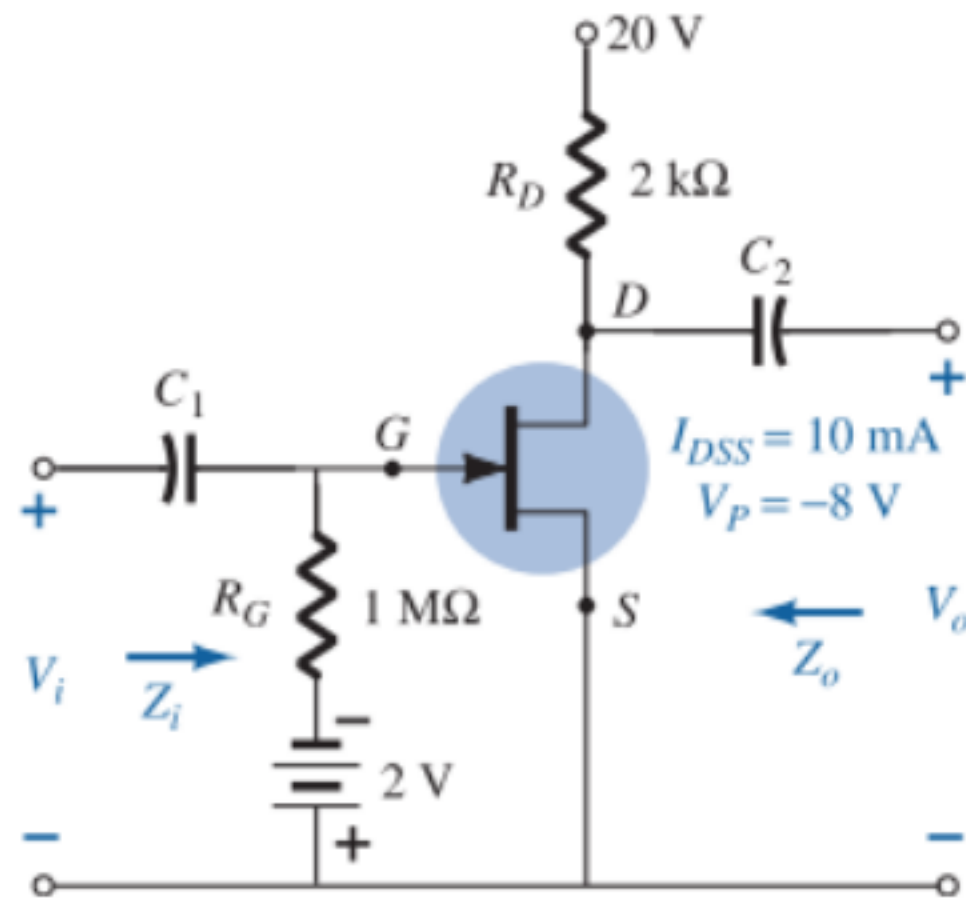
AC Analysis for Fixed-Bias Configuration (2)



- Input impedance, $Z_i = R_G$
- Output impedance, $Z_o = R_D \parallel r_d$
- When $r_d \geq 10R_D$, $Z_o \cong R_D$
- Input Voltage, $v_i = V_{gs}$
- Output Voltage, $v_o = -g_m V_{gs} (R_D \parallel r_d) = -g_m v_i (R_D \parallel r_d)$
- Voltage gain, $A_v = \frac{v_o}{v_i} = -g_m (R_D \parallel r_d)$
- The negative sign in A_v clearly reveals a phase shift of 180° between input and output voltages.



Problem 17: Consider $V_{GSQ} = -2V$, $I_{DQ} = 5.625 \text{ mA}$ & $y_{os} = 40 \mu S$. Determine g_m , r_d , Z_i , Z_o , A_v

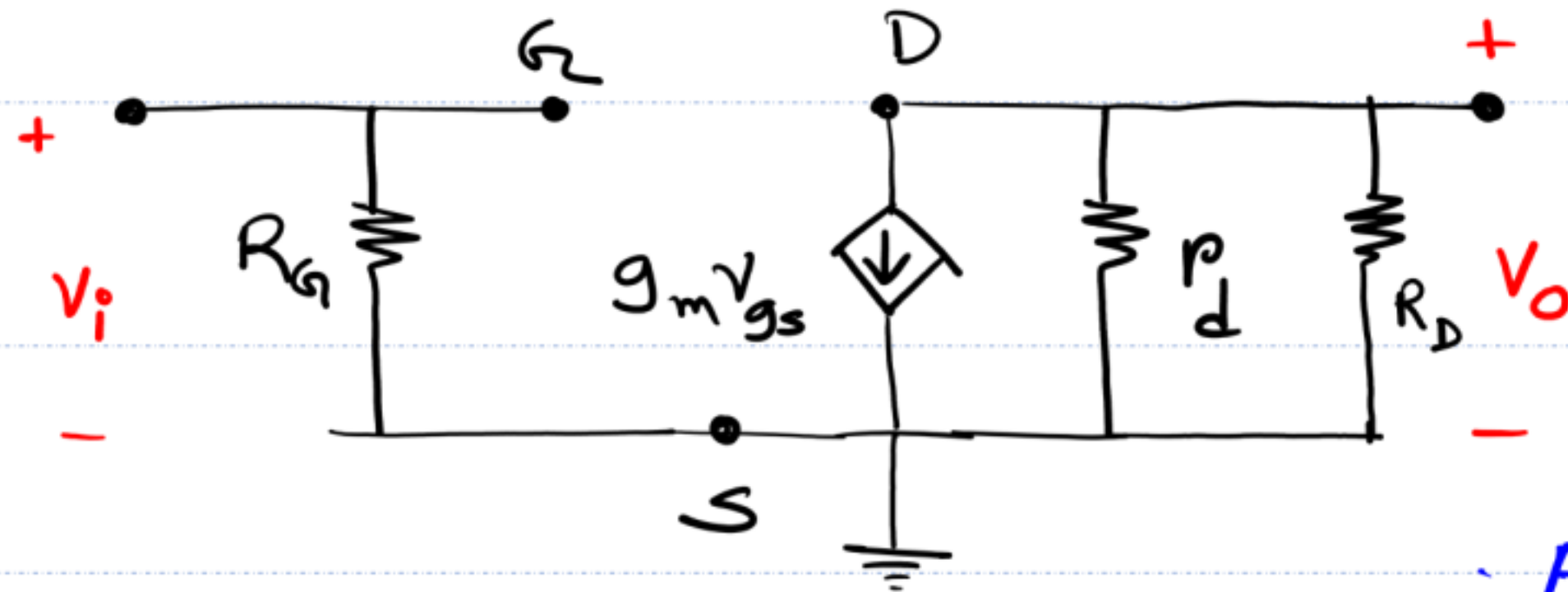


$$r_d = \frac{1}{y_{os}} = 25 \text{ k}\Omega$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right)$$

$$= \frac{2 \times 10 \text{ mA}}{8} \left(1 - \frac{-2}{-8}\right)$$

$$= 1.88 \text{ mS}$$



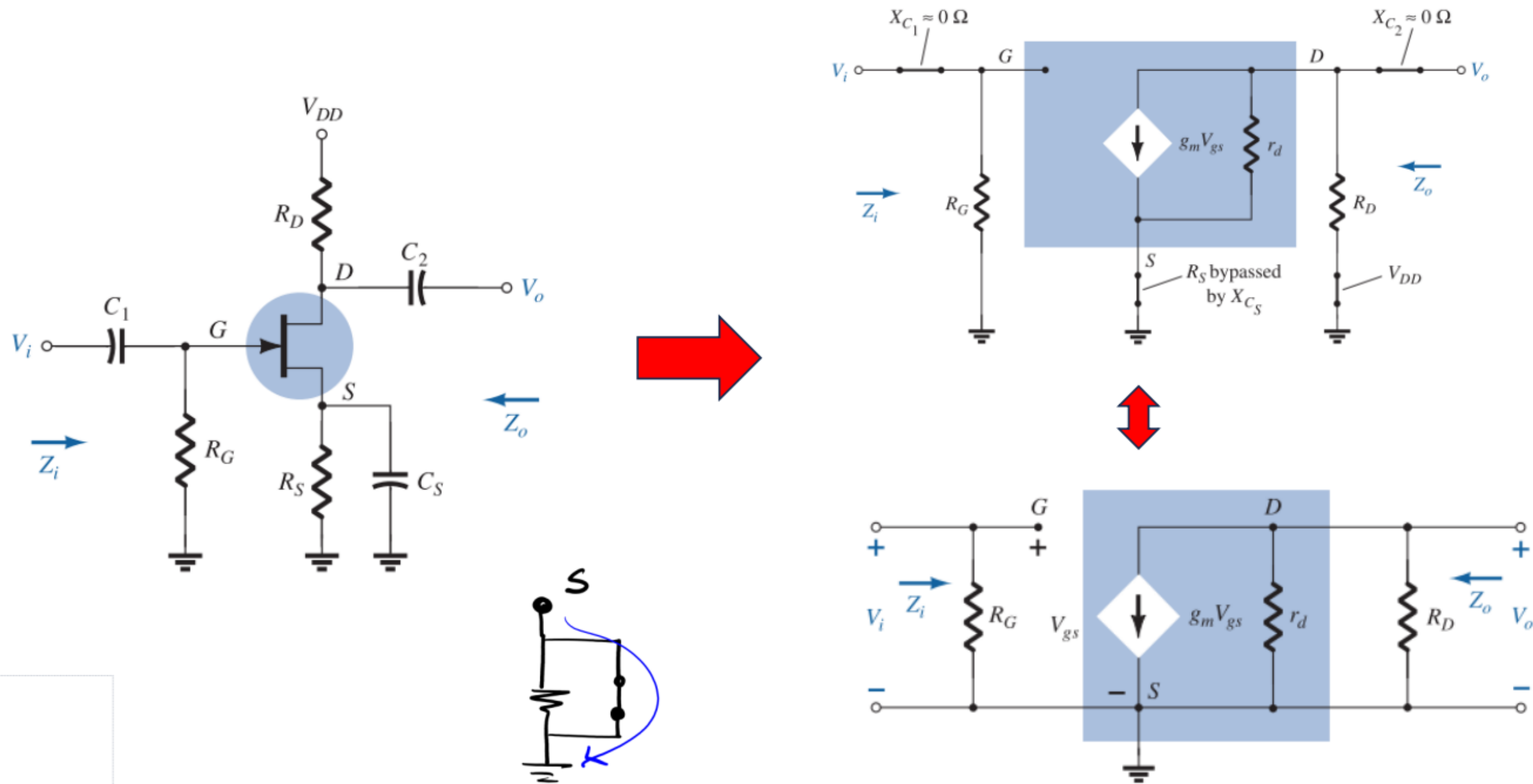
$$Z_i = R_G = 1 \text{ M}\Omega$$

$$Z_o = r_d \parallel R_D = 1.85 \text{ k}\Omega$$

$$A_v = \frac{V_o}{V_i} = -g_m (r_d \parallel R_D)$$

$$\therefore A_v = -g_m r_d \parallel R_D = -3.48$$

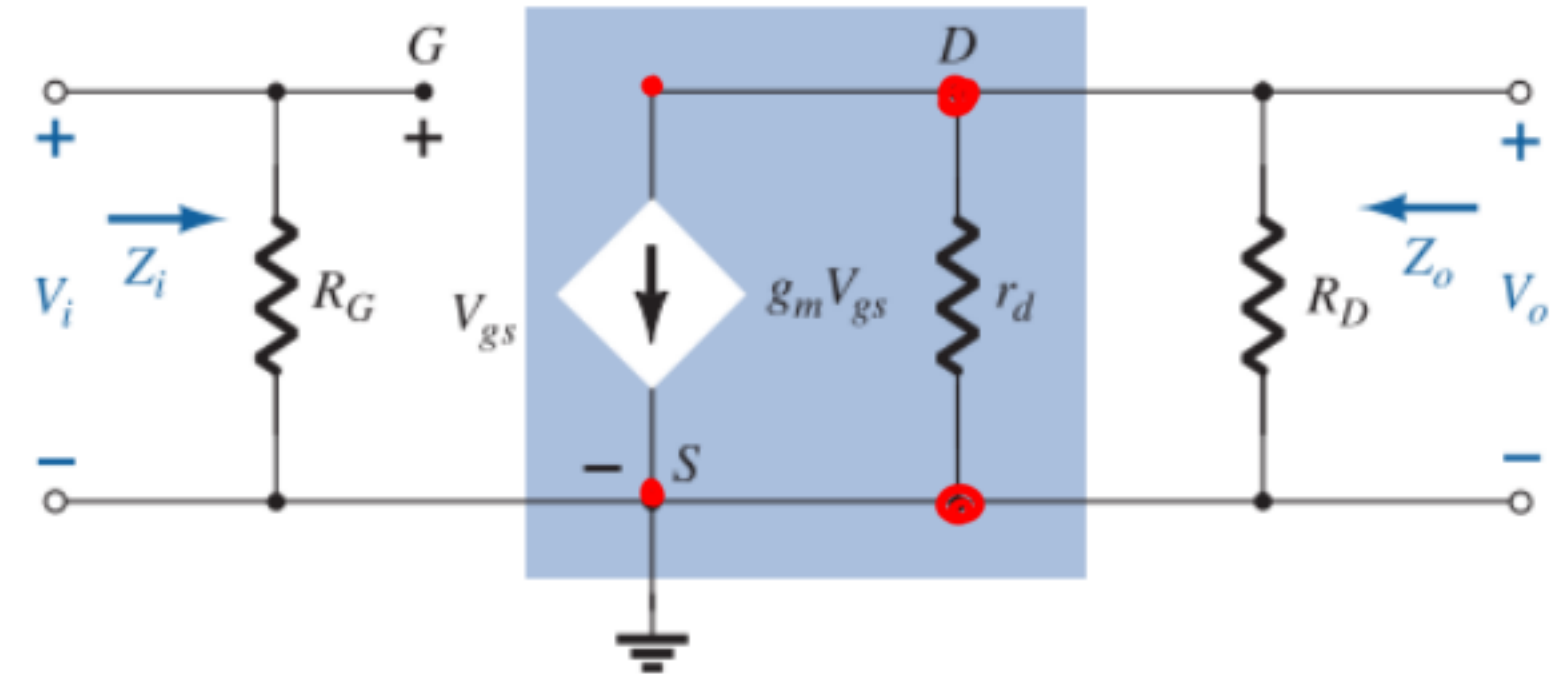
AC Analysis for Self-Bias Configuration (1) : R_S bypassed



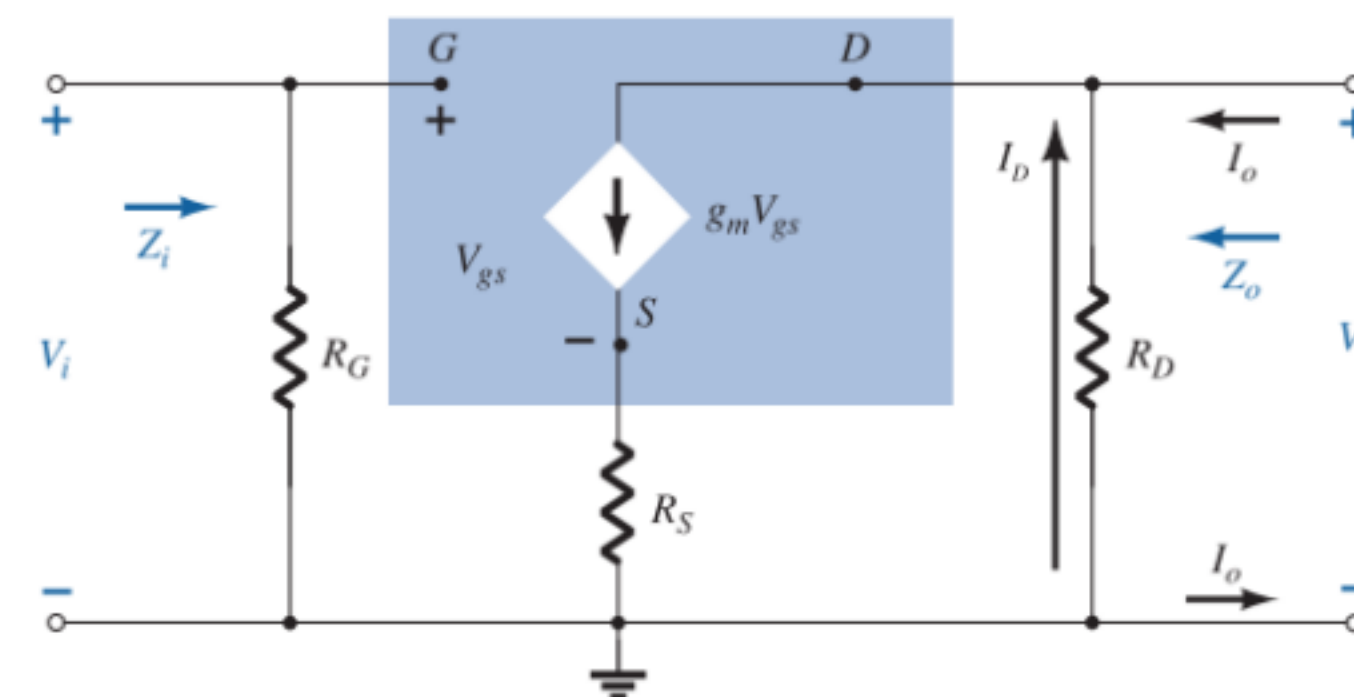
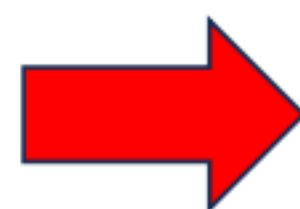
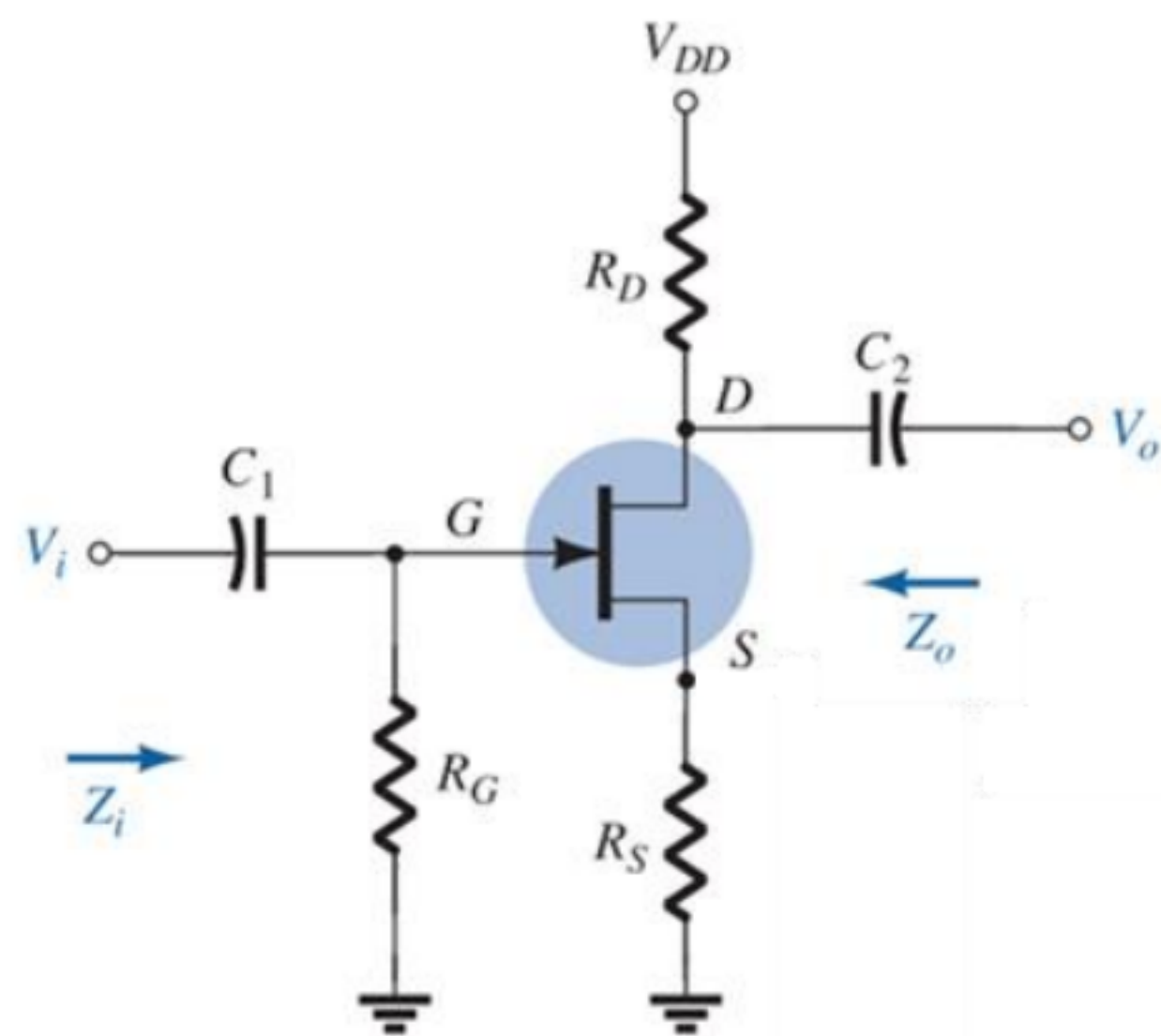
AC Analysis for Self-Bias Configuration (1) : R_S bypassed



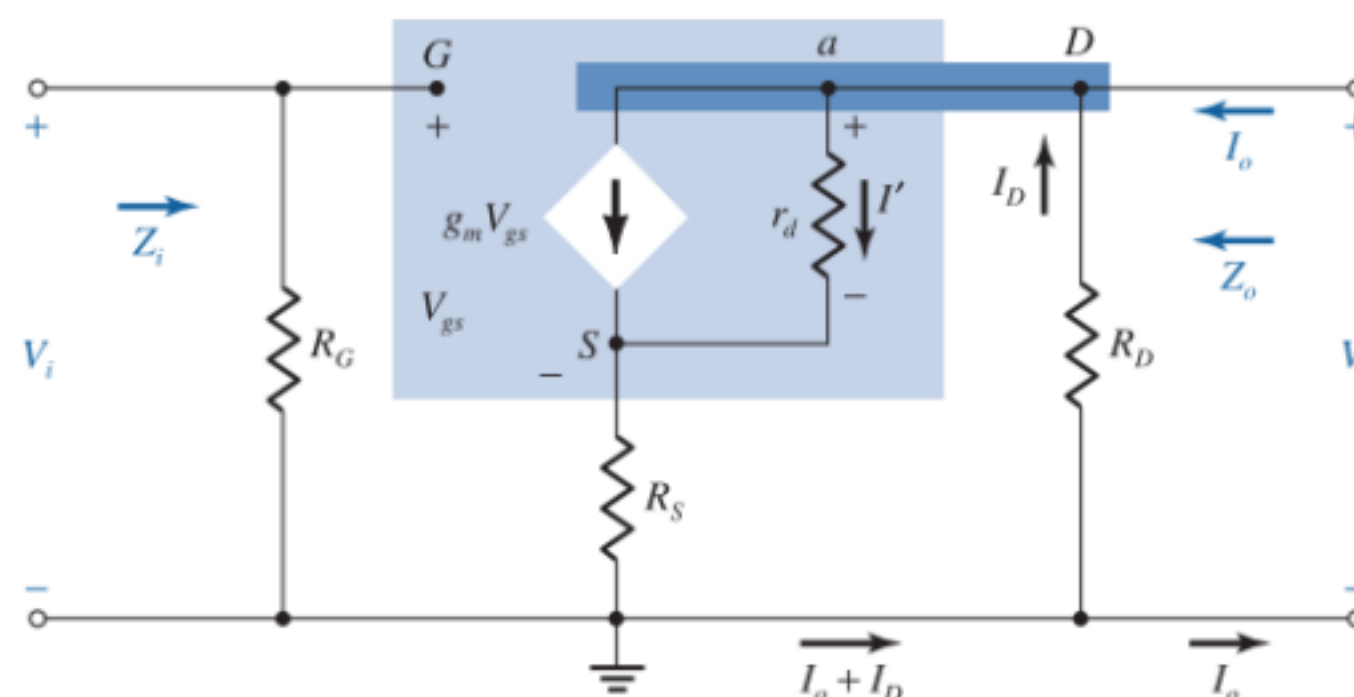
- Input impedance, $Z_i = R_G$
- Output impedance, $Z_o = R_D \parallel r_d$
- For cases when $r_d \geq 10R_D$, $Z_o \cong R_D$
- Input Voltage, $v_i = V_{gs}$
- Output Voltage, $v_o = -g_m V_{gs} (R_D \parallel r_d) = -g_m v_i (R_D \parallel r_d)$
- Voltage gain, $A_v = \frac{v_o}{v_i} = -g_m (R_D \parallel r_d)$
- The negative sign in A_v clearly reveals a phase shift of 180° between input and output voltages.



AC Analysis for Self-Bias Configuration (1) : R_S unbypassed

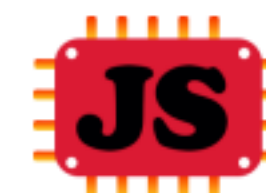


ignoring r_d



Without ignoring r_d

r_d



AC Analysis for Self-Bias Configuration (1) : R_S unbypassed



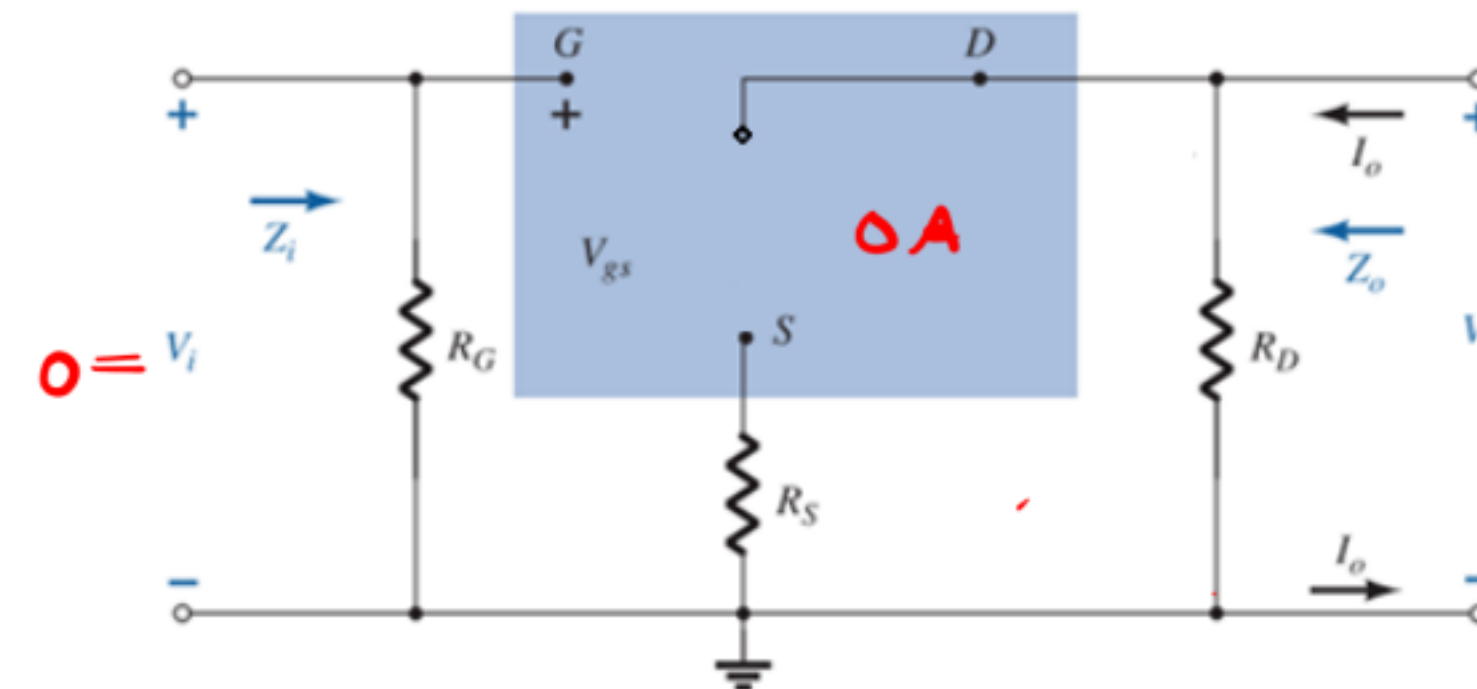
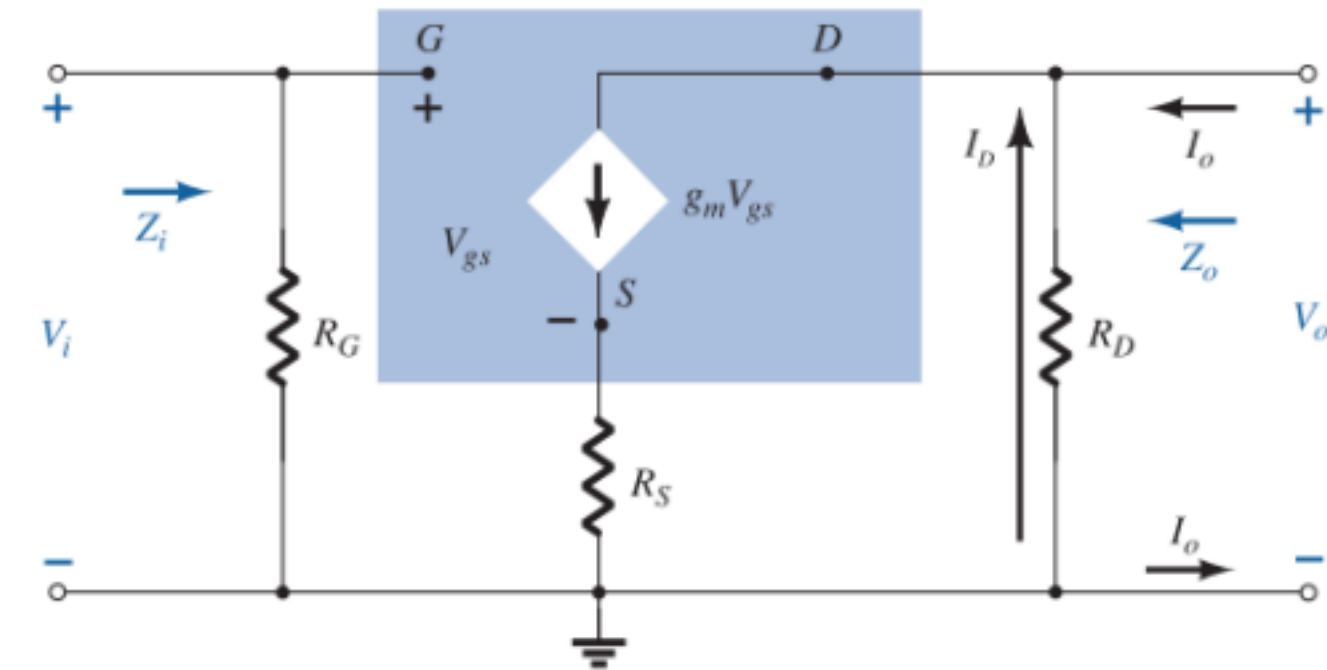
(IGNORING r_D)

- Input impedance, $Z_i = R_G$
- Output impedance, $Z_o = \frac{V_o}{I_o}$ (when $V_i = V_{gs} = 0$)

To find Z_o :

- Input voltage, V_i is set to zero. Thus, $V_{gs} = 0$ also.
- Current source, $I_D = g_m V_{gs} = 0$
- Thus, the only current in the output side is I_o
- I_o flows only through R_D as the D-S branch is open!

$$\therefore Z_o = \frac{V_o}{I_o} = R_D$$



AC Analysis for Self-Bias Configuration (1) : R_S unbypassed

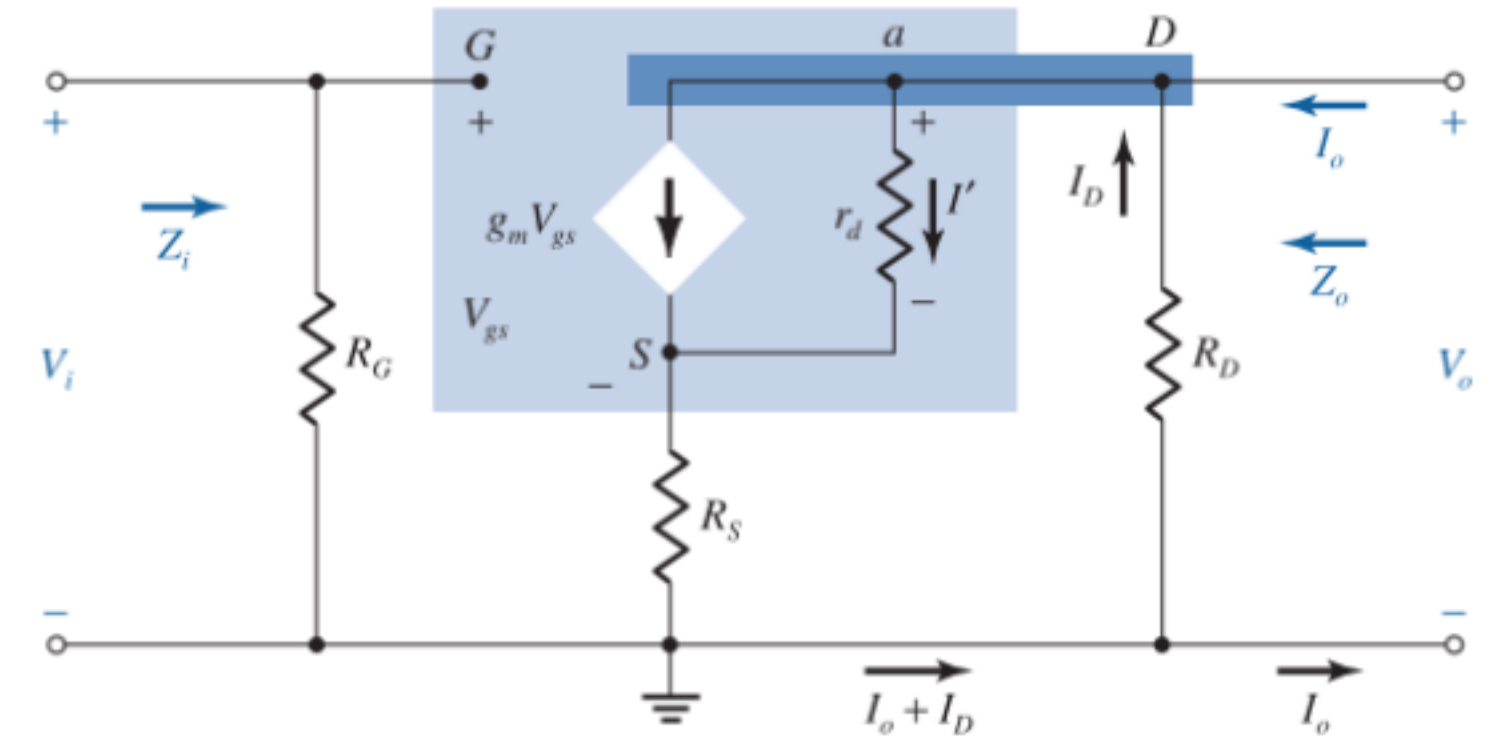


(WITHOUT IGNORING r_D)

- Input impedance, $Z_i = R_G$
- Output impedance, $Z_o = \frac{V_o}{I_o}$ (when $V_i = V_{gs} = 0$)
- Finding Z_o involves complex calculations & is tedious.
- So, only the result is shown here:

$$Z_o = \frac{\left[1 + g_m R_S + \frac{R_S}{r_d} \right]}{\left[1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d} \right]} R_D$$

- When $r_d \geq 10R_D$, $Z_o \cong R_D$



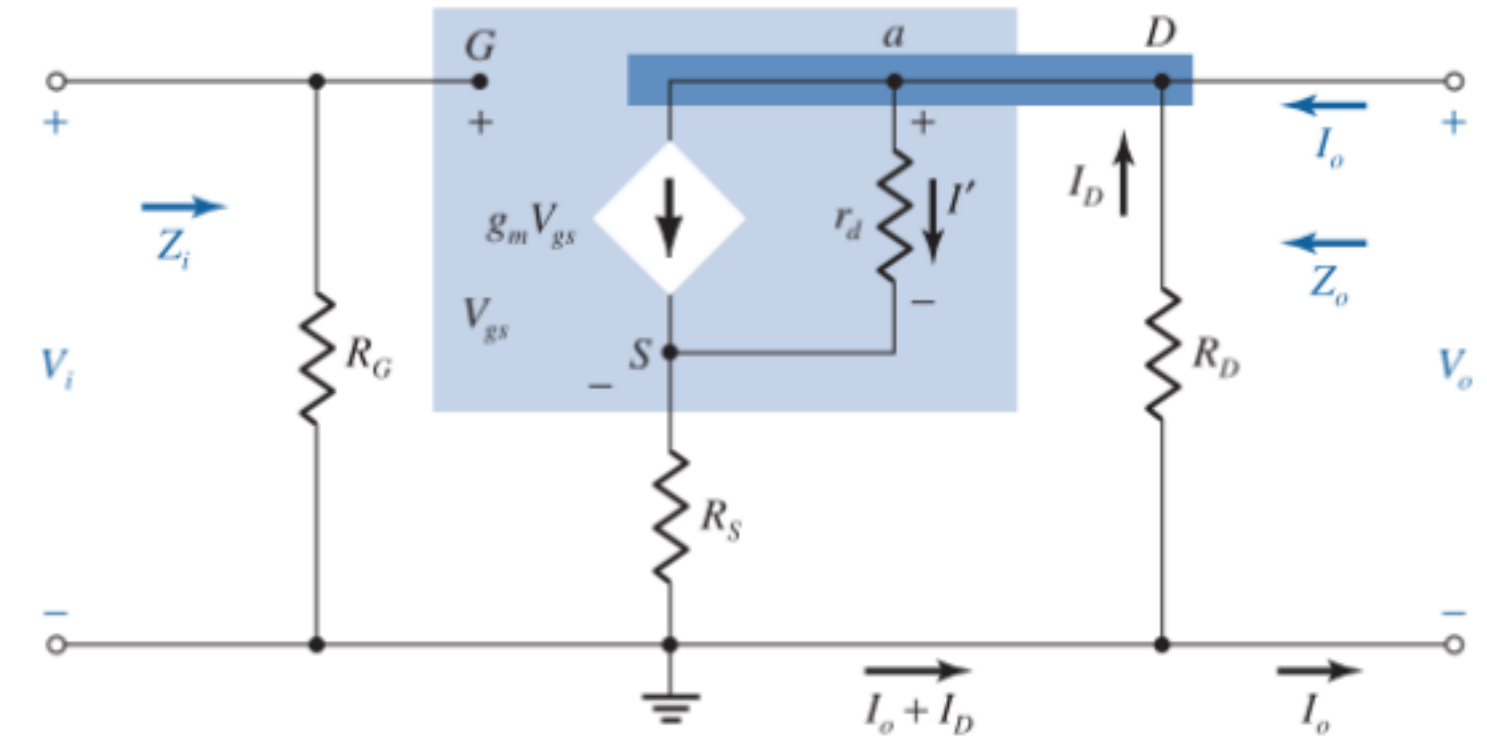
AC Analysis for Self-Bias Configuration (1) : R_S unbypassed



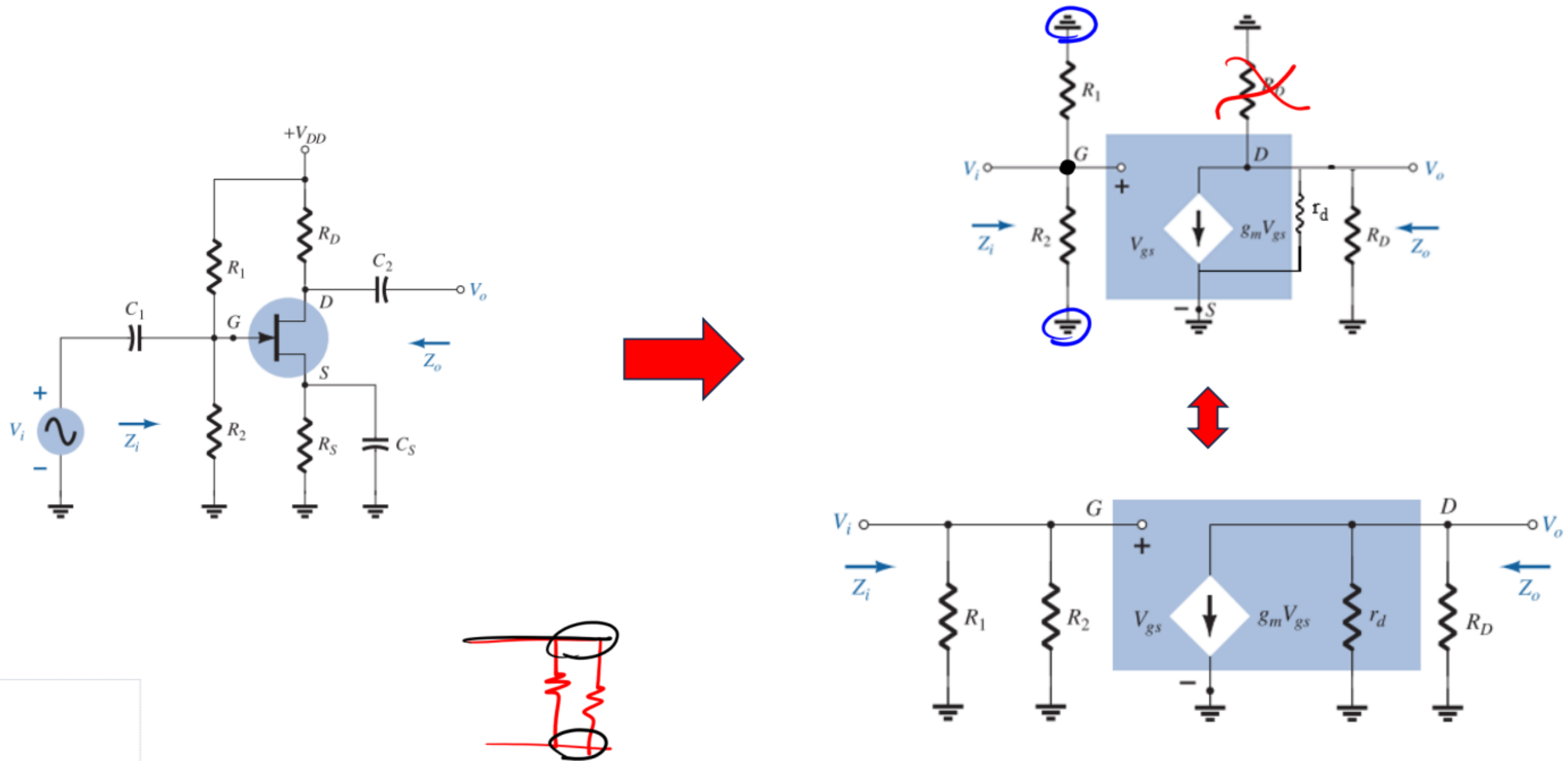
- **Voltage gain, $A_v = \frac{v_o}{v_i}$**
- Finding A_v requires many calculations & is tedious.
- So, only the result is shown here:

$$A_v = \frac{V_o}{V_i} = -\frac{g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

- When $r_d \geq 10(R_D + R_S)$, $A_v \cong -\frac{g_m R_D}{1 + g_m R_S}$



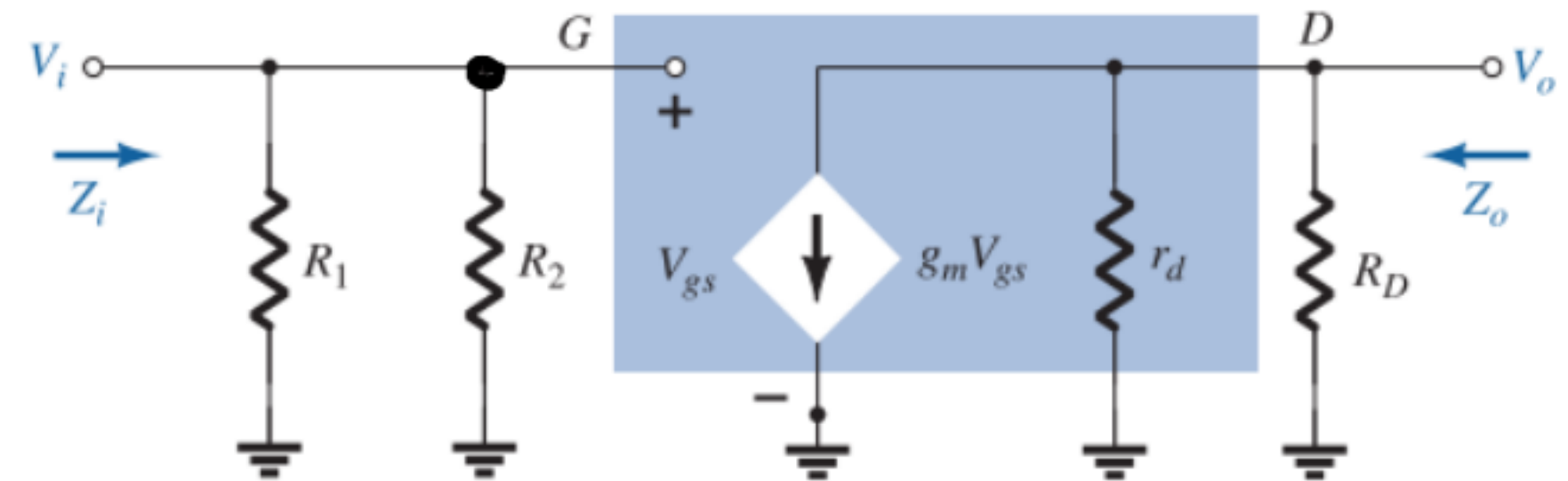
AC Analysis for Voltage-Divider Bias Configuration (1)



AC Analysis for Voltage-Divider Bias Configuration (2)



- Input impedance, $Z_i = R_1 \parallel R_2$
- Output impedance, $Z_o = R_D \parallel r_d$
- For cases when $r_d \geq 10R_D$, $Z_o \cong R_D$



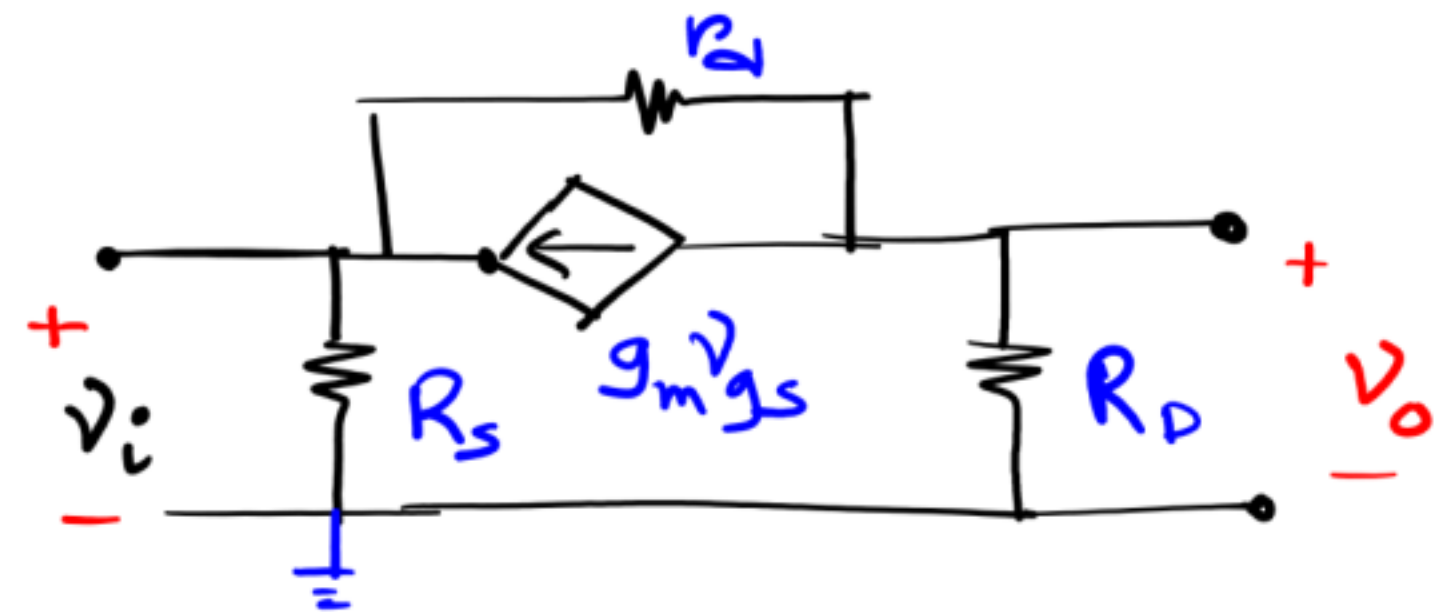
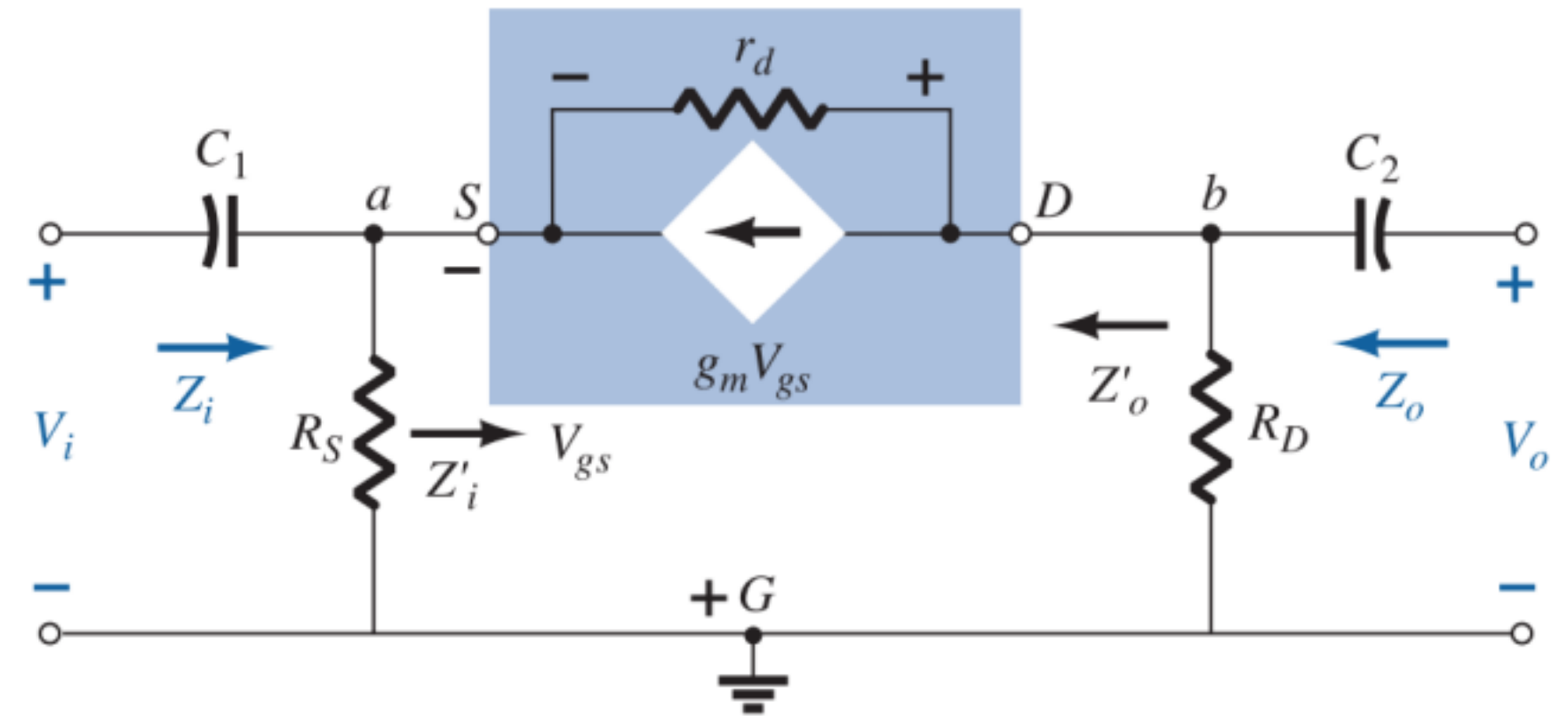
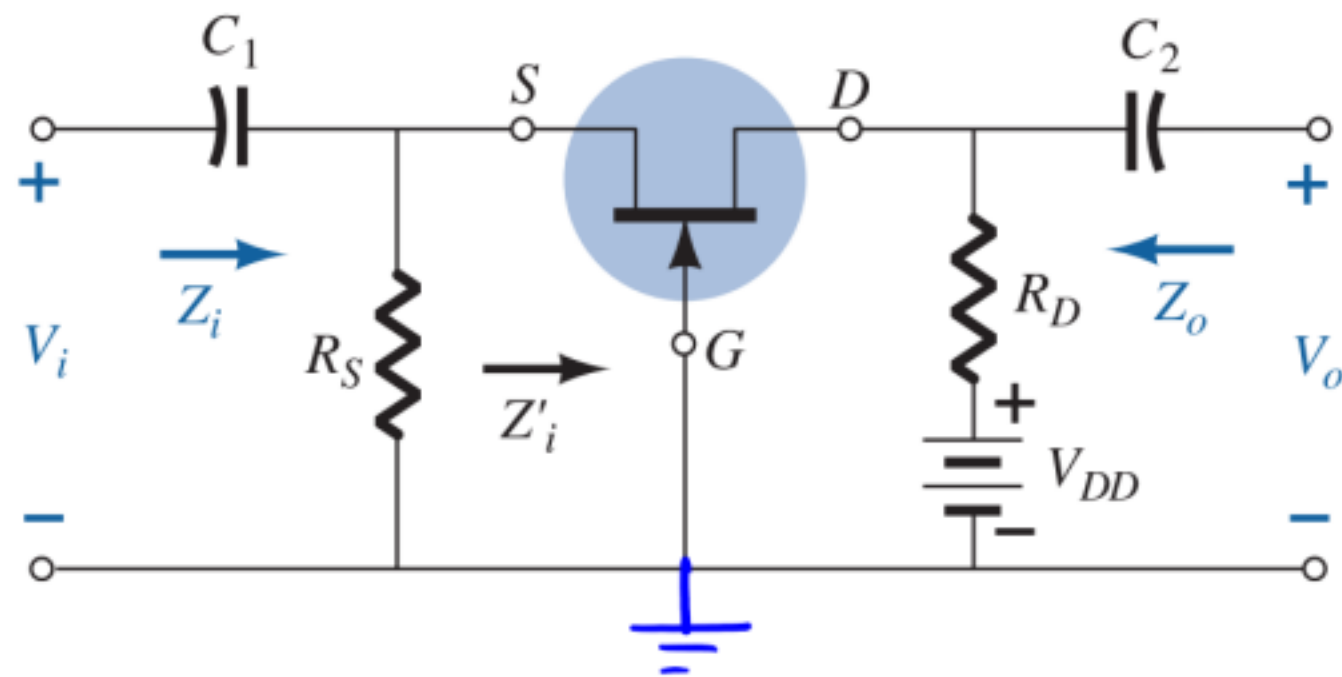
- Input Voltage, $v_i = V_{gs}$
- Output Voltage, $v_o = -g_m V_{gs} (R_D \parallel r_d) = -g_m v_i (R_D \parallel r_d)$
- Voltage gain, $A_v = \frac{v_o}{v_i} = -g_m (R_D \parallel r_d)$
- The negative sign in A_v clearly reveals a phase shift of 180° between input and output voltages.



AC Analysis for Common-Gate Configuration (1)



- Here, Gate is grounded. Input is in the Source, Output is taken from the Drain



AC Analysis for Common-Gate Configuration (2)

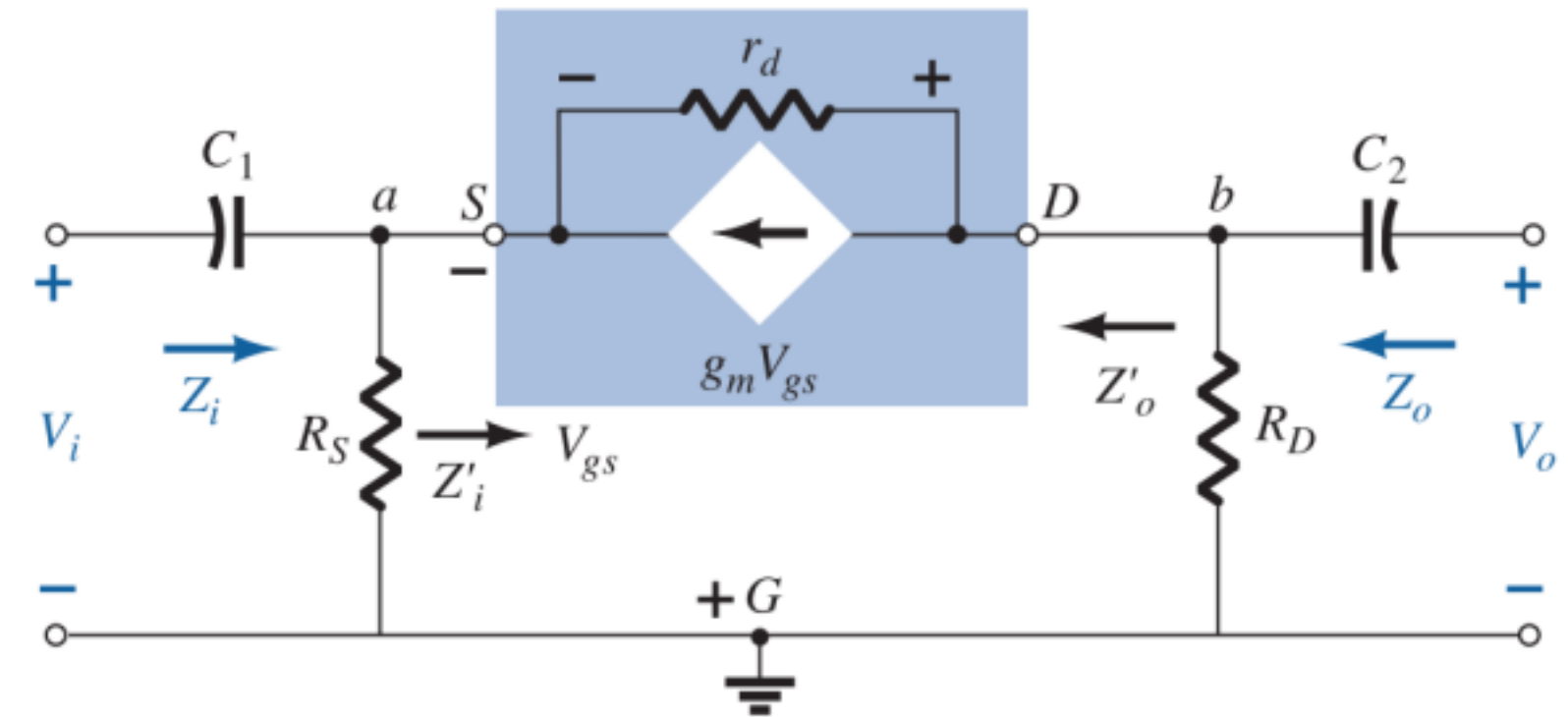


Input impedance, Z_i

- Finding Z_i requires many calculations & is tedious.
- So, only the result is shown here:

$$Z_i = R_S \parallel \left[\frac{r_d + R_D}{1 + g_m r_d} \right]$$

- When $r_d \geq 10R_D$, $Z_o \cong R_S \parallel \frac{1}{g_m}$



AC Analysis for Common-Gate Configuration (3)



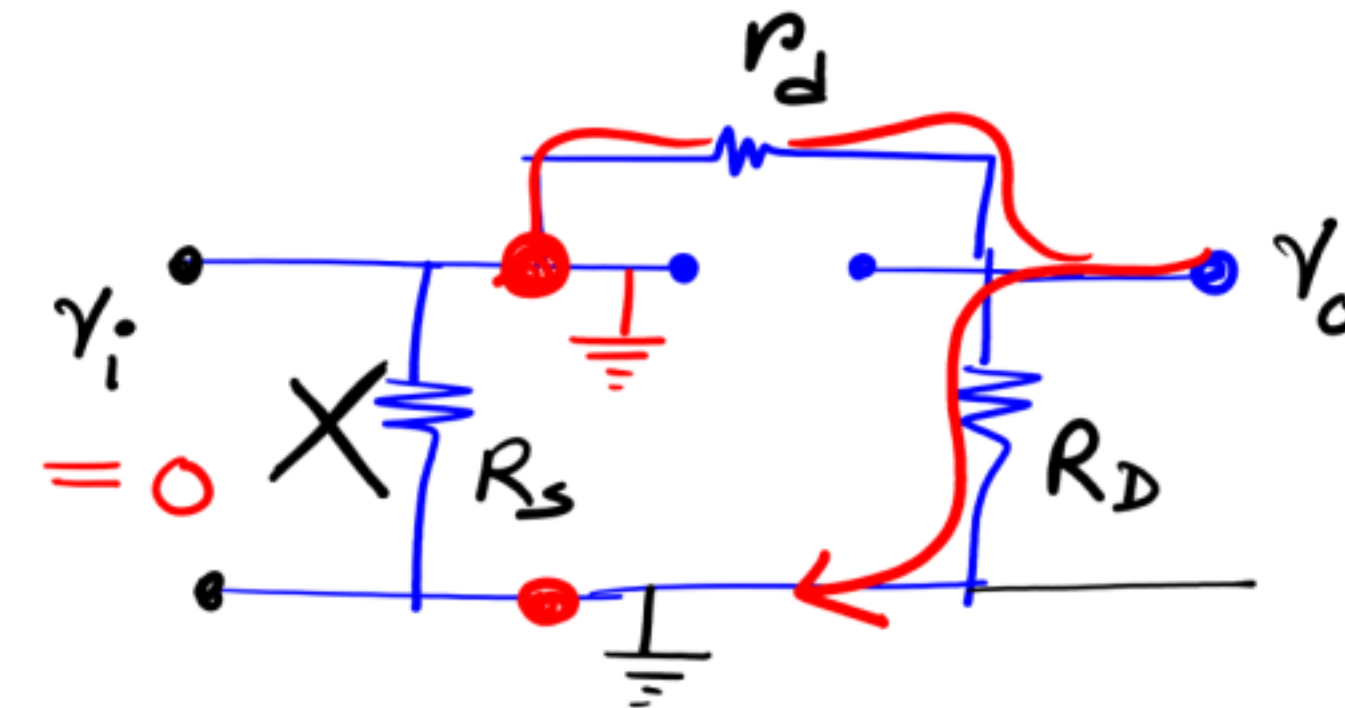
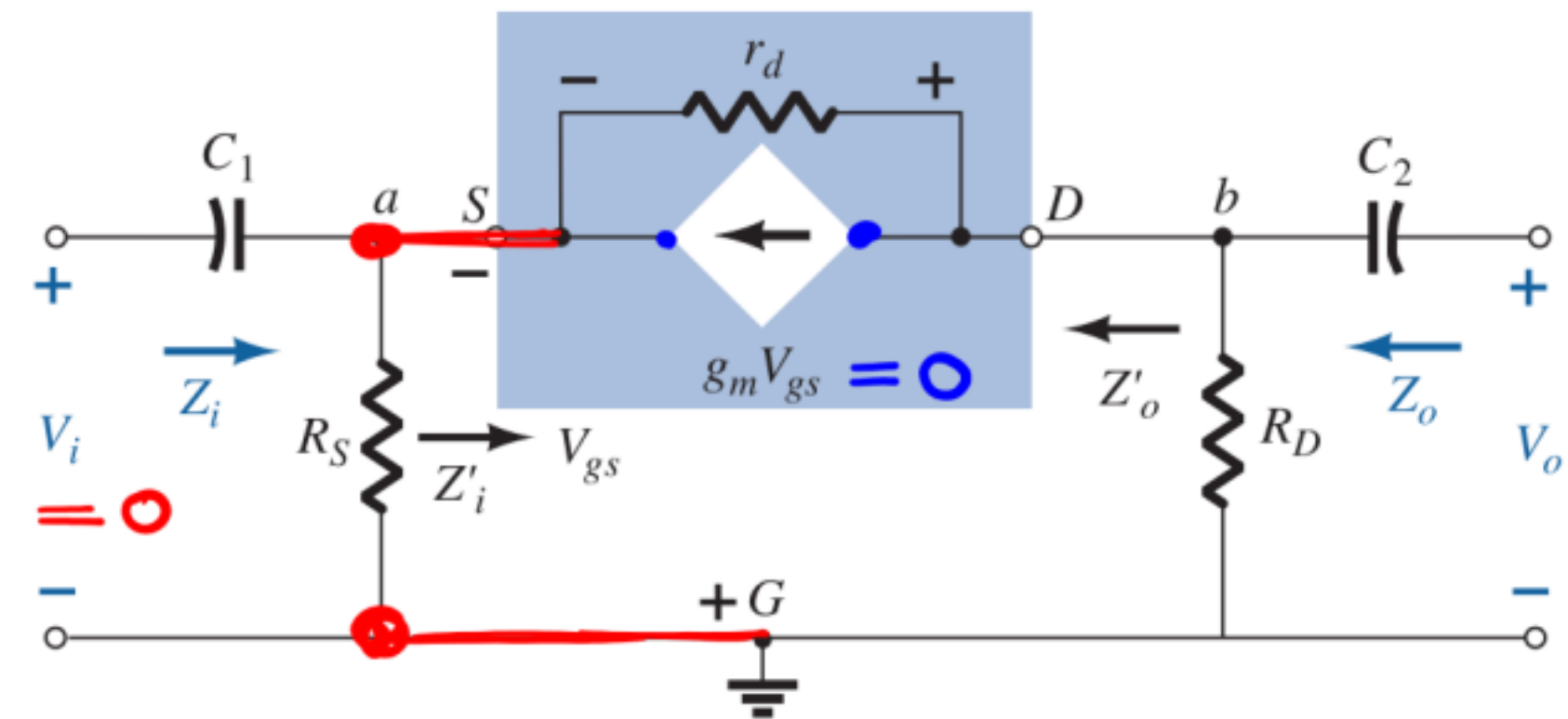
Output Impedance, Z_o

- Output impedance, $Z_o = \frac{V_o}{I_o}$ (**when $V_i = V_{gs} = 0$**)
- Setting $V_i = V_{gs} = 0$ makes R_S in-effective.
- As $V_{gs} = 0$, $I_d = g_m V_{gs} = 0 \rightarrow$ becomes open-circuit.

➤ r_d becomes parallel to R_D as $g_m V_{gs} = 0$

$$\therefore Z_o = R_D \parallel r_d$$

- For cases when $r_d \geq 10R_D$, $Z_o \cong R_D$



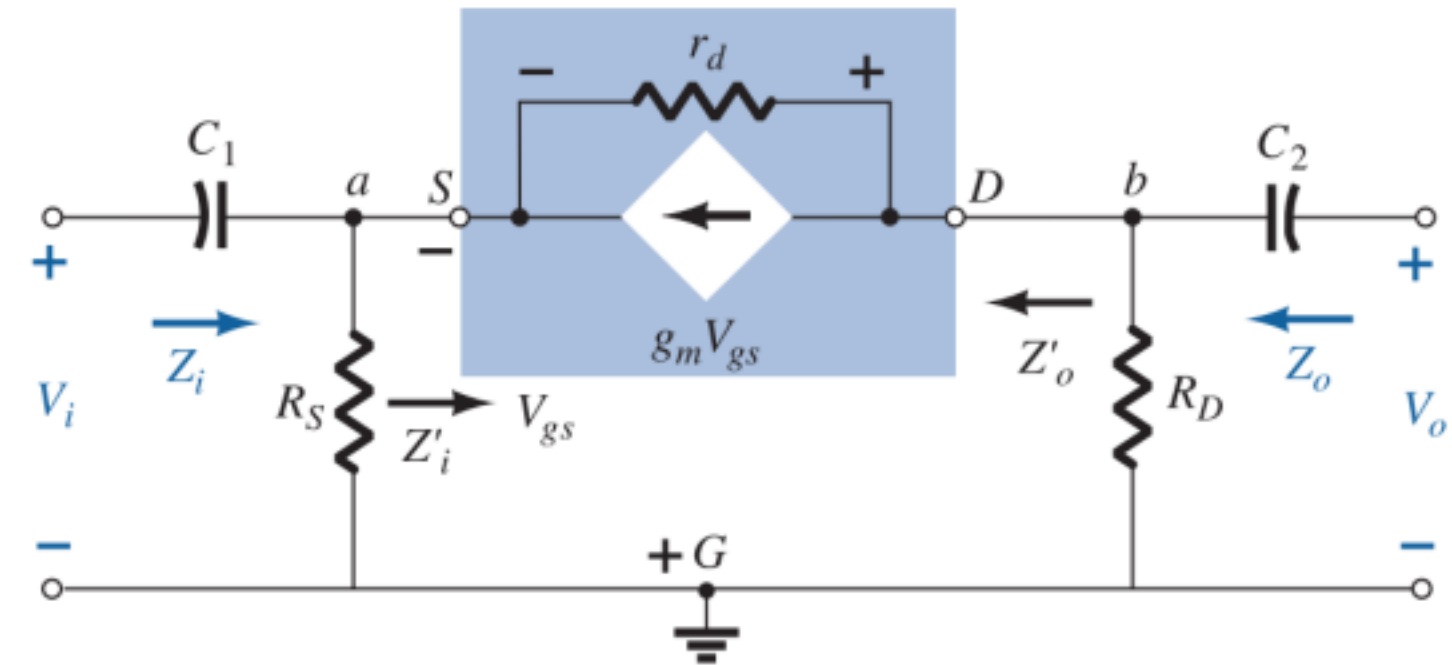
AC Analysis for Common-Gate Configuration (3)



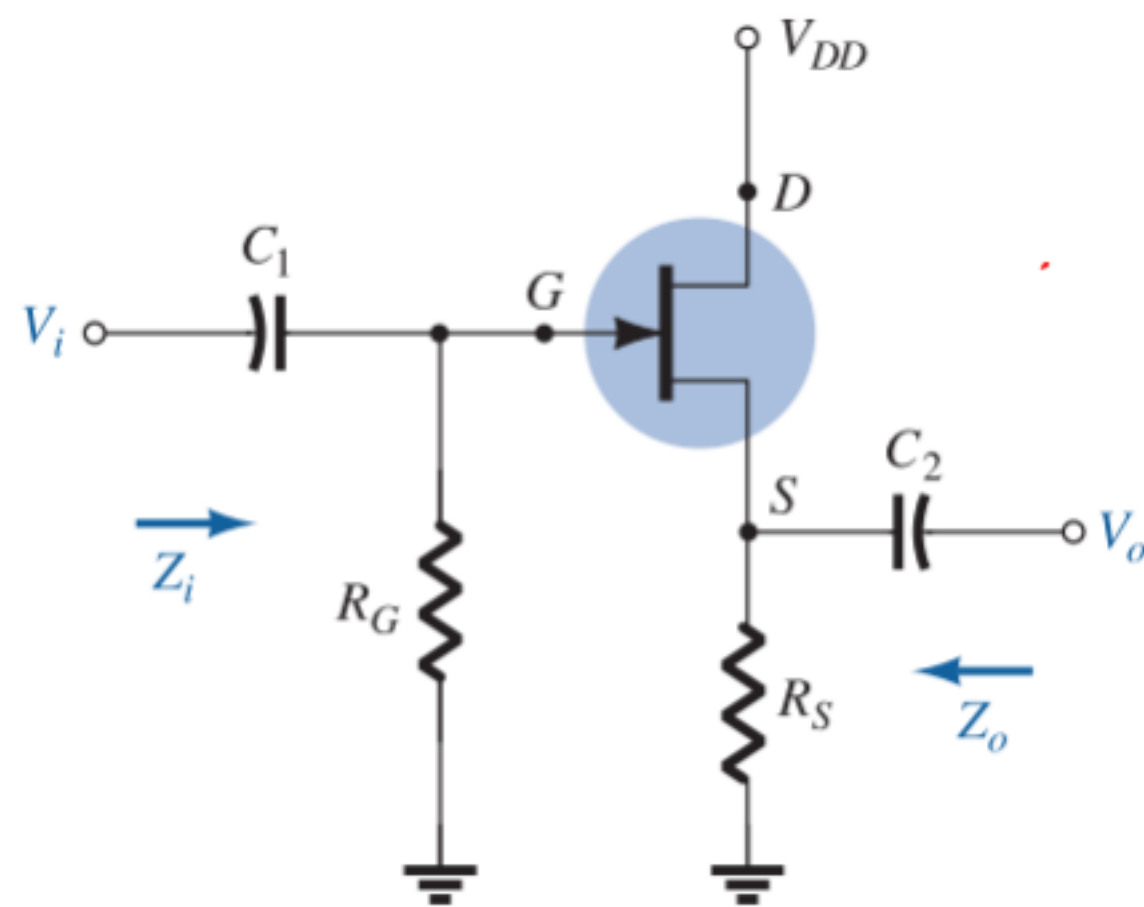
- **Voltage gain, $A_v = \frac{v_o}{v_i}$**
- Finding A_v requires many calculations & is tedious.
- So, only the result is shown here:

$$A_v = \frac{V_o}{V_i} = \frac{\left[g_m R_D + \frac{R_D}{r_d} \right]}{\left[1 + \frac{R_D}{r_d} \right]}$$

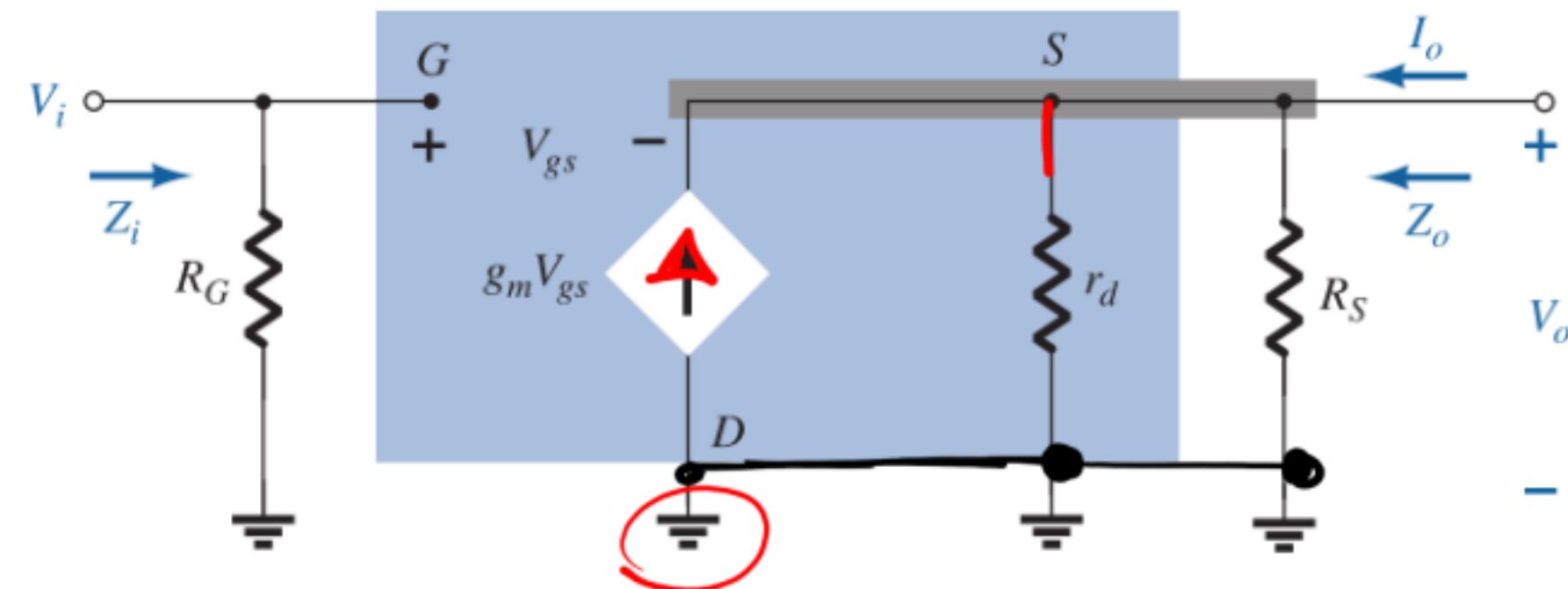
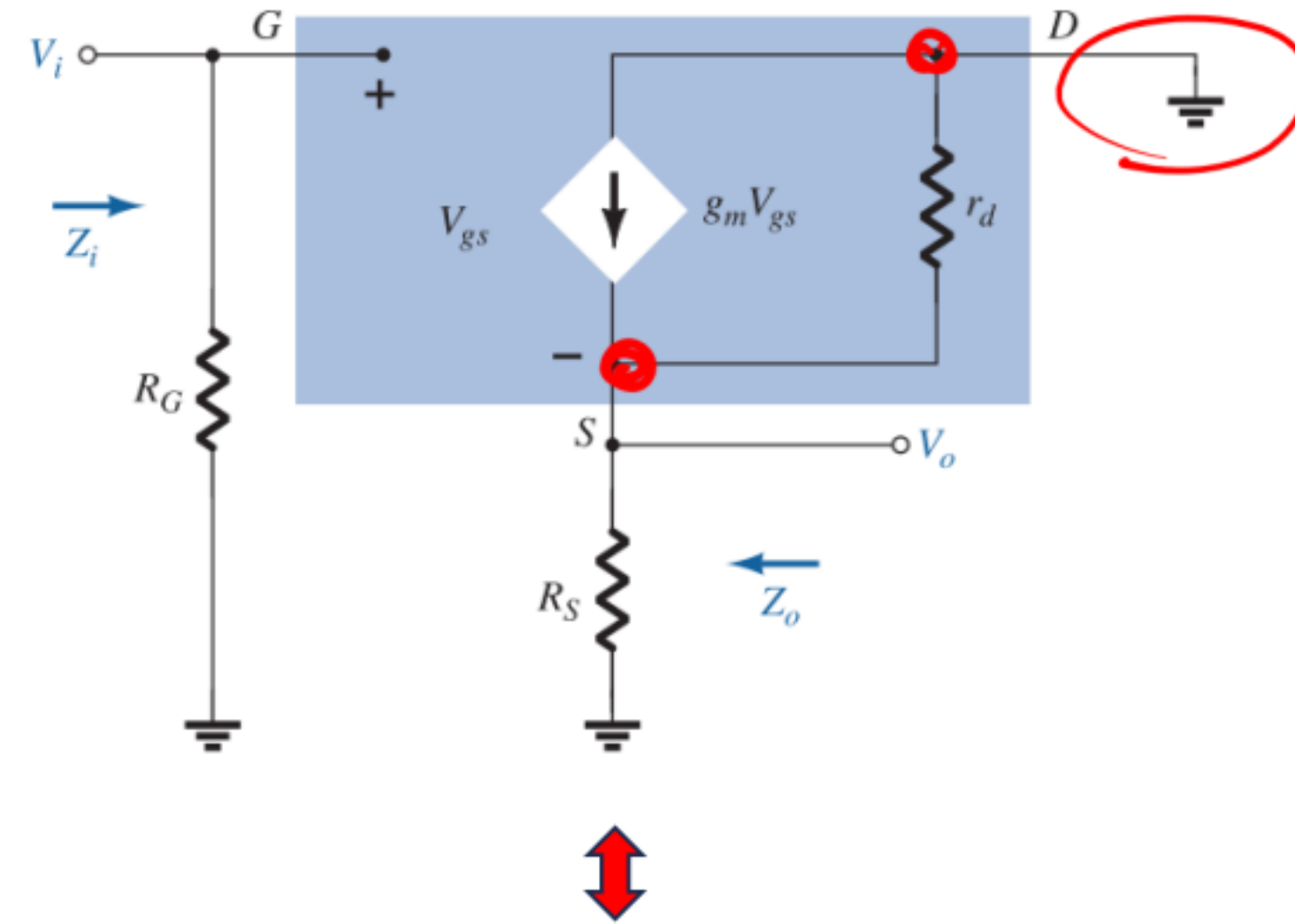
- When $r_d \geq 10(R_D)$, $A_v \cong g_m R_D$
- The fact that A_v is positive results in an in-phase relationship between V_i & V_o for the common-gate configuration.



AC Analysis for Common-Drain Configuration (1)



V_i
 V_o
gnd
GSD



AC Analysis for Common-Drain Configuration (2)



- Input impedance, $Z_i = R_G$
- Output Impedance, $Z_o = ?$

$$\hat{I}_o = \hat{I}_1 + \hat{I}_2 = \frac{V_o}{r_d \parallel R_S} + (-g_m V_{gs})$$

$$= \frac{V_o}{r_d \parallel R_S} + g_m V_s \rightarrow V_o$$

$$\hat{I}_o = \frac{V_o}{r_d \parallel R_S} + g_m V_o$$

$$\therefore Z_o = \frac{V_o}{\hat{I}_o}$$

$$= \frac{1}{\frac{1}{r_d \parallel R_S} + \frac{1}{1/g_m}}$$

$$= V_o \left(\frac{1}{r_d \parallel R_S} + \frac{1}{1/g_m} \right)$$

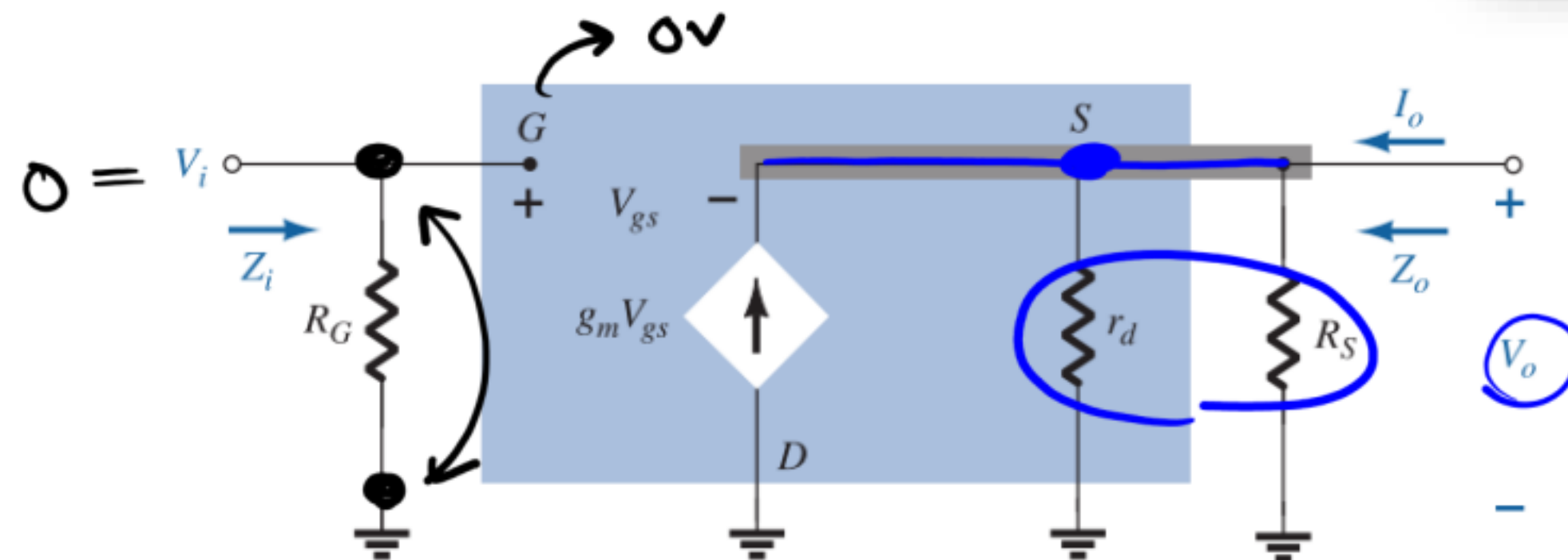
$$Z_o = r_d \parallel R_S \parallel 1/g_m$$

$\rightarrow$

$$Z_o \cong R_S \parallel 1/g_m$$

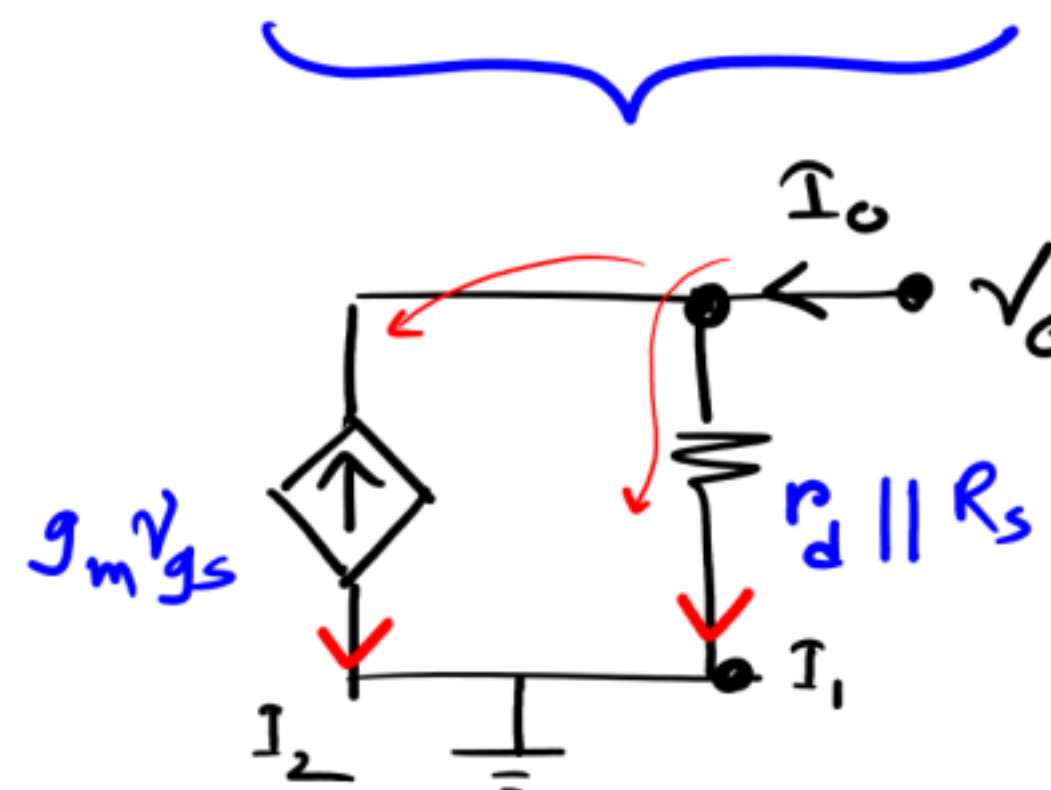
$r_d \geq 10R_S$

$$\frac{1}{\frac{1}{R_x} + \frac{1}{R_y}} = \frac{1}{\frac{1}{R_p}} = R_p$$



$$V_{gs} = -V_s$$

$$\approx V_g = 0$$



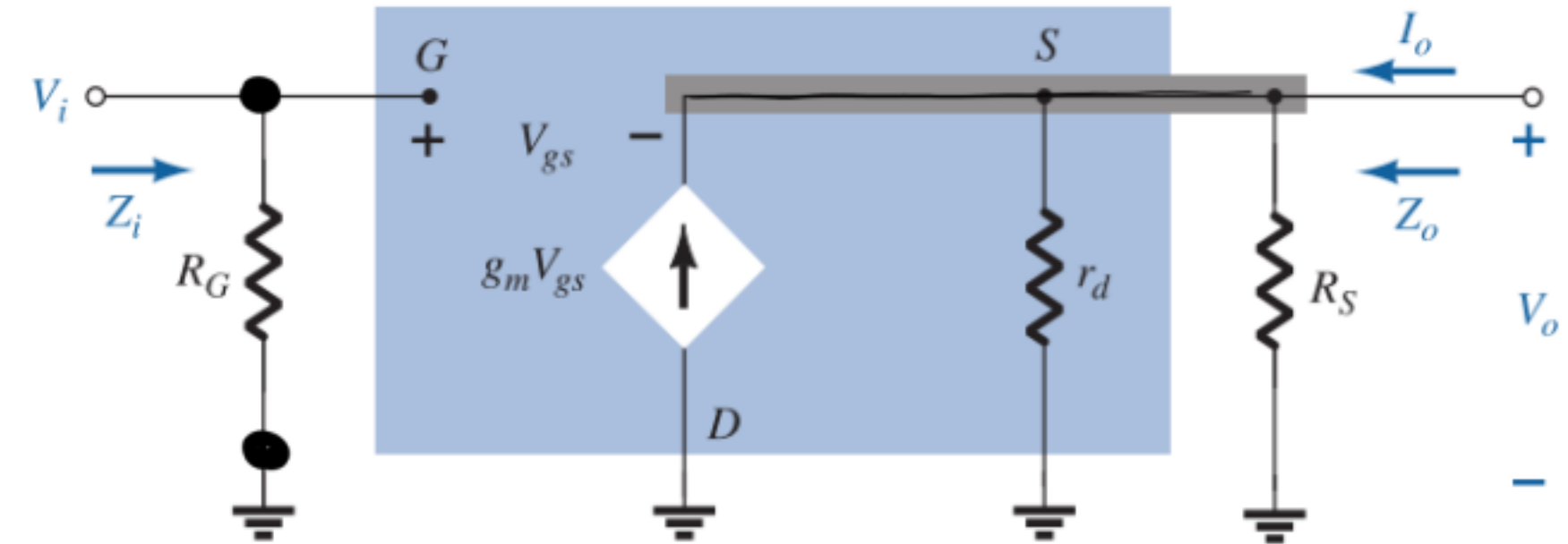
AC Analysis for Common-Drain Configuration (3)



Voltage gain, A_v : $v_i = v_{gs}$ ——— ①

$$\begin{aligned} v_o &= g_m v_{gs} (r_d \parallel R_S) \\ &= g_m (v_g - v_s) (r_d \parallel R_S) \\ &= g_m (v_i - v_o) (r_d \parallel R_S) \end{aligned}$$

$$\begin{aligned} v_o &= k (v_i - v_o) \longrightarrow \therefore v_o (1 + k) = k v_i \quad \therefore \frac{v_o}{v_i} = \frac{k}{1 + k} \\ &= \frac{g_m (r_d \parallel R_S)}{1 + g_m (r_d \parallel R_S)} \end{aligned}$$



$$A_v = \frac{V_o}{V_i} = \frac{g_m (r_d \parallel R_S)}{1 + g_m (r_d \parallel R_S)}$$

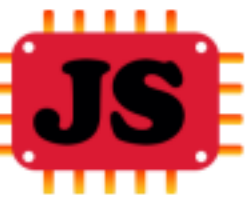
$\longrightarrow$

$$A_v = \frac{V_o}{V_i} \cong \frac{g_m R_S}{1 + g_m R_S}$$

$r_d \geq 10 R_S$



THE END



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